

Where the Seconds Go: Supplement

The prediction scorecard

Every prediction committed before a GPU job started, split into its separable clauses and scored under the rule of the paper’s appendix “Prediction log”. Jobs 073–098: generated by `scripts/scorecard.py` from `prereg/scorecard.clauses.json`. Job 099, the host panel: scored by machine by `scripts/panel_099.py` (`prereg/scorecard_099.json`); outcome notes in `prereg/panel_outcome_099.md`. Job 100: scored by machine by `scripts/job100.py` (`prereg/scorecard_100.json`); jobs 101 and 102 by `scripts/job101.py` and `scripts/job102.py`; jobs 103 and 104 by `scripts/job103.py`; job 105 by `scripts/job105.py`; job 106 by `scripts/job106.py`; job 107 by `scripts/job107.py`; job 108 by `scripts/job108.py`; jobs 109–111 by `scripts/job109.py`; job 112 by `scripts/job112.py`; job 113 by `scripts/job113.py`; jobs 114 and 115 by `scripts/job114.py`.

Table S1: Prediction scorecard, jobs 073–098: every committed prediction split into its separable clauses and scored under one rule. *Held*: the point estimate is on the predicted side and the 95% paired-bootstrap interval excludes the threshold (for a band, lies inside it). *Held (point)*: the point estimate is on the predicted side but the interval includes the threshold, or no interval exists. *Untested*: the clause could not be evaluated. *Void*: the job was voided (084, a throttled card; its predictions are scored on 084b). Types: sign (a direction), threshold (a numeric bar), band (an interval), equality (a deterministic outcome). Measured values are ratios of mean speeds with 95% intervals unless the clause says otherwise; `prereg/scorecard.clauses.json` names the source of every number. Totals: 565 clauses, 218 held, 193 held (point), 142 failed, 8 untested, 4 void.

Job	Clause	Type	Measured	Status
Jobs 073–075 : 20 clauses, 7 held, 3 held (point), 9 failed, 1 untested, 0 void				
073 P1a	ours+FETCH $\geq$ 5% ahead of FreeToken, 11%	threshold	1.109 [1.098, 1.120]	held
073 P1b	ours+FETCH $\geq$ 5% ahead of FreeToken, 25%	threshold	1.084 [1.076, 1.092]	held
073 P1c	ours+FETCH within $\pm 5\%$ of FreeToken, 40%	band	0.941 [0.933, 0.951]	failed
073 P2	ours+FETCH $\geq 1.8\times$ llama.cpp, every budget	threshold	1.832 [1.808, 1.856]; 2.358 [2.322, 2.396]; 2.647 [2.605, 2.686]	held
073 P3a	same-problem warm-up reads 5-15% faster, the two caches	band	0.991 [0.962, 1.028]; 0.997 [0.986, 1.008]	failed
073 P3b	same-problem warm-up within $\pm 2\%$, llama.cpp	band	0.962 [0.934, 0.979]	failed
073 P4	$\geq 80\%$ of greedy outputs identical to llama.cpp’s, every system	equality	ours (6 configurations): 0.000-0.033; FreeToken (6 configurations): 0.000; llama.cpp -ncmoe 32 / 22: 0.700 / 0.400	failed
074 P1	ec-bench launch time ≤ 0.25 ms, C51	threshold	0.201	held (point)
074 P2a	ours v2 ≥ 127 tok/s, C51	threshold	95.1	failed
074 P2b	ours v2 ahead of FreeToken offload, C51	sign	0.831 [0.817, 0.845]	failed
074 P3a	ours v2 ≥ 59 tok/s, C14	threshold	41.5	failed
074 P3b	ours v2 ≥ 91 tok/s, C32	threshold	65.3	failed
074 P4	half-life 64 within $\pm 2\%$ of half-life 16, C51	band	1.012 [0.997, 1.027]	held (point)
074b P1	GPU sampling cuts between-step host time by ≥ 1 ms	threshold	1.6	held (point)
074b P2a	ours v2 $\geq 10\%$ faster with GPU sampling, C51	threshold	1.117 [1.109, 1.125]	held
074b P2b	stock llama.cpp $\geq 3\%$ faster with GPU sampling	threshold	–	<i>untested</i>
074b P3	ours v2+FETCH with GPU sampling $\geq$ FreeToken offload, C51	sign	1.074 [1.063, 1.085]	held
075 P1	ours ahead of FreeToken at 40% (RTX PRO 6000)	sign	1.122 [1.107, 1.137]	held
075 P2	ours within -3 to +10% of FreeToken at 60%	band	1.118 [1.109, 1.127]	failed
075 P3	ours at 60% $\geq 55\%$ of the all-in-VRAM speed	threshold	0.726 [0.717, 0.735]	held
Jobs 076–081 : 23 clauses, 19 held, 4 held (point), 0 failed, 0 untested, 0 void				
076 P1	ours ahead of FreeToken at 11/25/40%	sign	1.205 [1.190, 1.220]; 1.217 [1.201, 1.234]; 1.118 [1.108, 1.129]	held

continued

Job	Clause	Type	Measured	Status
076 P2	ours $\geq 1.8\times$ llama.cpp, every budget	threshold	1.924 [1.894, 1.956]; 2.683 [2.631, 2.735]; 3.191 [3.130, 3.250]	held
076 P3	the law within 8% on own-text runs (four configurations)	band	1.009 [0.962, 1.066]; 1.006 [0.947, 1.086]; 1.052 [0.960, 1.164]; 0.960 [0.909, 1.024]	held (point)
077 P1a	both entrants run end to end on gpt-oss-120b	equality	30 of 30 problems at every budget for both	held
077 P1b	both entrants faster than llama.cpp at 25/40%	sign	pipeshard 25 / 40%: 1.348 [1.336, 1.360]; 1.417 [1.399, 1.435]; leloch 25 / 40%: 1.446 [1.431, 1.461]; 1.280 [1.269, 1.291]	held
077 P2	ours ahead of both entrants, every budget	sign	1.421 [1.403, 1.439]; 1.896 [1.871, 1.921]; 2.343 [2.293, 2.391]	held
078 P1	ours ahead of FreeToken at 12.5/25% (Qwen3)	sign	1.071 [1.063, 1.080]; 1.067 [1.043, 1.094]	held
078 P2	ours ahead of KTransformers, every budget	sign	2.168 [2.140, 2.197]; 3.059 [2.975, 3.162]; 4.420 [4.310, 4.550]	held
078 P3	ours $\geq 1.5\times$ llama.cpp, every budget	threshold	2.122 [2.092, 2.151]; 3.051 [2.967, 3.152]; 4.470 [4.350, 4.609]	held
079 P1a	host-side time per step < 1 ms	threshold	0.266	held (point)
079 P1b	host-side time < 10% of the token time	threshold	0.027	held (point)
079 P2	$\geq 1.5\times$ FreeToken’s kernels per token	threshold	1.606	held (point)
079 P3	ours within $\pm 5\%$ of FreeToken at 43.75%	band	0.991 [0.975, 1.009]	held
080 P1	law table $\geq 3\%$ over the current table, 43.75%	threshold	1.048 [1.041, 1.055]	held
080 P2	law table within 2% of the sweep’s best, 43.75%	band	0.991 [0.986, 0.996]	held
080 P3	best table leads FreeToken, 43.75%	sign	1.049 [1.036, 1.064]	held
080 P4	law table beats the current table, 25%	sign	1.067 [1.062, 1.071]	held
081 P1	law table beats current, gpt-oss, every budget	sign	1.079 [1.072, 1.085]; 1.054 [1.041, 1.066]; 1.033 [1.021, 1.044]	held
081 P2	ours leads FreeToken, gpt-oss 11/25/40%	sign	1.294 [1.278, 1.312]; 1.275 [1.254, 1.295]; 1.154 [1.134, 1.172]	held
081 P3a	law table beats current, Qwen3 12.5%	sign	1.080 [1.073, 1.084]	held
081 P3b	ours leads FreeToken, Qwen3 12.5%	sign	1.032 [1.022, 1.042]	held
081 P4a	ours $\geq 1.8\times$ llama.cpp, gpt-oss, every budget	threshold	2.001 [1.964, 2.041]; 2.731 [2.668, 2.795]; 3.196 [3.133, 3.259]	held
081 P4b	ours $\geq 2\times$ llama.cpp, Qwen3 12.5%	threshold	2.051 [2.026, 2.077]	held
Jobs 082–087: 43 clauses, 24 held, 4 held (point), 10 failed, 1 untested, 4 void				
082 P1	steady-state lead over FreeToken, gpt-oss, held-out prompts	sign	–	untested
082 P2	FreeToken gains more after 256 tokens than ours, every budget	sign	FreeToken: 1.017 / 1.065 / 1.071; ours: 1.038 / 1.094 / 1.117	failed
082 P3	FreeToken ahead in steady state, Qwen3 43.75%	sign	1.054 [1.025, 1.086]	failed
082 P4a	every ablation step $\geq -1\%$	threshold	0.723 [0.712, 0.733]; 0.744 [0.740, 0.748]	failed
082 P4b	ours $\geq 2\times$ stock at 25%, both models	threshold	2.658 [2.566, 2.749]; 2.800 [2.743, 2.858]	held
082 P5a	parity: $\geq 98\%$ top-1 agreement, both models	threshold	0.986 / 0.993	held
082 P5b	parity: $ \Delta\text{NLL} \leq 1\%$, both models	threshold	-0.0022 / 0.0031	held
082 P6	llama.cpp Qwen3 25/43.75% within 5% of the law	band	0.966 [0.962, 0.969]; 0.962 [0.958, 0.965]	held
083 P1	still reasoning at 2,048 tokens on ≥ 8 of 10, both systems	threshold	ours: 9 / 9 / 10; FreeToken: 10 / 9 / 10	held
083 P2	ours within 15% of its Table 1 rate	band	1.046 [1.019, 1.072]; 1.037 [1.013, 1.060]; 1.033 [1.003, 1.061]	held
083 P3a	ours leads FreeToken, all tokens, 11/25/40%	sign	1.358 [1.338, 1.378]; 1.322 [1.303, 1.342]; 1.191 [1.172, 1.212]	held
083 P3b	ours leads FreeToken, tokens 257-2,048	sign	1.366 [1.343, 1.389]; 1.328 [1.310, 1.350]; 1.194 [1.175, 1.215]	held
083 P4	lead $\geq 15\%$ at 11 and 25%	threshold	1.358 [1.338, 1.378]; 1.322 [1.303, 1.342]	held
084 P1	Qwen3 all in VRAM at 120-190 tok/s	band	–	void
084 P2	gpt-oss all in VRAM within 5% of 261.4 tok/s	band	–	void
084 P3	ours with 128 slots within 5% of all in VRAM, both	band	–	void
084 P4	sim. hit rate within 5 points; ours at 20-50% of the limit	band	–	void
084b P1	Qwen3 all in VRAM at 120-190 tok/s	band	177.8 [177.1, 178.4]	held
084b P2	gpt-oss all in VRAM within 5% of 261.4 tok/s	band	0.977 [0.971, 0.982]	held

continued

Job	Clause	Type	Measured	Status
084b P3	ours with 128 slots within 5% of all in VRAM, both	band	0.997 [0.991, 1.004]; 1.047 [1.043, 1.052]	held (point)
084b P4a	sim. hit rate within 5 points of the engine’s, every budget	band	-0.014 / -0.008 / -0.0003 / -0.053 / -0.026 / -0.006	failed
084b P4b	ours at 20-50% of the speed limit, every budget	band	0.413 / 0.256 / 0.294 / 0.433 / 0.302 / 0.352	held (point)
084c	<i>No predictions of its own: it supplies the gpt-oss routing trace that 084b-P4a and 084b-P4b are scored on.</i>	–	–	–
085 P1	law’s tables fetch less than on the headline machine	equality	gpt-oss: 0,0,0,1,1 (two entries lower); Qwen3: 0,0,0,1,1,2,2,3,3 (six entries lower)	held
085 P2a	law beats fixed, gpt-oss, every budget	sign	1.340 [1.327, 1.355]; 1.274 [1.256, 1.292]; 1.211 [1.187, 1.235]	held
085 P2b	law’s gain larger than headline at ≥ 2 of 3 (pro rata of 4 of 6)	threshold	3 of 3 (every interval excludes the headline value); pooled with 085b: 6 of 6	held
085 P3	ours leads FreeToken, gpt-oss, every budget	sign	2.030 [1.993, 2.068]; 1.699 [1.662, 1.737]; 1.428 [1.391, 1.463]	held
085 P4	lead larger than headline at ≥ 2 of 3 (pro rata of 4 of 6)	threshold	3 of 3 (every interval excludes the headline value); pooled with 085b (1 of 3): 4 of 6	held
085 P5	llama.cpp at 25% within 5% of the law, gpt-oss	band	0.818 [0.817, 0.819]	failed
085b P1	law’s table fetches less than on the headline machine	equality	0,0,1,1,2,2,3,3,4 (three entries lower)	held
085b P2a	law beats fixed, Qwen3, every budget	sign	1.184 [1.177, 1.190]; 1.149 [1.142, 1.157]; 1.094 [1.086, 1.101]	held
085b P2b	law’s gain larger than headline at ≥ 2 of 3	threshold	3 of 3 (every interval excludes the headline value)	held
085b P3	ours leads FreeToken, Qwen3, every budget	sign	1.067 [1.058, 1.076]; 1.076 [1.063, 1.089]; 1.038 [1.017, 1.060]	held
085b P4	lead larger than headline at ≥ 2 of 3	threshold	1 of 3: 12.5% 1.067 [1.058, 1.076] > 1.032; 25% 1.076 [1.063, 1.089] < 1.153; 43.75% 1.038 [1.017, 1.060] below 1.049 with the interval covering it	failed
085b P5	llama.cpp at 25% within 5% of the law, Qwen3	band	0.974 [0.972, 0.976]	held
086 P1	FreeToken as shipped within 15% of its paper’s rate	band	93.8 [91.3, 96.5]	held (point)
086 P2	ours leads FreeToken at 12.5/25/37.5%	sign	1.030 [1.023, 1.038]; 1.010 [0.998, 1.021]; 1.003 [0.994, 1.013]	failed
086 P3	ours $\geq 2\times$ llama.cpp, every budget	threshold	1.976 [1.950, 2.005]; 2.311 [2.264, 2.361]; 2.481 [2.441, 2.521]	failed
086 P4	law beats fixed, every budget	sign	1.071 [1.068, 1.074]; 1.060 [1.057, 1.063]; 1.057 [1.052, 1.061]	held
087 P1	profiled static cache $\geq 1.3\times$ stock, both models	threshold	1.307 [1.236, 1.379]; 1.889 [1.803, 1.969]	held (point)
087 P2	replacement adds less than placement, both models	sign	0.268 [0.122, 0.434]; -0.765 [-0.898, -0.619]	failed
087 P3	decayed frequency beats the profiled static cache, both	sign	1.575 [1.497, 1.674]; 1.124 [1.070, 1.187]	held
087 P4a	every step after the static cache $\geq -1\%$	threshold	0.877 [0.835, 0.924]	failed
087 P4b	ours $\geq 2\times$ stock, both models	threshold	2.680 [2.595, 2.763]; 2.815 [2.754, 2.877]	held
Jobs 088–098: 479 clauses, 168 held, 182 held (point), 123 failed, 6 untested, 0 void				
088 P1	FreeToken hybrid on 8 threads $\geq 1.3\times$ job 085’s hybrid, every budget	threshold	3.12 / 2.04 / 1.4	held (point)
088 P2	FreeToken’s best improves on job 085’s best at ≥ 2 of 3 budgets	threshold	1.52 / 1.29 / 1.08	held (point)
088 P3a	ours leads FreeToken’s best, 11% (CI > 1)	sign	1.582 [1.564, 1.603]	held
088 P3b	ours leads FreeToken’s best, 25% (CI > 1)	sign	1.519 [1.472, 1.559]	held
088 P3c	ours leads FreeToken’s best, 40% (CI > 1)	sign	1.506 [1.476, 1.537]	held
088 P3d	ours / FreeToken between $1.10\times$ and $1.50\times$, 11%	band	1.582 [1.564, 1.603]	failed
088 P3e	ours / FreeToken between $1.10\times$ and $1.50\times$, 25%	band	1.519 [1.472, 1.559]	failed
088 P3f	ours / FreeToken between $1.10\times$ and $1.50\times$, 40%	band	1.506 [1.476, 1.537]	failed
088 P4a	llama.cpp -t 8 within 10% of the law	band	-9.4	held (point)

continued

Job	Clause	Type	Measured	Status
088 P4b	llama.cpp -t 8 faster than -t 24	sign	1.219 [1.213, 1.225]	held
088 P5	the law's table is 0,0,0,1,1 or fetches less	equality	0,0,0,1,1	held
089 P1a	the card reports a 14,001 MHz memory clock	equality	14001 MHz	held
089 P1b	device read $\geq 1,500$ GB/s (not throttled)	threshold	1694	held
089 P2a	ours / FreeToken within ± 0.06 of Table 1, gpt-oss 11%	band	-0.088 [-0.099, -0.075]	failed
089 P2b	ours / FreeToken within ± 0.06 of Table 1, gpt-oss 25%	band	-0.078 [-0.112, -0.051]	failed
089 P2c	ours / FreeToken within ± 0.06 of Table 1, gpt-oss 40%	band	-0.061 [-0.077, -0.043]	failed
089 P2d	ours / FreeToken within ± 0.06 of Table 1, Qwen3 12.5%	band	-0.006 [-0.014, 0.004]	held
089 P2e	ours / FreeToken within ± 0.06 of Table 1, Qwen3 25%	band	-0.105 [-0.116, -0.094]	failed
089 P2f	ours / FreeToken within ± 0.06 of Table 1, Qwen3 43.75%	band	-0.075 [-0.087, -0.062]	failed
089 P3a	ours' speed within $\pm 12\%$ of host B's, gpt-oss 11%	band	-17.1	failed
089 P3b	ours' speed within $\pm 12\%$ of host B's, gpt-oss 25%	band	-13.8	failed
089 P3c	ours' speed 5-20% below host B's, gpt-oss 40%	band	-12.4	held (point)
089 P3d	ours' speed within $\pm 12\%$ of host B's, Qwen3 12.5%	band	-16.0	failed
089 P3e	ours' speed within $\pm 12\%$ of host B's, Qwen3 25%	band	-14.6	failed
089 P3f	ours' speed 5-20% below host B's, Qwen3 43.75%	band	-11.5	held (point)
089 P4	ours-first and FreeToken-first orders agree within 3%, every cell	band	-1.0 / 0.2 / 0.7 / 0.3 / -0.2 / 0.1	held (point)
089 P5a	LRU 10-30% slower than decayed frequency, gpt-oss 11%	band	18.5 [17.7, 19.3]	held
089 P5b	LRU 10-30% slower than decayed frequency, gpt-oss 25%	band	12.8 [11.5, 14.1]	held
089 P5c	LRU 10-30% slower than decayed frequency, gpt-oss 40%	band	7.8 [6.5, 9.1]	failed
089 P5d	LRU 10-30% slower than decayed frequency, Qwen3 12.5%	band	20.9 [20.6, 21.3]	held
089 P5e	LRU 10-30% slower than decayed frequency, Qwen3 25%	band	15.4 [14.9, 15.9]	held
089 P5f	LRU 10-30% slower than decayed frequency, Qwen3 43.75%	band	6.4 [5.9, 6.9]	failed
089 P5g	LRU trails FreeToken at Qwen3 12.5%	sign	0.814 [0.808, 0.821]	held
089 P5h	LRU still leads FreeToken at gpt-oss 11%	sign	0.973 [0.962, 0.986]	failed
089 P5i	LRU still leads FreeToken at gpt-oss 25%	sign	1.045 [1.016, 1.068]	held
089 P6a	prefill at 2,048 tokens within $\pm 20\%$ of stock, gpt-oss	band	0.800	held (point)
089 P6b	prefill at 2,048 tokens within $\pm 20\%$ of stock, Qwen3	band	0.825	held (point)
089 P6c	decode at 2 and 4 sequences below stock's, gpt-oss	sign	0.778 / 0.778 / 0.773 / 0.784	held (point)
089 P6d	decode at 2 and 4 sequences below stock's, Qwen3	sign	0.779 / 0.789 / 0.775 / 0.792	held (point)
090 P1a	mean KL ≤ 0.01 nats, gpt-oss	threshold	0.0019	held
090 P1b	mean KL ≤ 0.01 nats, Qwen3	threshold	0.0005	held
090 P1c	99.9th-percentile KL ≤ 0.5 nats, gpt-oss	threshold	0.121	held
090 P1d	99.9th-percentile KL ≤ 0.5 nats, Qwen3	threshold	0.059	held
090 P2a	top-1 agreement $\geq 98\%$, gpt-oss	threshold	98.5	held
090 P2b	top-1 agreement $\geq 98\%$, Qwen3	threshold	99.3	held
090 P2c	stock's -n-cpu-moe agreement within 1 point of ours, gpt-oss	band	0.33	held
090 P2d	stock's -n-cpu-moe agreement within 1 point of ours, Qwen3	band	-0.07	held
090 P3a	mean NLL within $\pm 0.5\%$ of the reference, gpt-oss	band	0.16	held
090 P3b	mean NLL within $\pm 0.5\%$ of the reference, Qwen3	band	0.19	held
091 P1a	ours leads FreeToken, gpt-oss 11% (CI > 1)	sign	1.371 [1.346, 1.395]	held
091 P1b	ours leads FreeToken, gpt-oss 25% (CI > 1)	sign	1.296 [1.265, 1.326]	held
091 P1c	ours leads FreeToken, Qwen3 12.5% (CI > 1)	sign	1.033 [1.021, 1.045]	held

continued

Job	Clause	Type	Measured	Status
091 P1d	ours leads FreeToken, Qwen3 25% (CI > 1)	sign	1.045 [1.028, 1.063]	held
091 P1e	lead in 10-35%, gpt-oss 11%	band	37.1 [34.6, 39.5]	failed
091 P1f	lead in 10-35%, gpt-oss 25%	band	29.6 [26.5, 32.6]	held
091 P1g	lead in 0-15%, Qwen3 12.5%	band	3.3 [2.1, 4.5]	held
091 P1h	lead in 0-15%, Qwen3 25%	band	4.5 [2.8, 6.3]	held
091 P2a	ours $\geq 1.8 \times$ llama.cpp, gpt-oss 11%	threshold	1.810 [1.760, 1.860]	held (point)
091 P2b	ours $\geq 1.8 \times$ llama.cpp, gpt-oss 25%	threshold	2.620 [2.520, 2.740]	held
091 P2c	ours $\geq 1.8 \times$ llama.cpp, Qwen3 12.5%	threshold	1.780 [1.760, 1.810]	failed
091 P2d	ours $\geq 1.8 \times$ llama.cpp, Qwen3 25%	threshold	2.540 [2.490, 2.610]	held
091 P3a	the law's gpt-oss table fetches no more than the headline machine's	equality	0,0,1,2,2	failed
091 P3b	the law's Qwen3 table fetches no more than the headline machine's	equality	0,0,1,1,2,3,4,4,5	failed
091 P4a	ours at 25-45% of the limit, gpt-oss 11%	band	44.4 [43.3, 45.5]	held (point)
091 P4b	ours at 25-45% of the limit, gpt-oss 25%	band	29.9 [29.1, 30.8]	held
091 P4c	ours at 25-45% of the limit, Qwen3 12.5%	band	45.7 [45.1, 46.3]	failed
091 P4d	ours at 25-45% of the limit, Qwen3 25%	band	34.3 [33.5, 35.2]	held
091 P5	ours-first and FreeToken-first orders agree within 3%, every cell	band	1.6 / 2.0 / -0.6 / 0.1	held (point)
092 P1a	ours leads FreeToken, gpt-oss 11% (CI > 1)	sign	4.460 [4.250, 4.690]	held
092 P1b	ours leads FreeToken, Qwen3 12.5% (CI > 1)	sign	2.120 [2.020, 2.210]	held
092 P1c	ours leads FreeToken, Qwen3 25% (CI > 1)	sign	2.680 [2.530, 2.830]	held
092 P1d	lead in 10-35%, gpt-oss 11%	band	346 [325, 369]	failed
092 P1e	lead in 0-15%, Qwen3 12.5%	band	112 [102, 121]	failed
092 P1f	lead in 0-15%, Qwen3 25%	band	168 [153, 183]	failed
092 P2a	ours $\geq 1.8 \times$ llama.cpp, gpt-oss 11%	threshold	1.640 [1.600, 1.670]	failed
092 P2b	ours $\geq 1.8 \times$ llama.cpp, Qwen3 12.5%	threshold	1.560 [1.480, 1.630]	failed
092 P2c	ours $\geq 1.8 \times$ llama.cpp, Qwen3 25%	threshold	2.180 [2.090, 2.270]	held
092 P3a	ours $\geq 1.3 \times$ llama.cpp on Mixtral-8x7B, C = 2	threshold	1.015 [1.008, 1.024]	failed
092 P3b	ours $\geq 1.3 \times$ llama.cpp on Mixtral-8x7B, C = 4	threshold	–	<i>untested</i>
092 P4a	the law's gpt-oss table fetches no more than the headline machine's	equality	0,0,0,0,0	held
092 P4b	the law's Qwen3 table fetches no more than the headline machine's	equality	0,0,0,0,0,0,0,0,0	held
092 P5a	ours-first and FreeToken-first orders agree within 3%, Qwen3 12.5%	band	7.0	failed
092 P5b	ours-first and FreeToken-first orders agree within 3%, Qwen3 25%	band	2.6	held (point)
093 P1a	oracle (W = all) faster than base, gpt-oss 11%	sign	0.923 [0.914, 0.933]	failed
093 P1b	oracle (W = all) faster than base, gpt-oss 25%	sign	1.398 [1.384, 1.412]	held
093 P1c	oracle (W = all) faster than base, Qwen3 12.5%	sign	0.916 [0.911, 0.922]	failed
093 P1d	oracle (W = all) faster than base, Qwen3 25%	sign	1.247 [1.235, 1.259]	held
093 P2a	oracle gain 20-80%, gpt-oss 11%	band	-7.7 [-8.6, -6.7]	failed
093 P2b	oracle gain 20-80%, gpt-oss 25%	band	39.8 [38.4, 41.2]	held
093 P2c	oracle gain 20-80%, Qwen3 12.5%	band	-8.4 [-8.9, -7.8]	failed
093 P2d	oracle gain 20-80%, Qwen3 25%	band	24.7 [23.5, 25.9]	held
093 P3a	W = 16 captures $\geq$ half of the W = 0 gain, gpt-oss 11%	threshold	–	<i>untested</i>
093 P3b	W = 16 captures $\geq$ half of the W = 0 gain, gpt-oss 25%	threshold	87	held (point)
093 P3c	W = 16 captures $\geq$ half of the W = 0 gain, Qwen3 12.5%	threshold	–	<i>untested</i>
093 P3d	W = 16 captures $\geq$ half of the W = 0 gain, Qwen3 25%	threshold	100	held (point)
093 P3e	W = 4 captures $\geq$ half of the W = 0 gain, gpt-oss 11%	threshold	–	<i>untested</i>
093 P3f	W = 4 captures $\geq$ half of the W = 0 gain, Qwen3 12.5%	threshold	–	<i>untested</i>
093 P4a	ours with the oracle at 45-65% of the limit, gpt-oss 11%	band	44.0 [43.2, 44.7]	failed
093 P4b	ours with the oracle at 45-65% of the limit, gpt-oss 25%	band	43.7 [42.5, 44.8]	failed
093 P4c	ours with the oracle at 45-65% of the limit, Qwen3 12.5%	band	44.8 [44.1, 45.5]	failed
093 P4d	ours with the oracle at 45-65% of the limit, Qwen3 25%	band	44.5 [43.3, 45.8]	failed
093 P5a	oracle hit rate within 3 points of the optimum's, gpt-oss 11%	band	-5.9	failed

continued

Job	Clause	Type	Measured	Status
093 P5b	oracle hit rate within 3 points of the optimum's, gpt-oss 25%	band	3.1	failed
093 P5c	oracle hit rate within 3 points of the optimum's, Qwen3 12.5%	band	-9.2	failed
093 P5d	oracle hit rate within 3 points of the optimum's, Qwen3 25%	band	0.4	held
093 P5e	oracle admissions within 30% of the optimum's reads, gpt-oss 11%	band	-9	held
093 P5f	oracle admissions within 30% of the optimum's reads, gpt-oss 25%	band	31	failed
093 P5g	oracle admissions within 30% of the optimum's reads, Qwen3 12.5%	band	-35	failed
093 P5h	oracle admissions within 30% of the optimum's reads, Qwen3 25%	band	7	held
093 P6a	no-overlap 5-25% slower than base, gpt-oss 11%	band	-0.5 [-1.0, -0.1]	failed
093 P6b	no-overlap 5-25% slower than base, gpt-oss 25%	band	0.2 [0.1, 0.2]	failed
093 P6c	no-overlap 5-25% slower than base, Qwen3 12.5%	band	-0.0 [-0.1, 0.0]	failed
093 P6d	no-overlap 5-25% slower than base, Qwen3 25%	band	0.1 [0.0, 0.2]	failed
093 P6e	all-CPU 0-10% slower than base, gpt-oss 11%	band	7.8 [7.2, 8.4]	held
093 P6f	all-CPU 0-10% slower than base, gpt-oss 25%	band	9.3 [8.7, 9.8]	held
093 P6g	all-CPU 0-10% slower than base, Qwen3 12.5%	band	14.7 [14.1, 15.3]	failed
093 P6h	all-CPU 0-10% slower than base, Qwen3 25%	band	14.6 [14.0, 15.2]	failed
093 P7a	oracle gain below 15% at the GPU-bound cell, gpt-oss 40%	threshold	48.5 [46.2, 50.9]	failed
093 P7b	oracle gain below 15% at the GPU-bound cell, Qwen3 43.75%	threshold	56.5 [53.3, 59.2]	failed
094 P1a	bypass oracle's reads within 15% of the optimum's, gpt-oss 11%	band	30.8	failed
094 P1b	bypass oracle's reads within 15% of the optimum's, gpt-oss 25%	band	31.1	failed
094 P1c	bypass oracle's reads within 15% of the optimum's, gpt-oss 40%	band	29.0	failed
094 P1d	bypass oracle's reads within 15% of the optimum's, Qwen3 12.5%	band	43.4	failed
094 P1e	bypass oracle's reads within 15% of the optimum's, Qwen3 25%	band	46.6	failed
094 P1f	bypass oracle's reads within 15% of the optimum's, Qwen3 43.75%	band	38.5	failed
094 P2a	bypass oracle faster than base, gpt-oss 11%	sign	1.033 [1.029, 1.038]	held
094 P2b	bypass oracle faster than base, gpt-oss 25%	sign	1.181 [1.175, 1.187]	held
094 P2c	bypass oracle faster than base, gpt-oss 40%	sign	1.232 [1.220, 1.243]	held
094 P2d	bypass oracle faster than base, Qwen3 12.5%	sign	1.002 [0.997, 1.006]	held (point)
094 P2e	bypass oracle faster than base, Qwen3 25%	sign	1.131 [1.122, 1.139]	held
094 P2f	bypass oracle faster than base, Qwen3 43.75%	sign	1.189 [1.178, 1.200]	held
094 P2g	bypass gain 10-45%, gpt-oss 11%	band	3.3 [2.9, 3.8]	failed
094 P2h	bypass gain 10-45%, gpt-oss 25%	band	18.1 [17.5, 18.7]	held
094 P2i	bypass gain 10-45%, Qwen3 12.5%	band	0.2 [-0.3, 0.6]	failed
094 P2j	bypass gain 10-45%, Qwen3 25%	band	13.1 [12.2, 13.9]	held
094 P3a	bypass hit rate within 4 points of the optimum's, gpt-oss 11%	band	-8.2	failed
094 P3b	bypass hit rate within 4 points of the optimum's, gpt-oss 25%	band	-3.2	held
094 P3c	bypass hit rate within 4 points of the optimum's, gpt-oss 40%	band	-1.4	held
094 P3d	bypass hit rate within 4 points of the optimum's, Qwen3 12.5%	band	-10.9	failed
094 P3e	bypass hit rate within 4 points of the optimum's, Qwen3 25%	band	-5.3	failed
094 P3f	bypass hit rate within 4 points of the optimum's, Qwen3 43.75%	band	-1.3	held
094 P4a	law over-predicts the bypass oracle by 10-40%, gpt-oss 11%	band	6.9	failed
094 P4b	law over-predicts the bypass oracle by 10-40%, gpt-oss 25%	band	5.8	failed
094 P4c	law over-predicts the bypass oracle by 10-40%, Qwen3 12.5%	band	4.7	failed
094 P4d	law over-predicts the bypass oracle by 10-40%, Qwen3 25%	band	8.5	failed

continued

Job	Clause	Type	Measured	Status
094 P5a	bypass beats prefetch by $\geq 15\%$, gpt-oss 11%	threshold	6.9	failed
094 P5b	bypass beats prefetch by $\geq 15\%$, Qwen3 12.5%	threshold	5.9	failed
094 P5c	bypass and prefetch within 15%, gpt-oss 25%	band	-27.6	failed
094 P5d	bypass and prefetch within 15%, gpt-oss 40%	band	-21.3	failed
094 P5e	bypass and prefetch within 15%, Qwen3 25%	band	-19.7	failed
094 P5f	bypass and prefetch within 15%, Qwen3 43.75%	band	-26.6	failed
094 P6a	W = 16 keeps $\geq 70\%$ of the W = all gain, gpt-oss 11%	threshold	176 [165, 190]	held
094 P6b	W = 16 keeps $\geq 70\%$ of the W = all gain, gpt-oss 25%	threshold	63	failed
094 P6c	W = 16 keeps $\geq 70\%$ of the W = all gain, gpt-oss 40%	threshold	34	failed
094 P6d	W = 16 keeps $\geq 70\%$ of the W = all gain, Qwen3 12.5%	threshold	–	<i>untested</i>
094 P6e	W = 16 keeps $\geq 70\%$ of the W = all gain, Qwen3 25%	threshold	94 [91, 96]	held
094 P6f	W = 16 keeps $\geq 70\%$ of the W = all gain, Qwen3 43.75%	threshold	53	failed
095 P1a	fetch oracle’s reads within 10% of the simulation, gpt-oss 11%	band	-0.2	held
095 P1b	fetch oracle’s reads within 10% of the simulation, gpt-oss 25%	band	-6.2	held
095 P1c	fetch oracle’s reads within 10% of the simulation, gpt-oss 40%	band	-17.2	failed
095 P1d	fetch oracle’s reads within 10% of the simulation, Qwen3 12.5%	band	-0.5	held
095 P1e	fetch oracle’s reads within 10% of the simulation, Qwen3 25%	band	-4.4	held
095 P1f	fetch oracle’s reads within 10% of the simulation, Qwen3 43.75%	band	-16.9	failed
095 P1g	fetch oracle’s hit rate within 3 points of the simulation, gpt-oss 11%	band	0.1	held
095 P1h	fetch oracle’s hit rate within 3 points of the simulation, gpt-oss 25%	band	0.8	held
095 P1i	fetch oracle’s hit rate within 3 points of the simulation, gpt-oss 40%	band	1.1	held
095 P1j	fetch oracle’s hit rate within 3 points of the simulation, Qwen3 12.5%	band	0.1	held
095 P1k	fetch oracle’s hit rate within 3 points of the simulation, Qwen3 25%	band	0.5	held
095 P1l	fetch oracle’s hit rate within 3 points of the simulation, Qwen3 43.75%	band	0.8	held
095 P2a	fetch oracle faster than base, gpt-oss 11%	sign	1.286 [1.279, 1.294]	held
095 P2b	fetch oracle faster than base, gpt-oss 25%	sign	1.316 [1.307, 1.326]	held
095 P2c	fetch oracle faster than base, gpt-oss 40%	sign	1.254 [1.242, 1.265]	held
095 P2d	fetch oracle faster than base, Qwen3 12.5%	sign	1.345 [1.338, 1.351]	held
095 P2e	fetch oracle faster than base, Qwen3 25%	sign	1.379 [1.368, 1.390]	held
095 P2f	fetch oracle faster than base, Qwen3 43.75%	sign	1.264 [1.251, 1.276]	held
095 P2g	law within 10% on the fetch oracle, gpt-oss 11%	band	-7.6 [-9.0, -6.3]	held
095 P2h	law within 10% on the fetch oracle, gpt-oss 25%	band	-13.4	failed
095 P2i	law within 10% on the fetch oracle, gpt-oss 40%	band	-13.6	failed
095 P2j	law within 10% on the fetch oracle, Qwen3 12.5%	band	-8.9 [-10.0, -7.8]	held
095 P2k	law within 10% on the fetch oracle, Qwen3 25%	band	-14.9	failed
095 P2l	law within 10% on the fetch oracle, Qwen3 43.75%	band	-16.9	failed
095 P3a	both2’s reads within 25% of the simulation, gpt-oss 11%	band	-3.7	held
095 P3b	both2’s reads within 25% of the simulation, gpt-oss 25%	band	-3.9	held
095 P3c	both2’s reads within 25% of the simulation, gpt-oss 40%	band	-12.3	held
095 P3d	both2’s reads within 25% of the simulation, Qwen3 12.5%	band	-1.9	held
095 P3e	both2’s reads within 25% of the simulation, Qwen3 25%	band	-4.9	held
095 P3f	both2’s reads within 25% of the simulation, Qwen3 43.75%	band	-8.7	held

continued

Job	Clause	Type	Measured	Status
095 P3g	both2's hit rate within 5 points of the simulation, gpt-oss 11%	band	-0.7	held
095 P3h	both2's hit rate within 5 points of the simulation, gpt-oss 25%	band	-1.6	held
095 P3i	both2's hit rate within 5 points of the simulation, gpt-oss 40%	band	-0.5	held
095 P3j	both2's hit rate within 5 points of the simulation, Qwen3 12.5%	band	0.2	held
095 P3k	both2's hit rate within 5 points of the simulation, Qwen3 25%	band	-0.4	held
095 P3l	both2's hit rate within 5 points of the simulation, Qwen3 43.75%	band	-0.3	held
095 P4a	both2 at 55-85% of the limit, gpt-oss 11%	band	63.9 [62.8, 65.0]	held
095 P4b	both2 at 55-85% of the limit, gpt-oss 25%	band	44.4	failed
095 P4c	both2 at 55-85% of the limit, Qwen3 12.5%	band	68.0 [67.1, 69.0]	held
095 P4d	both2 at 55-85% of the limit, Qwen3 25%	band	50.9	failed
095 P4e	both2 faster than the hit-optimal oracle, gpt-oss 11%	sign	1.532 [1.519, 1.546]	held
095 P4f	both2 faster than the hit-optimal oracle, gpt-oss 25%	sign	1.310 [1.291, 1.329]	held
095 P4g	both2 faster than the hit-optimal oracle, Qwen3 12.5%	sign	2.045 [2.016, 2.075]	held
095 P4h	both2 faster than the hit-optimal oracle, Qwen3 25%	sign	1.388 [1.374, 1.402]	held
095 P5a	law over-predicts both2 by 15-50%, gpt-oss 11%	band	12.7	failed
095 P5b	law over-predicts both2 by 15-50%, gpt-oss 25%	band	25.7 [23.2, 28.0]	held
095 P5c	law over-predicts both2 by 15-50%, Qwen3 12.5%	band	7.1	failed
095 P5d	law over-predicts both2 by 15-50%, Qwen3 25%	band	20.8 [17.8, 24.1]	held
095 P6a	both2 gains $\geq 25\%$ over base, gpt-oss 11%	threshold	39.4 [38.6, 40.3]	held
095 P6b	both2 gains $\geq 25\%$ over base, Qwen3 12.5%	threshold	42.1 [41.3, 42.9]	held
095 P7a	paced both3p within 10% of both2, gpt-oss 11%	band	4.2 [3.7, 4.7]	held
095 P7b	paced both3p within 10% of both2, gpt-oss 25%	band	11.8	failed
095 P7c	paced both3p within 10% of both2, gpt-oss 40%	band	7.3 [6.8, 7.8]	held
095 P7d	paced both3p within 10% of both2, Qwen3 12.5%	band	-0.5 [-0.8, -0.2]	held
095 P7e	paced both3p within 10% of both2, Qwen3 25%	band	13.0	failed
095 P7f	paced both3p within 10% of both2, Qwen3 43.75%	band	6.8 [6.1, 7.5]	held
096O4 P1a-gpt-oss11	fetch reads within 6% of the simulation, gpt-oss 11% (O4)	band	1.5	held (point)
096O4 P1c-gpt-oss11	fetch hit rate within 2 points of the simulation, gpt-oss 11% (O4)	band	-0.4	held (point)
096O4 P1b-gpt-oss11	both2 reads within 15% of the simulation, gpt-oss 11% (O4)	band	1.6	held (point)
096O4 P3a-gpt-oss11	foa reads 3-22% fewer than base, gpt-oss 11% (O4)	band	6.4	held (point)
096O4 P2-gpt-oss11	fetch $1.15\text{-}1.45\times$ base, gpt-oss 11% (O4)	band	1.361 [1.353, 1.369]	held
096O4 P3b-gpt-oss11	foa $0.97\text{-}1.12\times$ base, gpt-oss 11% (O4)	band	1.020 [1.019, 1.022]	held
096O4 P4-gpt-oss11	best prefetch > fetch > foa, gpt-oss 11% (O4)	sign	0.122	held
096O4 P7b-gpt-oss11	lead2 faster than bypass, gpt-oss 11% (O4)	sign	1.300	held (point)
096O4 P5a-gpt-oss11	nb2 reads $1.25\text{-}1.9\times$ lead2, gpt-oss 11% (O4)	band	1.25	failed
096O4 P5b-gpt-oss11	nb2 slower than lead2, gpt-oss 11% (O4)	sign	0.788	held (point)
096O4 P5c-gpt-oss11	nb2 at most $1.05\times$ base, gpt-oss 11% (O4)	threshold	1.049 [1.036, 1.062]	held (point)
096O4 P7a-gpt-oss11	bypass gains at most 8%, gpt-oss 11% (O4)	threshold	1.024 [1.019, 1.028]	held

continued

Job	Clause	Type	Measured	Status
096O4 P6b-gpt-oss11	nb2 within 10% of hitopt, gpt-oss 11% (O4)	band	0.7	held (point)
096O4 P9a-gpt-oss11	bytes share 25-55% of the gap, gpt-oss 11% (O4)	band	51.6	held (point)
096O4 P9b-gpt-oss11	overlap share 5-30% of the gap, gpt-oss 11% (O4)	band	12.8	held (point)
096O4 P1a-gpt-oss25	fetch reads within 6% of the simulation, gpt-oss 25% (O4)	band	3.1	held (point)
096O4 P1c-gpt-oss25	fetch hit rate within 2 points of the simulation, gpt-oss 25% (O4)	band	-0.4	held (point)
096O4 P1b-gpt-oss25	both2 reads within 15% of the simulation, gpt-oss 25% (O4)	band	-0.3	held (point)
096O4 P3a-gpt-oss25	foa reads 3-22% fewer than base, gpt-oss 25% (O4)	band	13.4	held (point)
096O4 P2-gpt-oss25	fetch 1.15-1.45 $\times$ base, gpt-oss 25% (O4)	band	1.426 [1.414, 1.438]	held
096O4 P3b-gpt-oss25	foa 0.97-1.12 $\times$ base, gpt-oss 25% (O4)	band	1.030 [1.029, 1.032]	held
096O4 P4-gpt-oss25	best prefetch > fetch > foa, gpt-oss 25% (O4)	sign	0.360	held
096O4 P7b-gpt-oss25	lead2 faster than bypass, gpt-oss 25% (O4)	sign	1.353	held (point)
096O4 P5a-gpt-oss25	nb2 reads 1.25-1.9 $\times$ lead2, gpt-oss 25% (O4)	band	1.36	held (point)
096O4 P6b-gpt-oss25	nb2 within 10% of hitopt, gpt-oss 25% (O4)	band	1.6	held (point)
096O4 P8-gpt-oss25	hitoptp faster than hitopt, gpt-oss 25% (O4)	sign	1.200	held (point)
096O4 P9a-gpt-oss25	bytes share 25-55% of the gap, gpt-oss 25% (O4)	band	44.1	held (point)
096O4 P9b-gpt-oss25	overlap share 5-30% of the gap, gpt-oss 25% (O4)	band	21.9	held (point)
096O4 P1a-gpt-oss40	fetch reads within 6% of the simulation, gpt-oss 40% (O4)	band	2.9	held (point)
096O4 P1c-gpt-oss40	fetch hit rate within 2 points of the simulation, gpt-oss 40% (O4)	band	-0.2	held (point)
096O4 P1b-gpt-oss40	both2 reads within 15% of the simulation, gpt-oss 40% (O4)	band	-9.8	held (point)
096O4 P3a-gpt-oss40	foa reads 3-22% fewer than base, gpt-oss 40% (O4)	band	21.0	held (point)
096O4 P5a-gpt-oss40	nb2 reads 1.25-1.9 $\times$ lead2, gpt-oss 40% (O4)	band	1.29	held (point)
096O4 P6b-gpt-oss40	nb2 within 10% of hitopt, gpt-oss 40% (O4)	band	2.0	held (point)
096O4 P1a-Qwen312.5	fetch reads within 6% of the simulation, Qwen3 12.5% (O4)	band	0.4	held (point)
096O4 P1c-Qwen312.5	fetch hit rate within 2 points of the simulation, Qwen3 12.5% (O4)	band	-0.2	held (point)
096O4 P1b-Qwen312.5	both2 reads within 15% of the simulation, Qwen3 12.5% (O4)	band	0.3	held (point)
096O4 P3a-Qwen312.5	foa reads 3-22% fewer than base, Qwen3 12.5% (O4)	band	4.0	held (point)
096O4 P2-Qwen312.5	fetch 1.15-1.45 $\times$ base, Qwen3 12.5% (O4)	band	1.438 [1.430, 1.445]	held
096O4 P3b-Qwen312.5	foa 0.97-1.12 $\times$ base, Qwen3 12.5% (O4)	band	1.015 [1.014, 1.016]	held
096O4 P4-Qwen312.5	best prefetch > fetch > foa, Qwen3 12.5% (O4)	sign	0.067	held

continued

Job	Clause	Type	Measured	Status
096O4 P7b-Qwen312.5	lead2 faster than bypass, Qwen3 12.5% (O4)	sign	1.252	held (point)
096O4 P5a-Qwen312.5	nb2 reads 1.25-1.9× lead2, Qwen3 12.5% (O4)	band	1.09	failed
096O4 P5b-Qwen312.5	nb2 slower than lead2, Qwen3 12.5% (O4)	sign	0.923	held (point)
096O4 P5c-Qwen312.5	nb2 at most 1.05× base, Qwen3 12.5% (O4)	threshold	1.149 [1.140, 1.158]	failed
096O4 P7a-Qwen312.5	bypass gains at most 8%, Qwen3 12.5% (O4)	threshold	0.995 [0.992, 0.998]	held
096O4 P6a-Qwen312.5	nb2 faster than hitopt, Qwen3 12.5% (O4)	sign	1.542	held (point)
096O4 P9a-Qwen312.5	bytes share 25-55% of the gap, Qwen3 12.5% (O4)	band	60.2	failed
096O4 P9b-Qwen312.5	overlap share 5-30% of the gap, Qwen3 12.5% (O4)	band	6.9	held (point)
096O4 P1a-Qwen325	fetch reads within 6% of the simulation, Qwen3 25% (O4)	band	0.7	held (point)
096O4 P1c-Qwen325	fetch hit rate within 2 points of the simulation, Qwen3 25% (O4)	band	-0.1	held (point)
096O4 P1b-Qwen325	both2 reads within 15% of the simulation, Qwen3 25% (O4)	band	0.2	held (point)
096O4 P3a-Qwen325	foa reads 3-22% fewer than base, Qwen3 25% (O4)	band	6.9	held (point)
096O4 P2-Qwen325	fetch 1.15-1.45× base, Qwen3 25% (O4)	band	1.513 [1.500, 1.525]	failed
096O4 P3b-Qwen325	foa 0.97-1.12× base, Qwen3 25% (O4)	band	1.016 [1.015, 1.017]	held
096O4 P4-Qwen325	best prefetch > fetch > foa, Qwen3 25% (O4)	sign	0.271	held
096O4 P7b-Qwen325	lead2 faster than bypass, Qwen3 25% (O4)	sign	1.410	held (point)
096O4 P5a-Qwen325	nb2 reads 1.25-1.9× lead2, Qwen3 25% (O4)	band	1.28	held (point)
096O4 P8-Qwen325	hitoptp faster than hitopt, Qwen3 25% (O4)	sign	1.048	held (point)
096O4 P9a-Qwen325	bytes share 25-55% of the gap, Qwen3 25% (O4)	band	53.4	held (point)
096O4 P9b-Qwen325	overlap share 5-30% of the gap, Qwen3 25% (O4)	band	16.8	held (point)
096O4 P1a-Qwen343.75	fetch reads within 6% of the simulation, Qwen3 43.75% (O4)	band	1.4	held (point)
096O4 P1c-Qwen343.75	fetch hit rate within 2 points of the simulation, Qwen3 43.75% (O4)	band	-0.0	held (point)
096O4 P1b-Qwen343.75	both2 reads within 15% of the simulation, Qwen3 43.75% (O4)	band	-4.6	held (point)
096O4 P3a-Qwen343.75	foa reads 3-22% fewer than base, Qwen3 43.75% (O4)	band	16.3	held (point)
096O4 P5a-Qwen343.75	nb2 reads 1.25-1.9× lead2, Qwen3 43.75% (O4)	band	1.31	held (point)
096O5 P1a-gpt-oss11	fetch reads within 6% of the simulation, gpt-oss 11% (O5)	band	1.5	held (point)
096O5 P1c-gpt-oss11	fetch hit rate within 2 points of the simulation, gpt-oss 11% (O5)	band	-0.4	held (point)
096O5 P1b-gpt-oss11	both2 reads within 15% of the simulation, gpt-oss 11% (O5)	band	1.6	held (point)
096O5 P3a-gpt-oss11	foa reads 3-22% fewer than base, gpt-oss 11% (O5)	band	6.6	held (point)
096O5 P2-gpt-oss11	fetch 1.15-1.45× base, gpt-oss 11% (O5)	band	1.159 [1.154, 1.165]	held

continued

Job	Clause	Type	Measured	Status
096O5 P3b-gpt-oss11	foa 0.97-1.12× base, gpt-oss 11% (O5)	band	0.998 [0.994, 1.001]	held
096O5 P4-gpt-oss11	best prefetch > fetch > foa, gpt-oss 11% (O5)	sign	0.130	held
096O5 P7b-gpt-oss11	lead2 faster than bypass, gpt-oss 11% (O5)	sign	1.042	held (point)
096O5 P5a-gpt-oss11	nb2 reads 1.25-1.9× lead2, gpt-oss 11% (O5)	band	1.27	held (point)
096O5 P5b-gpt-oss11	nb2 slower than lead2, gpt-oss 11% (O5)	sign	0.624	held (point)
096O5 P5c-gpt-oss11	nb2 at most 1.05× base, gpt-oss 11% (O5)	threshold	0.644 [0.635, 0.652]	held
096O5 P7a-gpt-oss11	bypass gains at most 8%, gpt-oss 11% (O5)	threshold	0.990 [0.986, 0.994]	held
096O5 P6b-gpt-oss11	nb2 within 10% of hitopt, gpt-oss 11% (O5)	band	0.9	held (point)
096O5 P9a-gpt-oss11	bytes share 25-55% of the gap, gpt-oss 11% (O5)	band	25.8	held (point)
096O5 P9b-gpt-oss11	overlap share 5-30% of the gap, gpt-oss 11% (O5)	band	17.2	held (point)
096O5 P1a-gpt-oss25	fetch reads within 6% of the simulation, gpt-oss 25% (O5)	band	3.1	held (point)
096O5 P1c-gpt-oss25	fetch hit rate within 2 points of the simulation, gpt-oss 25% (O5)	band	-0.4	held (point)
096O5 P1b-gpt-oss25	both2 reads within 15% of the simulation, gpt-oss 25% (O5)	band	-0.3	held (point)
096O5 P3a-gpt-oss25	foa reads 3-22% fewer than base, gpt-oss 25% (O5)	band	15.3	held (point)
096O5 P2-gpt-oss25	fetch 1.15-1.45× base, gpt-oss 25% (O5)	band	1.227 [1.220, 1.234]	held
096O5 P3b-gpt-oss25	foa 0.97-1.12× base, gpt-oss 25% (O5)	band	0.974 [0.971, 0.978]	held
096O5 P4-gpt-oss25	best prefetch > fetch > foa, gpt-oss 25% (O5)	sign	0.245	held
096O5 P7b-gpt-oss25	lead2 faster than bypass, gpt-oss 25% (O5)	sign	1.020	held (point)
096O5 P5a-gpt-oss25	nb2 reads 1.25-1.9× lead2, gpt-oss 25% (O5)	band	1.36	held (point)
096O5 P6b-gpt-oss25	nb2 within 10% of hitopt, gpt-oss 25% (O5)	band	1.7	held (point)
096O5 P8-gpt-oss25	hitoptp faster than hitopt, gpt-oss 25% (O5)	sign	1.420	held (point)
096O5 P9a-gpt-oss25	bytes share 25-55% of the gap, gpt-oss 25% (O5)	band	26.6	held (point)
096O5 P9b-gpt-oss25	overlap share 5-30% of the gap, gpt-oss 25% (O5)	band	21.3	held (point)
096O5 P1a-gpt-oss40	fetch reads within 6% of the simulation, gpt-oss 40% (O5)	band	2.9	held (point)
096O5 P1c-gpt-oss40	fetch hit rate within 2 points of the simulation, gpt-oss 40% (O5)	band	-0.2	held (point)
096O5 P1b-gpt-oss40	both2 reads within 15% of the simulation, gpt-oss 40% (O5)	band	-9.8	held (point)
096O5 P3a-gpt-oss40	foa reads 3-22% fewer than base, gpt-oss 40% (O5)	band	22.3	failed
096O5 P5a-gpt-oss40	nb2 reads 1.25-1.9× lead2, gpt-oss 40% (O5)	band	1.29	held (point)

continued

Job	Clause	Type	Measured	Status
096O5 P6b-gpt-oss40	nb2 within 10% of hitopt, gpt-oss 40% (O5)	band	2.5	held (point)
096O5 P1a-Qwen312.5	fetch reads within 6% of the simulation, Qwen3 12.5% (O5)	band	0.4	held (point)
096O5 P1c-Qwen312.5	fetch hit rate within 2 points of the simulation, Qwen3 12.5% (O5)	band	-0.2	held (point)
096O5 P1b-Qwen312.5	both2 reads within 15% of the simulation, Qwen3 12.5% (O5)	band	0.3	held (point)
096O5 P3a-Qwen312.5	foa reads 3-22% fewer than base, Qwen3 12.5% (O5)	band	4.6	held (point)
096O5 P2-Qwen312.5	fetch 1.15-1.45 $\times$ base, Qwen3 12.5% (O5)	band	1.199 [1.193, 1.205]	held
096O5 P3b-Qwen312.5	foa 0.97-1.12 $\times$ base, Qwen3 12.5% (O5)	band	1.006 [1.004, 1.007]	held
096O5 P4-Qwen312.5	best prefetch > fetch > foa, Qwen3 12.5% (O5)	sign	0.048	held
096O5 P7b-Qwen312.5	lead2 faster than bypass, Qwen3 12.5% (O5)	sign	1.084	held (point)
096O5 P5a-Qwen312.5	nb2 reads 1.25-1.9 $\times$ lead2, Qwen3 12.5% (O5)	band	1.14	failed
096O5 P5b-Qwen312.5	nb2 slower than lead2, Qwen3 12.5% (O5)	sign	0.753	held (point)
096O5 P5c-Qwen312.5	nb2 at most 1.05 $\times$ base, Qwen3 12.5% (O5)	threshold	0.766 [0.760, 0.773]	held
096O5 P7a-Qwen312.5	bypass gains at most 8%, Qwen3 12.5% (O5)	threshold	0.938 [0.934, 0.943]	held
096O5 P6a-Qwen312.5	nb2 faster than hitopt, Qwen3 12.5% (O5)	sign	1.413	held (point)
096O5 P9a-Qwen312.5	bytes share 25-55% of the gap, Qwen3 12.5% (O5)	band	31.6	held (point)
096O5 P9b-Qwen312.5	overlap share 5-30% of the gap, Qwen3 12.5% (O5)	band	6.9	held (point)
096O5 P1a-Qwen325	fetch reads within 6% of the simulation, Qwen3 25% (O5)	band	0.7	held (point)
096O5 P1c-Qwen325	fetch hit rate within 2 points of the simulation, Qwen3 25% (O5)	band	-0.1	held (point)
096O5 P1b-Qwen325	both2 reads within 15% of the simulation, Qwen3 25% (O5)	band	0.2	held (point)
096O5 P3a-Qwen325	foa reads 3-22% fewer than base, Qwen3 25% (O5)	band	8.4	held (point)
096O5 P2-Qwen325	fetch 1.15-1.45 $\times$ base, Qwen3 25% (O5)	band	1.256 [1.247, 1.265]	held
096O5 P3b-Qwen325	foa 0.97-1.12 $\times$ base, Qwen3 25% (O5)	band	0.999 [0.997, 1.001]	held
096O5 P4-Qwen325	best prefetch > fetch > foa, Qwen3 25% (O5)	sign	0.182	held
096O5 P7b-Qwen325	lead2 faster than bypass, Qwen3 25% (O5)	sign	1.004	held (point)
096O5 P5a-Qwen325	nb2 reads 1.25-1.9 $\times$ lead2, Qwen3 25% (O5)	band	1.30	held (point)
096O5 P8-Qwen325	hitoptp faster than hitopt, Qwen3 25% (O5)	sign	1.415	held (point)
096O5 P9a-Qwen325	bytes share 25-55% of the gap, Qwen3 25% (O5)	band	31.1	held (point)
096O5 P9b-Qwen325	overlap share 5-30% of the gap, Qwen3 25% (O5)	band	16.3	held (point)
096O5 P1a-Qwen343.75	fetch reads within 6% of the simulation, Qwen3 43.75% (O5)	band	1.4	held (point)

continued

Job	Clause	Type	Measured	Status
096O5 P1c-Qwen343.75	fetch hit rate within 2 points of the simulation, Qwen3 43.75% (O5)	band	-0.0	held (point)
096O5 P1b-Qwen343.75	both2 reads within 15% of the simulation, Qwen3 43.75% (O5)	band	-4.6	held (point)
096O5 P3a-Qwen343.75	foa reads 3-22% fewer than base, Qwen3 43.75% (O5)	band	18.8	held (point)
096O5 P5a-Qwen343.75	nb2 reads 1.25-1.9× lead2, Qwen3 43.75% (O5)	band	1.32	held (point)
096 P10-gpt-oss11	fetch/base differs by at most 0.15 between hosts, gpt-oss 11%	threshold	0.201	failed
096 P10-gpt-oss25	fetch/base differs by at most 0.15 between hosts, gpt-oss 25%	threshold	0.199	failed
096 P10-gpt-oss40	fetch/base differs by at most 0.15 between hosts, gpt-oss 40%	threshold	0.135	held (point)
096 P10-Qwen312.5	fetch/base differs by at most 0.15 between hosts, Qwen3 12.5%	threshold	0.239	failed
096 P10-Qwen325	fetch/base differs by at most 0.15 between hosts, Qwen3 25%	threshold	0.257	failed
096 P10-Qwen343.75	fetch/base differs by at most 0.15 between hosts, Qwen3 43.75%	threshold	0.152	failed
098 P1-gpt-oss11	FreeToken’s best at most 1.10× its Table 1 setting, gpt-oss 11%	threshold	1.042 [1.038, 1.049]	held
098 P2-gpt-oss11	ours leads FreeToken’s best by at least 5%, gpt-oss 11%	threshold	1.173 [1.157, 1.187]	held
098 P3-gpt-oss11-cap1	fetch cap 1 within 8% of the calibrated split, gpt-oss 11%	band	-0.9	held (point)
098 P3-gpt-oss11-cap2	fetch cap 2 within 8% of the calibrated split, gpt-oss 11%	band	-5.2	held (point)
098 P4-gpt-oss11	8 threads slower than default or within 3%, gpt-oss 11%	threshold	4.2	failed
098 P5-gpt-oss11	FreeToken’s best backend is hybrid, gpt-oss 11%	equality	1	held
098 P6-gpt-oss11	FreeToken 27-30 GiB, ours below it, gpt-oss 11%	band	28.4	held
098 P1-gpt-oss25	FreeToken’s best at most 1.10× its Table 1 setting, gpt-oss 25%	threshold	1.030 [1.016, 1.045]	held
098 P2-gpt-oss25	ours leads FreeToken’s best by at least 5%, gpt-oss 25%	threshold	1.167 [1.147, 1.189]	held
098 P3-gpt-oss25-cap1	fetch cap 1 within 8% of the calibrated split, gpt-oss 25%	band	-0.3	held (point)
098 P3-gpt-oss25-cap2	fetch cap 2 within 8% of the calibrated split, gpt-oss 25%	band	-3.5	held (point)
098 P4-gpt-oss25	8 threads slower than default or within 3%, gpt-oss 25%	threshold	1.6	held (point)
098 P5-gpt-oss25	FreeToken’s best backend is hybrid, gpt-oss 25%	equality	0	failed
098 P6-gpt-oss25	FreeToken 27-30 GiB, ours below it, gpt-oss 25%	band	28.4	held
098 P1-gpt-oss40	FreeToken’s best at most 1.10× its Table 1 setting, gpt-oss 40%	threshold	1.000 [1.000, 1.000]	held
098 P2-gpt-oss40	ours / FreeToken-best in 0.95-1.15, gpt-oss 40%	band	1.094 [1.078, 1.112]	held
098 P3-gpt-oss40-cap1	fetch cap 1 within 8% of the calibrated split, gpt-oss 40%	band	0.3	held (point)
098 P3-gpt-oss40-cap2	fetch cap 2 within 8% of the calibrated split, gpt-oss 40%	band	-0.8	held (point)
098 P4-gpt-oss40	8 threads slower than default or within 3%, gpt-oss 40%	threshold	0.2	held (point)
098 P5-gpt-oss40	FreeToken’s best backend is offload, gpt-oss 40%	equality	1	held
098 P6-gpt-oss40	FreeToken 27-30 GiB, ours below it, gpt-oss 40%	band	28.4	held
098 P1-Qwen312.5	FreeToken’s best at most 1.10× its Table 1 setting, Qwen3 12.5%	threshold	1.011 [1.009, 1.014]	held
098 P2-Qwen312.5	ours / FreeToken-best in 0.90-1.12, Qwen3 12.5%	band	1.020 [1.012, 1.028]	held

continued

Job	Clause	Type	Measured	Status
098 P3-Qwen312.5-cap1	fetch cap 1 within 8% of the calibrated split, Qwen3 12.5%	band	-9.4	failed
098 P3-Qwen312.5-cap2	fetch cap 2 within 8% of the calibrated split, Qwen3 12.5%	band	-6.5	held (point)
098 P4-Qwen312.5	8 threads slower than default or within 3%, Qwen3 12.5%	threshold	1.1	held (point)
098 P5-Qwen312.5	FreeToken’s best backend is hybrid, Qwen3 12.5%	equality	1	held
098 P6-Qwen312.5	FreeToken 27-30 GiB, ours below it, Qwen3 12.5%	band	28.6	held
098 P1-Qwen325	FreeToken’s best at most $1.10\times$ its Table 1 setting, Qwen3 25%	threshold	1.014 [1.010, 1.019]	held
098 P2-Qwen325	ours / FreeToken-best in 0.90-1.12, Qwen3 25%	band	1.044 [1.032, 1.056]	held
098 P3-Qwen325-cap1	fetch cap 1 within 8% of the calibrated split, Qwen3 25%	band	-8.0	held (point)
098 P3-Qwen325-cap2	fetch cap 2 within 8% of the calibrated split, Qwen3 25%	band	-8.7	failed
098 P4-Qwen325	8 threads slower than default or within 3%, Qwen3 25%	threshold	1.4	held (point)
098 P5-Qwen325	FreeToken’s best backend is hybrid, Qwen3 25%	equality	1	held
098 P6-Qwen325	FreeToken 27-30 GiB, ours below it, Qwen3 25%	band	28.6	held
098 P1-Qwen343.75	FreeToken’s best at most $1.10\times$ its Table 1 setting, Qwen3 43.75%	threshold	1.000 [1.000, 1.000]	held
098 P2-Qwen343.75	ours / FreeToken-best in 0.90-1.12, Qwen3 43.75%	band	0.983 [0.970, 0.996]	held
098 P3-Qwen343.75-cap1	fetch cap 1 within 8% of the calibrated split, Qwen3 43.75%	band	-5.3	held (point)
098 P3-Qwen343.75-cap2	fetch cap 2 within 8% of the calibrated split, Qwen3 43.75%	band	-6.8	held (point)
098 P4-Qwen343.75	8 threads slower than default or within 3%, Qwen3 43.75%	threshold	1.1	held (point)
098 P5-Qwen343.75	FreeToken’s best backend is offload, Qwen3 43.75%	equality	1	held
098 P6-Qwen343.75	FreeToken 27-30 GiB, ours below it, Qwen3 43.75%	band	30.1	failed
098 P7-gpt-oss	CPU-only: llama.cpp at least $2.0\times$ FreeToken’s cpu backend, gpt-oss	threshold	1.128 [1.126, 1.130]	failed
098 P7-Qwen3	CPU-only: llama.cpp at least $1.2\times$ FreeToken’s cpu backend, Qwen3	threshold	0.988 [0.987, 0.989]	failed
097O4 P1	six model files hash to the replay’s (O4)	equality	6	held
097O4 P2a-gpt-oss11	learned reads fewer than foa, gpt-oss 11% (O4)	sign	5.9	held (point)
097O4 P2b-gpt-oss11	learned’s read reduction within 5 points of the replay’s, gpt-oss 11% (O4)	band	0.1	held (point)
097O4 P3-gpt-oss11	learned $1.01\text{-}1.18\times$ foa, gpt-oss 11% (O4)	band	1.011 [1.001, 1.020]	held (point)
097O4 P5a-gpt-oss11	learned faster than base, gpt-oss 11% (O4)	sign	1.030 [1.023, 1.039]	held
097O4 P5b-gpt-oss11	learned recovers 10-45% of fetch’s gain, gpt-oss 11% (O4)	band	0.084	failed
097O4 P6-gpt-oss11	learned host time at most 300 us per token, gpt-oss 11% (O4)	threshold	315	failed
097O4 P7-foa-gpt-oss11	foa/base within 0.02 of job 096 on the same machine, gpt-oss 11% (O4)	band	-0.001	held (point)
097O4 P7-fetch-gpt-oss11	fetch/base within 0.05 of job 096 on the same machine, gpt-oss 11% (O4)	band	-0.001	held (point)

continued

Job	Clause	Type	Measured	Status
097O4 P7-both3p-gpt-oss11	both3p/base within 0.08 of job 096 on the same machine, gpt-oss 11% (O4)	band	-0.005	held (point)
097O4 P2a-gpt-oss25	learned reads fewer than foa, gpt-oss 25% (O4)	sign	11.5	held (point)
097O4 P2b-gpt-oss25	learned's read reduction within 5 points of the replay's, gpt-oss 25% (O4)	band	3.9	held (point)
097O4 P3-gpt-oss25	learned 1.01-1.18 $\times$ foa, gpt-oss 25% (O4)	band	1.030 [1.018, 1.041]	held
097O4 P5a-gpt-oss25	learned faster than base, gpt-oss 25% (O4)	sign	1.053 [1.043, 1.063]	held
097O4 P5b-gpt-oss25	learned recovers 10-45% of fetch's gain, gpt-oss 25% (O4)	band	0.129	held (point)
097O4 P6-gpt-oss25	learned host time at most 300 us per token, gpt-oss 25% (O4)	threshold	286	held (point)
097O4 P7-foa-gpt-oss25	foa/base within 0.02 of job 096 on the same machine, gpt-oss 25% (O4)	band	-0.007	held (point)
097O4 P7-fetch-gpt-oss25	fetch/base within 0.05 of job 096 on the same machine, gpt-oss 25% (O4)	band	-0.013	held (point)
097O4 P7-both3p-gpt-oss25	both3p/base within 0.08 of job 096 on the same machine, gpt-oss 25% (O4)	band	-0.007	held (point)
097O4 P2a-gpt-oss40	learned reads fewer than foa, gpt-oss 40% (O4)	sign	16.1	held (point)
097O4 P2b-gpt-oss40	learned's read reduction within 5 points of the replay's, gpt-oss 40% (O4)	band	4.0	held (point)
097O4 P4-gpt-oss40	learned at least 0.99 $\times$ foa, gpt-oss 40% (O4)	threshold	1.045 [1.037, 1.052]	held
097O4 P6-gpt-oss40	learned host time at most 300 us per token, gpt-oss 40% (O4)	threshold	240	held (point)
097O4 P7-foa-gpt-oss40	foa/base within 0.02 of job 096 on the same machine, gpt-oss 40% (O4)	band	0.005	held (point)
097O4 P7-fetch-gpt-oss40	fetch/base within 0.05 of job 096 on the same machine, gpt-oss 40% (O4)	band	0.007	held (point)
097O4 P7-both3p-gpt-oss40	both3p/base within 0.08 of job 096 on the same machine, gpt-oss 40% (O4)	band	0.023	held (point)
097O4 P2a-Qwen312.5	learned reads fewer than foa, Qwen3 12.5% (O4)	sign	5.6	held (point)
097O4 P2b-Qwen312.5	learned's read reduction within 5 points of the replay's, Qwen3 12.5% (O4)	band	-6.4	failed
097O4 P3-Qwen312.5	learned 1.01-1.18 $\times$ foa, Qwen3 12.5% (O4)	band	1.018 [1.015, 1.022]	held
097O4 P5a-Qwen312.5	learned faster than base, Qwen3 12.5% (O4)	sign	1.032 [1.029, 1.036]	held
097O4 P5b-Qwen312.5	learned recovers 10-45% of fetch's gain, Qwen3 12.5% (O4)	band	0.074	failed
097O4 P6-Qwen312.5	learned host time at most 600 us per token, Qwen3 12.5% (O4)	threshold	580	held (point)
097O4 P7-foa-Qwen312.5	foa/base within 0.02 of job 096 on the same machine, Qwen3 12.5% (O4)	band	-0.001	held (point)
097O4 P7-fetch-Qwen312.5	fetch/base within 0.05 of job 096 on the same machine, Qwen3 12.5% (O4)	band	-0.002	held (point)
097O4 P7-both3p-Qwen312.5	both3p/base within 0.08 of job 096 on the same machine, Qwen3 12.5% (O4)	band	0.004	held (point)
097O4 P2a-Qwen325	learned reads fewer than foa, Qwen3 25% (O4)	sign	11.3	held (point)

continued

Job	Clause	Type	Measured	Status
097O4 P2b-Qwen325	learned's read reduction within 5 points of the replay's, Qwen3 25% (O4)	band	0.2	held (point)
097O4 P3-Qwen325	learned 1.01-1.18× foa, Qwen3 25% (O4)	band	1.034 [1.026, 1.040]	held
097O4 P5a-Qwen325	learned faster than base, Qwen3 25% (O4)	sign	1.052 [1.045, 1.059]	held
097O4 P5b-Qwen325	learned recovers 10-45% of fetch's gain, Qwen3 25% (O4)	band	0.101	held (point)
097O4 P6-Qwen325	learned host time at most 600 us per token, Qwen3 25% (O4)	threshold	547	held (point)
097O4 P7-foa-Qwen325	foa/base within 0.02 of job 096 on the same machine, Qwen3 25% (O4)	band	0.002	held (point)
097O4 P7-fetch-Qwen325	fetch/base within 0.05 of job 096 on the same machine, Qwen3 25% (O4)	band	0.000	held (point)
097O4 P7-both3p-Qwen325	both3p/base within 0.08 of job 096 on the same machine, Qwen3 25% (O4)	band	-0.003	held (point)
097O4 P2a-Qwen343.75	learned reads fewer than foa, Qwen3 43.75% (O4)	sign	17.1	held (point)
097O4 P2b-Qwen343.75	learned's read reduction within 5 points of the replay's, Qwen3 43.75% (O4)	band	8.1	failed
097O4 P4-Qwen343.75	learned at least 0.99× foa, Qwen3 43.75% (O4)	threshold	1.024 [1.017, 1.030]	held
097O4 P6-Qwen343.75	learned host time at most 600 us per token, Qwen3 43.75% (O4)	threshold	460	held (point)
097O4 P7-foa-Qwen343.75	foa/base within 0.02 of job 096 on the same machine, Qwen3 43.75% (O4)	band	0.003	held (point)
097O4 P7-fetch-Qwen343.75	fetch/base within 0.05 of job 096 on the same machine, Qwen3 43.75% (O4)	band	-0.003	held (point)
097O4 P7-both3p-Qwen343.75	both3p/base within 0.08 of job 096 on the same machine, Qwen3 43.75% (O4)	band	-0.006	held (point)
097O6 P1	six model files hash to the replay's (O6)	equality	6	held
097O6 P2a-gpt-oss11	learned reads fewer than foa, gpt-oss 11% (O6)	sign	6.0	held (point)
097O6 P2b-gpt-oss11	learned's read reduction within 5 points of the replay's, gpt-oss 11% (O6)	band	0.2	held (point)
097O6 P3-gpt-oss11	learned 1.01-1.18× foa, gpt-oss 11% (O6)	band	0.942 [0.937, 0.947]	failed
097O6 P5a-gpt-oss11	learned faster than base, gpt-oss 11% (O6)	sign	0.936 [0.931, 0.941]	failed
097O6 P5b-gpt-oss11	learned recovers 10-45% of fetch's gain, gpt-oss 11% (O6)	band	-0.271	failed
097O6 P6-gpt-oss11	learned host time at most 300 us per token, gpt-oss 11% (O6)	threshold	292	held (point)
097O6 P2a-gpt-oss25	learned reads fewer than foa, gpt-oss 25% (O6)	sign	11.4	held (point)
097O6 P2b-gpt-oss25	learned's read reduction within 5 points of the replay's, gpt-oss 25% (O6)	band	3.8	held (point)
097O6 P3-gpt-oss25	learned 1.01-1.18× foa, gpt-oss 25% (O6)	band	0.939 [0.933, 0.945]	failed
097O6 P5a-gpt-oss25	learned faster than base, gpt-oss 25% (O6)	sign	0.926 [0.920, 0.933]	failed
097O6 P5b-gpt-oss25	learned recovers 10-45% of fetch's gain, gpt-oss 25% (O6)	band	-0.283	failed
097O6 P6-gpt-oss25	learned host time at most 300 us per token, gpt-oss 25% (O6)	threshold	275	held (point)

continued

Job	Clause	Type	Measured	Status
097O6 P2a-gpt-oss40	learned reads fewer than foa, gpt-oss 40% (O6)	sign	16.0	held (point)
097O6 P2b-gpt-oss40	learned's read reduction within 5 points of the replay's, gpt-oss 40% (O6)	band	3.9	held (point)
097O6 P4-gpt-oss40	learned at least $0.99\times$ foa, gpt-oss 40% (O6)	threshold	0.969 [0.964, 0.974]	failed
097O6 P6-gpt-oss40	learned host time at most 300 us per token, gpt-oss 40% (O6)	threshold	237	held (point)
097O6 P2a-Qwen312.5	learned reads fewer than foa, Qwen3 12.5% (O6)	sign	7.7	held (point)
097O6 P2b-Qwen312.5	learned's read reduction within 5 points of the replay's, Qwen3 12.5% (O6)	band	-4.3	held (point)
097O6 P3-Qwen312.5	learned $1.01-1.18\times$ foa, Qwen3 12.5% (O6)	band	0.915 [0.911, 0.919]	failed
097O6 P5a-Qwen312.5	learned faster than base, Qwen3 12.5% (O6)	sign	0.910 [0.906, 0.914]	failed
097O6 P5b-Qwen312.5	learned recovers 10-45% of fetch's gain, Qwen3 12.5% (O6)	band	-0.308	failed
097O6 P6-Qwen312.5	learned host time at most 600 us per token, Qwen3 12.5% (O6)	threshold	525	held (point)
097O6 P2a-Qwen325	learned reads fewer than foa, Qwen3 25% (O6)	sign	13.2	held (point)
097O6 P2b-Qwen325	learned's read reduction within 5 points of the replay's, Qwen3 25% (O6)	band	2.1	held (point)
097O6 P3-Qwen325	learned $1.01-1.18\times$ foa, Qwen3 25% (O6)	band	0.905 [0.899, 0.911]	failed
097O6 P5a-Qwen325	learned faster than base, Qwen3 25% (O6)	sign	0.896 [0.890, 0.901]	failed
097O6 P5b-Qwen325	learned recovers 10-45% of fetch's gain, Qwen3 25% (O6)	band	-0.342	failed
097O6 P6-Qwen325	learned host time at most 600 us per token, Qwen3 25% (O6)	threshold	585	held (point)
097O6 P2a-Qwen343.75	learned reads fewer than foa, Qwen3 43.75% (O6)	sign	17.9	held (point)
097O6 P2b-Qwen343.75	learned's read reduction within 5 points of the replay's, Qwen3 43.75% (O6)	band	8.9	failed
097O6 P4-Qwen343.75	learned at least $0.99\times$ foa, Qwen3 43.75% (O6)	threshold	0.933 [0.927, 0.939]	failed
097O6 P6-Qwen343.75	learned host time at most 600 us per token, Qwen3 43.75% (O6)	threshold	483	held (point)

S1 The calibrated time model

Our cache’s one per-machine decision, how many of a layer’s misses to fetch rather than run on the CPU, comes from a calibrated model of time per token. With X_c the bytes the CPU reads per token and X_p the bytes the link reads,

$$T = G_{\text{fit}} + \max\left(\frac{X_c}{B_c}, \frac{X_p}{B_p}, \frac{X_c + X_p}{B_{cp}}\right), \quad (\text{S1})$$

with one constant G_{fit} per engine and model, fitted once on one machine and frozen (it is neither the profiled G of Eq. (4) nor T_{GPU} of Eq. (2)). It is a calibrated model, not a law: G_{fit} is fitted, and it exceeds the whole all-in-VRAM token (5.17 against 3.92ms) because it absorbs per-token host work and the CPU helpers’ shortfall against the probe (below). As a predictor of time it ties a naive one-path model with the same G_{fit} (time = G_{fit} + all host bytes over the best probed rate):

Held-out set	Eq. (S1)		one path	
	median	90th pct.	median	90th pct.
blind test, 33 rows on 7 machines	3.4%	12%	3.6%	31%
frozen, 30 rows on 4 later machines	5.6%	16%	6.1%	11%

It over-predicts Qwen3’s speed on the later machines by +5 to +21%, a budget-dependent bias whose G_{fit} was borrowed from llama.cpp. Its value is the per-machine fetch table it picks (Eq. (S2) below), which gains 3.3–8.0% over a fixed table on host B and 9–34% on machines with half the link (Appendix J).

The blind test in full. On every machine of the blind test the job wrote the model’s prediction from its own probe before any model ran (Table S2; all of its cache rows are gpt-oss-120b on RTX 5090 machines). Over 29 configurations on 7 machines (33 measurements; one machine ran twice) the median error is 3.7% per configuration; without the 12-channel EPYC 9655 it is 3.3% over 24 configurations (90th percentile 7.5%). llama.cpp --n-cpu-moe is predicted within 4% on the Ryzen machines, including two Qwen3 configurations on host B (+3.6%, +4.0%; for Qwen3, G_{fit} =4.3ms, the mean residual of three llama.cpp runs on another machine), and misses by +10–+119% on the hybrid-core and server machines. With its constants frozen, the model scores the cache on the 4 RTX 5090 machines rented later (30 measurements at the six cells of Table 3, not written before the runs) at a median 5.6%, 90th percentile 16%, worst 21% (Qwen3-30B-A3B 43.75% on oracle host O1). Refit under leave-one-machine-out validation, the token-level max has a median error of 3.1% (90th percentile 9.4%), the additive form $G_{\text{fit}} + X_c/B_c + X_p/B_p$ 4.5% (11.3%) and a per-layer max summed over layers 3.2% (10.2%); the max beats the sum on the three rows that copy misses on demand behind a 53 GB/s link, where both paths carry most of the bytes, and on the EPYC 9655 rows, where both fail, the sum is closer on the two rows that copy behind a slow link, and elsewhere the two are within a few points (Appendix S1). Each failure has a measured cause (Appendix S1): on-demand copies behind a slow link share DRAM less evenly than the probe’s combined rate says; llama.cpp splits every matrix evenly across threads and runs at 46% of the model on a hybrid-core machine; and on a 12-channel EPYC host memory stops being the bottleneck and a per-layer latency floor takes over. G_{fit} (5.17ms refit) exceeds the whole all-in-VRAM token (3.92ms) because it absorbs a shortfall the probe does not see: the helpers read at 1.09–1.31× the probe’s time per expert, and a fitted rate factor is collinear with G_{fit} at this sample size.

What the model says, and the split it picks. Equation (S1) is a calibrated model of execution that reads host memory in sequence with its GPU work, as our cache and llama.cpp do. For such an engine every mechanism acts only through the bytes read from host memory per token and the number of read paths in use: a second path adds bandwidth on every machine of the blind test but the 12-channel EPYC (B_{cp} against $\max(B_c, B_p)$): 8–17% on its Ryzen machines, 1% on the Core Ultra, 33% on the Zen 2 EPYC; the RTX 3090 machine’s Core i9-11900KF, where the two paths together read less than the CPU alone, is a second exception), and that is the whole gain available to fetching or splitting misses between the CPU and PCIe. Prefetching never reduces the bytes; what it can buy is overlap of the reads with the step, which the model does not price: charging every read in sequence, it over-predicts the time per token of the read-ahead oracle by +1 to +38% (Section 5) and of a Belady prefetch by up to 53% (Appendix E). The model’s job in the system is the split. Its per-layer form,

$$T(n, f) = \max\left(\frac{fS}{B_p} + g, [n > f](L_c + \frac{(n-f)S}{B_c}), [0 < f < n] \frac{nS}{B_{cp}}\right), \quad (\text{S2})$$

with g the GPU’s expert work per layer (48 μ s for Qwen3, 37 μ s for gpt-oss, from profiles) and L_c =20 μ s the hand-off, picks for each count n of missed experts the number f to copy with the smallest T . On host B the model’s table beats

Table S2: The calibrated model predicted blind: each machine wrote its prediction from its own bandwidth probe, with constants frozen in the public repository, before any model run. Error of predicted over measured speed, gpt-oss-120b (cache rows) and llama.cpp --n-cpu-moe (last column, %).

Host CPU	Job	n	Median err %	Max err %	llama.cpp err %
Core Ultra 7 270K (8P+16E)	069c	5	1.5	11.9	+119
EPYC 7352 (Zen 2)	069c	5	3.2	11.7	+10
EPYC 9655 (12-ch. DDR5)	069c	5	18.7	31.9	+16
Ryzen 9 7900	072	5	4.9	8.2	+4
Ryzen 9 9950X3D (A)	073	8	1.6	4.5	—
Ryzen 9 7950X	074	1	4.0	4.0	—
Ryzen 9 9950X	076	4	2.5	5.0	—
Ryzen 9 9950X3D (B)	082	0	—	—	+4, +4

the fixed table we used before at every budget of both models, by 3.3–8.0%; on machines whose link we measured at 27–30 GB/s instead of 53 it fetches less and gains 21–34% on gpt-oss and 9–18% on Qwen3, the per-machine split mattering most where the link is slow (Appendix J). In the running example on host B the table is 0,0,1,1,2 ($f(0.4)$): one missed expert runs on the CPU, hidden behind the GPU’s other experts; from two misses the model copies one. The lead over FreeToken at that cell, $1.27\times$, is mostly the admission policy (77% of it by the LRU ablation of Appendix J), the rest the CPU path that FreeToken’s bandwidth model declines to use there.

Is the max the right form? In 19 of the 29 configurations nothing is copied, so they test one roof. We therefore refit the model and its alternatives on the same 33 measurements under leave-one-machine-out validation (each of the 7 machines predicted with constants fitted on the other six): the token-level max of Eq. (S1) with G_{fit} refit has a median error of 3.1% (90th percentile 9.4%); the additive form $G_{\text{fit}} + X_c/B_c + X_p/B_p$, 4.5% (11.3%); a per-layer max summed over layers from the engine’s per-layer miss histograms, 3.2% (10.2%). The additive form ties in sample but misses the three rows that copy misses on demand behind a 53 GB/s link, where both paths carry most of the bytes, by -11%, while the max is within $\pm 0.2\%$ there; on the slow-link copy rows the additive form is the closer of the two. The per-layer form neither beats the max nor explains its failures: the Jensen gap between a sum of per-layer maxima and the maximum of sums is a fifth of the Core Ultra’s residual, none of the EPYC 7352’s, and has the wrong sign on the EPYC 9655 and on the prefetch rows. We keep Eq. (S1), with its evidence stated: the max beats the sum on the three fast-link copy rows and on the EPYC 9655 rows, where both fail, the sum is closer on the two slow-link copy rows, and elsewhere the two are within a few points, so the form is settled only where both paths carry most of the bytes. FreeToken’s own runs are not predicted, because its counters are not exposed; the model covers our engine and llama.cpp.

Where it fails, and why. Each failure now has a measured cause. (i) Copying misses on demand (FETCH) at 25% is over-predicted by +11.7% on the Core Ultra 7 and +11.4% on the EPYC 7352 (links of 36 and 26 GB/s); +8.8% and +8.5% with G_{fit} refit. The engine’s per-layer timers show why: a CPU miss served beside a copy takes 20–62% longer than one served alone, so the two paths share DRAM less evenly than the probe’s combined rate says; a variant of the model that uses the probe’s per-path rates under contention and a per-layer hand-off constant predicts both rows within +3.1% and +4.1%, at a worse median (4.2%). (ii) llama.cpp on a Core Ultra 7 with 8 performance and 16 efficiency cores runs at 46% of the model’s prediction at its default thread count, because it splits every matrix evenly across its threads; our helpers claim work dynamically and stay on the model on the same machine. We did not test llama.cpp restricted to the performance cores. (iii) On a 12-channel EPYC 9655 (559 GB/s) host memory stops being the bottleneck (+16 to +29% with G_{fit} refit under leave-one-machine-out validation; Table S2 gives the blind errors): the per-layer timers show a floor of 55–70 μs per layer with CPU misses that no desktop machine has (their intercepts

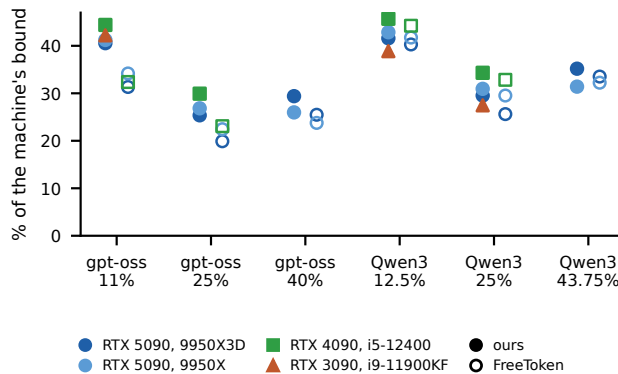


Figure S1: Ours (filled) and FreeToken (hollow) as a percentage of each machine’s own bound (exact optimum, the machine’s highest probed host rate, the card’s datasheet rate) at every cell of Table S3: two RTX 5090 machines, an RTX 4090 and an RTX 3090 (FreeToken’s 3090 cells, its carried-over backend running below llama.cpp, are omitted). The bound ranges from 41 to 519 tok/s across these cells; ours stays at 25–46% of it.

are at or below zero), the latency of the hand-off at a bandwidth where a 13 MB expert takes $24 \mu\text{s}$. (iv) Prefetch configurations are under-predicted by -8.7 to -3.4%: their copies overlap attention, which the model charges in full.

S2 The grid: other cards, and the scope of the gain

Table S3 lists every cell of Table 3’s protocol on the four machines we measured, each against its own bound, and Fig. S1 plots the fractions. The protocol was repeated on an RTX 4090 next to a Core i5-12400 (the host reads 37 GB/s, the link 25) and an RTX 3090 next to a Core i9-11900KF (45 GB/s; CPU and link together read *less* than the CPU alone, so the model’s table copies nothing), at the cells that fit 24 GB. On the 4090 ours runs $1.03\text{--}1.37\times$ FreeToken and $1.8\text{--}2.6\times$ llama.cpp at half host B’s absolute speeds, the host being the reason; on the 3090 it runs $1.6\text{--}2.2\times$ llama.cpp (FreeToken’s carried-over backend ran below llama.cpp there and is not counted). The cards’ own bandwidth never enters: every one of the seven cells on the 24 GB cards is host-bound in Eq. (1), and ours stands at 30–46% and 27–42% of it. Mixtral-8x7B (8 experts per layer, top-2; 26 GB of experts on the 3090) marks the scope: with 2 of 8 experts resident the cache hits 31% of reads where whole-layer pinning hits 25%, so there is little locality to earn, and ours ties llama.cpp (1.015 [1.008, 1.024]); on the 128-expert models the same cache hits 53–79% at 11–25% resident. Across eight machines the gpt-oss lead over FreeToken grows with the machine’s CPU-to-link ratio, from $1.09\text{--}1.22\times$ where the CPU reads $1.05\text{--}1.14\times$ its link’s rate to $1.51\text{--}1.58\times$ behind a link of half the bandwidth, in the direction the model’s split implies (the model does not predict FreeToken’s runs), and the Qwen3 lead stays at $0.95\text{--}1.15\times$ (Fig. S2, Appendix J).

The 24 GB cards run the cells that fit (the VRAM arithmetic, written on the machine from the GGUF header, skipped Qwen3 at 43.75%: 28.0 GB of slots, dense weights and KV against the 23 GB available). The predictions for jobs 091 and 092 and their outcomes are in `prereg/grid_outcome_091.md` and `092.md`: on the 4090 our lead over FreeToken at gpt-oss 11% (37%) exceeded the predicted 10–35% band, ours ran $1.78\times$ llama.cpp at Qwen3 12.5% against a predicted 1.8, the model’s tables copied one more expert than host B’s at one entry each (the prediction reasoned from the link alone; the CPU is slower in the same proportion), and ours’ fraction of the bound at Qwen3 12.5% (45.7%) exceeded the predicted 25–45%; on the 3090 FreeToken’s carried-over backend ran below llama.cpp, ours ran $1.56\text{--}1.64\times$ llama.cpp at two cells against a predicted 1.8, Mixtral tied at $C=2$ against a predicted $1.3\times$ and its $C=4$ cell was lost to the job’s deadline, and one launch-order check exceeded 3% (7%).

S3 The order-free accounting of the gap

Two choices separate the deployed cache from MIN-1R: *what to cache* (the deployed policy’s admissions or MIN’s) and *how to load* (by the CPU, two reads, or in the step, one). The engine ran all four combinations on O4, O5, the panel and the three new machines of jobs 100–101. In the running example, changing only what to cache closes 4.5% of the 10.6 ms gap, changing only how to load 3.9%, and both 51.5% (rounded): the choices interact, because MIN caches 4.6 times as many experts per token as the deployed policy and, loaded by the CPU, each is read twice, and because the engine’s in-step fetches put misses the schedule did not choose into slots, so the two-read path never holds MIN’s

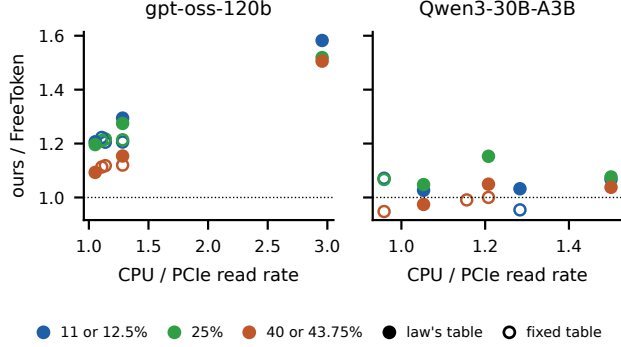


Figure S2: Ours over FreeToken’s better backend (carried over from host B on host S) in every same-machine comparison of the current engine on the two models of Table 3 (eight machines), against the machine’s CPU-to-PCIe read ratio (FreeToken’s own probe). Filled: FETCH table from the model on that machine; hollow: the fixed table. The rightmost gpt-oss points are a Core i9-14900K with FreeToken’s executor on its performance cores; the rightmost Qwen3 points are a Ryzen 9 7900.

set (job 113, Section 4: with those fetches off, Few-2R’s misses fall from 54.7–55.6 to 43.0 per token). We credit each choice with its effect averaged over the two orders, its Shapley value (Fawcett and Hoos, 2016; Jain, 1991) (26.1% and 25.5% here); Table S4 also lists each alone and the interaction. Reading ahead, from MIN-1R to the read-ahead oracle, is nested under MIN’s set (12.7%); what no oracle touches is the rest (35.7%). On the 13 machines whose link reads at least half as fast as the CPU, either change alone closes at most 26% of the gap and both together 14–60%; on the 2 below that line, loading in the step costs time at 3 of 4 machine-budgets, and at 3 of 4 on 2 more (job 103). These use the greedy set. With the fewest-admission set (job 104), both together close 6–49% at 11% on its three machines, against –26–39% with the greedy set, and the interaction remains (1–33%).

S4 The accounting as a model

Between the measured time of a cell and its bound lie five modelled fixes: overlap of host reads with GPU work (in Eq. (1)’s form, which also rebalances experts onto the CPU), the optimum’s reads instead of the best online policy’s (*foresight*), the best online policy’s reads at the probe’s highest rate instead of the engine’s through Eq. (S1) (policy and read paths), the engine’s per-token host work, and GPU kernels at the datasheet rate. Adding them in a fixed order makes the first one added look largest, so we report each fix’s Shapley value, its average marginal cost over every order (Fig. S3), a method with a precedent in performance debugging of microservices (Li et al., 2024). Two of the five terms are definitions rather than measurements: the GPU term is the residual, its factor calibrated per cell so that the model with no fix applied reproduces the measured time, and “overlap” is Eq. (1)’s form. On host B the model has two regimes. In the four host-bound cells missing foresight is the largest cost, 37–53% of the gap (26–31% of the token; in the running example 37%, where it edges overlap by 0.14 ms and is the largest marginal cost in 70 of the 120 orders); in the two GPU-bound cells host reads serialised with GPU work is largest, 39–46% of the gap, and foresight second at 23%. Policy and read paths cost 12–16%, per-token host work 3–10%, and GPU kernels below datasheet, net of the overlap the engine achieves, 4–16%. The regimes are a property of the machine: recomputed for the three oracle machines, whose host memory is weaker, foresight edges overlap at three of their six GPU-bound cells. The model prices foresight and overlap as nearly independent players, and the interior states it values are not realised by any online policy. The rest of this section measures them.

Table S3: The grid: Table 3’s protocol on every card and machine we measured, each cell against its own machine’s bound (the exact optimum’s reads on the same trace, the machine’s highest probed host rate, the card’s datasheet rate). *Binds*: what binds the bound; *host* when the optimum’s reads alone take longer than the GPU’s whole read, so the GPU has slack, *both* when the bound runs experts on the CPU until the two paths take equally long, which is where the all-in-VRAM speed can be exceeded. Ours and FreeToken are launch 2 (FreeToken’s backend carried over from host B’s selection); the 40 and 43.75% budgets do not fit a 24 GB card. [†]On the RTX 3090 FreeToken’s carried-over hybrid backend ran below llama.cpp; its own calibration recommends its offload backend there, which we did not run, so its speed is reported and no ratio against it is printed. Mixtral-8x7B (26 GB of experts, top-2 of 8) is ours against llama.cpp only, on the RTX 3090, launch 1.

Model	Experts on GPU	llama.cpp (tok/s)	FreeToken (tok/s)	Ours (tok/s)	Ours ÷ FreeToken (95% CI)	Ours ÷ llama.cpp	Bound (tok/s)	Binds	Ours, % of bound
RTX 5090 , 9950X3D (host B): B_{host} 88 GB/s, B_{gpu} 1792 GB/s; tables 0,0,1,1,2 / 0,0,1,1,2,3,3,4,5									
gpt-oss-120b	11%	34.9	54.0	69.9	1.29 [1.28, 1.31]	2.00	172	host	41
gpt-oss-120b	25%	40.0	85.6	109.2	1.27 [1.25, 1.29]	2.73	430	host	25
gpt-oss-120b	40%	47.7	132.2	152.5	1.15 [1.13, 1.17]	3.20	519	both	29
Qwen3-30B-A3B	12.5%	19.5	38.7	40.0	1.03 [1.02, 1.04]	2.05	96	host	42
Qwen3-30B-A3B	25%	22.1	54.7	63.1	1.15 [1.13, 1.17]	2.86	214	host	30
Qwen3-30B-A3B	43.75%	28.4	103.1	108.2	1.05 [1.04, 1.06]	3.81	308	both	35
RTX 5090 , 9950X (host S): B_{host} 71 GB/s, B_{gpu} 1792 GB/s; tables 0,0,1,2,3 / 0,0,1,2,2,3,4,5,6									
gpt-oss-120b	11%	28.9	48.0	57.9	1.21 [1.19, 1.22]	2.00	140	host	41
gpt-oss-120b	25%	33.6	78.7	94.1	1.20 [1.16, 1.22]	2.80	351	host	27
gpt-oss-120b	40%	40.1	122.2	133.6	1.09 [1.08, 1.11]	3.34	514	both	26
Qwen3-30B-A3B	12.5%	16.1	32.7	33.5	1.03 [1.02, 1.04]	2.08	78	host	43
Qwen3-30B-A3B	25%	18.5	51.4	53.9	1.05 [1.04, 1.06]	2.91	174	host	31
Qwen3-30B-A3B	43.75%	23.8	98.3	95.8	0.97 [0.96, 0.99]	4.02	305	both	31
RTX 4090 , i5-12400 (job 091): B_{host} 37 GB/s, B_{gpu} 1008 GB/s; tables 0,0,1,2,2 / 0,0,1,1,2,3,4,4,5									
gpt-oss-120b	11%	18.1	23.8	32.7	1.37 [1.35, 1.39]	1.81	74	host	44
gpt-oss-120b	25%	21.0	42.5	55.0	1.30 [1.26, 1.33]	2.62	184	host	30
Qwen3-30B-A3B	12.5%	10.5	18.2	18.8	1.03 [1.02, 1.05]	1.78	41	host	46
Qwen3-30B-A3B	25%	12.3	30.0	31.3	1.05 [1.03, 1.06]	2.54	91	host	34
RTX 3090 , i9-11900KF (job 092): B_{host} 45 GB/s, B_{gpu} 936 GB/s; tables 0,0,0,0,0 / 0,0,0,0,0,0,0,0,0									
gpt-oss-120b	11%	22.8	8.4 [†]	37.3	–	1.64	88	host	42
Qwen3-30B-A3B	12.5%	12.3	9.0 [†]	19.2	–	1.56	49	host	39
Qwen3-30B-A3B	25%	13.8	11.2 [†]	30.1	–	2.18	110	host	27
Mixtral-8x7B Q4_K_M $C=2$ (25%)		8.22	–	8.34	–	1.015 [1.008, 1.024]	–	–	–

Table S4: An order-free accounting of the gap between the deployed cache and the bound at the host-bound budgets, as percentages of the gap (ms per token). The engine ran both choices of what to cache (the deployed policy’s admissions or MIN’s) with both ways to load a cached expert (two reads: the CPU serves it, then a copy; one read: a copy in the step). *Load alone* and *cache alone* are each choice’s effect with the other at the deployed level; the *interaction* is how much more MIN’s set is worth loaded once than loaded twice. The Shapley values average each choice’s effect over the other’s two levels and add up, with reading ahead (the read-ahead oracle over MIN-1R, nested under MIN’s set, since it needs foresight) and the rest, to 100%. Panel: mean over its machines (number in parentheses) and, in small type, the range over its machines (not an interval); a fast link reads at least half as fast as the machine’s CPU, a slow one less. Last rows: the new machines of the follow-up launches, with their link-to-CPU ratio.

Machine	Budget	Gap (ms)	Load alone	Cache alone	Interaction	Shapley value cache	load	Read-ahead	Rest
O4	gpt-oss 11%	10.6	4	4	43	26	25	13	36
O4	gpt-oss 25%	8.5	4	22	17	31	13	22	34
O4	Qwen3 12.5%	18.4	3	−1	58	28	32	4	36
O4	Qwen3 25%	14.1	2	14	37	33	21	17	30
O5	gpt-oss 11%	11.4	0	−2	28	12	14	17	57
O5	gpt-oss 25%	9.1	−4	20	10	25	1	21	52
O5	Qwen3 12.5%	19.8	1	−13	43	9	23	7	62
O5	Qwen3 25%	15.3	0	14	17	22	9	16	53
Panel, fast link (8)	gpt-oss 11%	11.6	1 (−2 to 4)	3 (−2 to 6)	38 (30 to 47)	22 (14 to 26)	20 (14 to 27)	13 (−4 to 20)	45 (38 to 55)
Panel, fast link (8)	gpt-oss 25%	9.0	0 (−4 to 3)	22 (18 to 26)	15 (10 to 19)	29 (25 to 35)	8 (1 to 12)	20 (9 to 23)	43 (38 to 49)
Panel, slow link (2)	gpt-oss 11%	9.7	−6 (−9 to −3)	−4 (−9 to 0)	3 (−6 to 11)	−3 (−11 to 6)	−5 (−12 to 2)	17 (9 to 24)	91 (83 to 99)
Panel, slow link (2)	gpt-oss 25%	7.8	−17 (−22 to −11)	19 (16 to 23)	3 (3 to 3)	21 (17 to 25)	−15 (−21 to −10)	22 (17 to 26)	73 (68 to 78)
5700X3D (0.92)	gpt-oss 11%	14.9	4	−2	55	25	31	8	36
5700X3D (0.92)	gpt-oss 25%	11.5	4	22	24	34	17	18	31
9950X, x8 (0.54)	gpt-oss 11%	8.3	0	−3	17	5	8	28	59
9950X, x8 (0.54)	gpt-oss 25%	7.4	−15	13	17	22	−6	26	59
9800X3D, slower link (0.61)	gpt-oss 11%	10.8	−2	−4	29	11	12	20	57
9800X3D, slower link (0.61)	gpt-oss 25%	8.9	−5	20	12	26	1	20	54

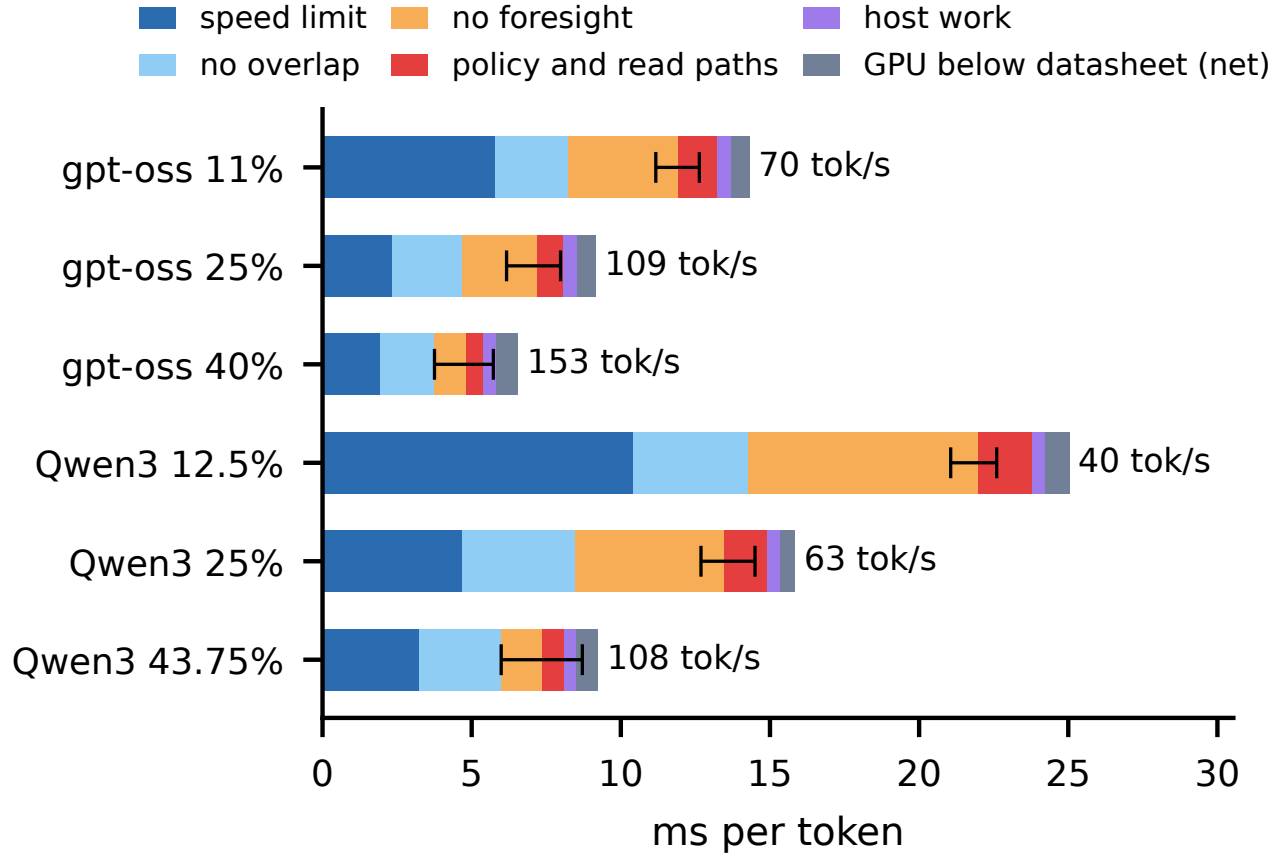


Figure S3: The accounting as a model: Shapley attribution of the gap between our cache’s measured time per token and the bound, for every cell of Table 3 on host B, averaged over all 120 orders in which the five fixes can be applied. Fixes: overlapping host reads with GPU work, which in Eq. (1)’s form also moves experts onto the CPU until the two paths balance; the optimum’s reads instead of the best online policy’s (foresight); the best online policy’s reads at the probe’s highest rate instead of the engine’s reads through Eq. (S1) (policy and read paths); the engine’s per-token host work; and GPU kernels at the datasheet rate, the term that closes the model on the measured time. Whiskers: the range of foresight’s marginal cost over the orders.

Table S5: Job 099 (the host panel), scored by machine by `scripts/panel_099.py` under the same rule. Totals: 273 clauses, 190 held, 42 held (point), 39 failed, 2 untested. Hosts Pa-Pj; cells gpt-oss 11% (g11) and 25% (g25), Qwen3 12.5% (P10). Measured: the point estimate and, where one exists, the 95% interval (bootstrap over problems within the host; pooled clauses have none).

Clause	Statement	Measured	Predicted	Status
Pa-P1-g11	engine reads within 4% of the replay (gpt-oss 11%, Pa)	0.016	≤ 0.04	held (point)
Pa-P3-w16-g11	w16 time share (gpt-oss 11%, Pa)	0.886 [0.880, 0.892]	≥ 0.80	held
Pa-P3-w4-g11	w4 time share (gpt-oss 11%, Pa)	0.426 [0.415, 0.438]	0.35-0.70	held
Pa-P3-w1-g11	w1 time share (gpt-oss 11%, Pa)	0.167 [0.160, 0.175]	≤ 0.30	held
Pa-P4-w8r5-g11	w8r5 time share (gpt-oss 11%, Pa)	0.240 [0.226, 0.253]	0.15-0.45	held
Pa-P4-allr5-g11	allr5 time share (gpt-oss 11%, Pa)	0.455 [0.442, 0.467]	0.30-0.60	held
Pa-P5a-g11	MIN's set worth more with one read than two (gpt-oss 11%, Pa)	3.873 [3.717, 4.038]	> 0 ms	held
Pa-P5b-g11	single read within 5% of deployed (gpt-oss 11%, Pa)	1.004 [1.003, 1.006]	0.95-1.05	held
Pa-P6-fetch-g11	fetch/base (gpt-oss 11%, Pa)	1.334 [1.325, 1.343]	1.1-1.55	held
Pa-P6-both3p-g11	both3p/base (gpt-oss 11%, Pa)	1.515 [1.501, 1.528]	1.25-1.9	held
Pa-P6-order-g11	both3p $>$ fetch $>$ foa (gpt-oss 11%, Pa)	1.000	ordering	held (point)
Pa-P9-g11	host plan at most 150 μ s per step (gpt-oss 11%, Pa)	261.2	$\leq 150 \mu$ s	failed
Pa-P7-g11	aa / foa in [0.92, 1.04] on a ≥ 40 GB/s link (gpt-oss 11%, Pa)	0.901 [0.899, 0.904]	0.92-1.04	failed
Pa-P1-g25	engine reads within 4% of the replay (gpt-oss 25%, Pa)	0.017	≤ 0.04	held (point)
Pa-P3-w16-g25	w16 time share (gpt-oss 25%, Pa)	0.573 [0.564, 0.582]	0.45-0.85	held
Pa-P3-w4-g25	w4 time share (gpt-oss 25%, Pa)	0.235 [0.230, 0.240]	0.12-0.40	held
Pa-P3-w1-g25	w1 time share (gpt-oss 25%, Pa)	0.084 [0.081, 0.088]	≤ 0.30	held
Pa-P4-w8r5-g25	w8r5 time share (gpt-oss 25%, Pa)	0.140 [0.136, 0.145]	≤ 0.30	held
Pa-P4-allr5-g25	allr5 time share (gpt-oss 25%, Pa)	0.402 [0.392, 0.413]	0.30-0.60	held
Pa-P5a-g25	MIN's set worth more with one read than two (gpt-oss 25%, Pa)	1.203 [1.127, 1.277]	> 0 ms	held
Pa-P5b-g25	single read within 5% of deployed (gpt-oss 25%, Pa)	0.997 [0.995, 0.999]	0.95-1.05	held
Pa-P6-fetch-g25	fetch/base (gpt-oss 25%, Pa)	1.332 [1.318, 1.344]	1.1-1.55	held
Pa-P6-both3p-g25	both3p/base (gpt-oss 25%, Pa)	1.687 [1.661, 1.710]	1.25-1.9	held
Pa-P6-order-g25	both3p $>$ fetch $>$ foa (gpt-oss 25%, Pa)	1.000	ordering	held (point)
Pa-P9-g25	host plan at most 150 μ s per step (gpt-oss 25%, Pa)	258.4	$\leq 150 \mu$ s	failed
Pa-P7-g25	aa / foa in [0.92, 1.04] on a ≥ 40 GB/s link (gpt-oss 25%, Pa)	0.928 [0.924, 0.932]	0.92-1.04	held
Pa-P10-w2	Qwen3 w2 time share (Pa)	0.414 [0.403, 0.425]	0.30-0.70	held
Pa-P10-w8r5	Qwen3 w8r5 time share (Pa)	0.382 [0.367, 0.395]	0.25-0.60	held
Pa-P10-fetch	Qwen3 fetch/base (Pa)	1.419 [1.410, 1.428]	1.15-1.55	held
Pb-P1-g11	engine reads within 4% of the replay (gpt-oss 11%, Pb)	0.016	≤ 0.04	held (point)
Pb-P3-w16-g11	w16 time share (gpt-oss 11%, Pb)	0.928 [0.923, 0.932]	≥ 0.80	held
Pb-P3-w4-g11	w4 time share (gpt-oss 11%, Pb)	0.468 [0.456, 0.480]	0.35-0.70	held
Pb-P3-w1-g11	w1 time share (gpt-oss 11%, Pb)	0.171 [0.159, 0.184]	≤ 0.30	held
Pb-P4-w8r5-g11	w8r5 time share (gpt-oss 11%, Pb)	0.265 [0.254, 0.276]	0.15-0.45	held
Pb-P4-allr5-g11	allr5 time share (gpt-oss 11%, Pb)	0.438 [0.422, 0.453]	0.30-0.60	held
Pb-P5a-g11	MIN's set worth more with one read than two (gpt-oss 11%, Pb)	4.823 [4.654, 4.997]	> 0 ms	held
Pb-P5b-g11	single read within 5% of deployed (gpt-oss 11%, Pb)	1.005 [1.003, 1.007]	0.95-1.05	held
Pb-P6-fetch-g11	fetch/base (gpt-oss 11%, Pb)	1.343 [1.334, 1.352]	1.1-1.55	held
Pb-P6-both3p-g11	both3p/base (gpt-oss 11%, Pb)	1.456 [1.444, 1.469]	1.25-1.9	held
Pb-P6-order-g11	both3p $>$ fetch $>$ foa (gpt-oss 11%, Pb)	1.000	ordering	held (point)
Pb-P9-g11	host plan at most 150 μ s per step (gpt-oss 11%, Pb)	202.5	$\leq 150 \mu$ s	failed
Pb-P7-g11	aa / foa in [0.92, 1.04] on a ≥ 40 GB/s link (gpt-oss 11%, Pb)	0.955 [0.952, 0.958]	0.92-1.04	held
Pb-P1-g25	engine reads within 4% of the replay (gpt-oss 25%, Pb)	0.016	≤ 0.04	held (point)
Pb-P3-w16-g25	w16 time share (gpt-oss 25%, Pb)	0.620 [0.612, 0.629]	0.45-0.85	held
Pb-P3-w4-g25	w4 time share (gpt-oss 25%, Pb)	0.240 [0.234, 0.245]	0.12-0.40	held
Pb-P3-w1-g25	w1 time share (gpt-oss 25%, Pb)	0.083 [0.079, 0.087]	≤ 0.30	held
Pb-P4-w8r5-g25	w8r5 time share (gpt-oss 25%, Pb)	0.141 [0.136, 0.147]	≤ 0.30	held
Pb-P4-allr5-g25	allr5 time share (gpt-oss 25%, Pb)	0.420 [0.409, 0.431]	0.30-0.60	held
Pb-P5a-g25	MIN's set worth more with one read than two (gpt-oss 25%, Pb)	1.605 [1.502, 1.712]	> 0 ms	held
Pb-P5b-g25	single read within 5% of deployed (gpt-oss 25%, Pb)	1.016 [1.012, 1.018]	0.95-1.05	held
Pb-P6-fetch-g25	fetch/base (gpt-oss 25%, Pb)	1.403 [1.387, 1.419]	1.1-1.55	held
Pb-P6-both3p-g25	both3p/base (gpt-oss 25%, Pb)	1.713 [1.690, 1.735]	1.25-1.9	held

continued

Clause	Statement	Measured	Predicted	Status
Pb-P6-order-g25	both3p > fetch > foa (gpt-oss 25%, Pb)	1.000	ordering	held (point)
Pb-P9-g25	host plan at most 150 μ s per step (gpt-oss 25%, Pb)	200.9	$\leq 150 \mu$ s	failed
Pb-P7-g25	aa / foa in [0.92, 1.04] on a ≥ 40 GB/s link (gpt-oss 25%, Pb)	0.974 [0.969, 0.980]	0.92-1.04	held
Pb-P10-w2	Qwen3 w2 time share (Pb)	0.449 [0.438, 0.459]	0.30-0.70	held
Pb-P10-w8r5	Qwen3 w8r5 time share (Pb)	0.391 [0.378, 0.404]	0.25-0.60	held
Pb-P10-fetch	Qwen3 fetch/base (Pb)	1.439 [1.430, 1.448]	1.15-1.55	held
Pc-P1-g11	engine reads within 4% of the replay (gpt-oss 11%, Pc)	0.016	≤ 0.04	held (point)
Pc-P3-w16-g11	w16 time share (gpt-oss 11%, Pc)	0.844 [0.836, 0.851]	≥ 0.80	held
Pc-P3-w4-g11	w4 time share (gpt-oss 11%, Pc)	0.362 [0.351, 0.375]	0.35-0.70	held
Pc-P3-w1-g11	w1 time share (gpt-oss 11%, Pc)	0.156 [0.149, 0.163]	≤ 0.30	held
Pc-P4-w8r5-g11	w8r5 time share (gpt-oss 11%, Pc)	0.222 [0.212, 0.231]	0.15-0.45	held
Pc-P4-allr5-g11	allr5 time share (gpt-oss 11%, Pc)	0.549 [0.540, 0.558]	0.30-0.60	held
Pc-P5a-g11	MIN's set worth more with one read than two (gpt-oss 11%, Pc)	3.322 [3.167, 3.483]	> 0 ms	held
Pc-P5b-g11	single read within 5% of deployed (gpt-oss 11%, Pc)	0.996 [0.992, 1.000]	0.95-1.05	held
Pc-P6-fetch-g11	fetch/base (gpt-oss 11%, Pc)	1.168 [1.160, 1.176]	1.1-1.55	held
Pc-P6-both3p-g11	both3p/base (gpt-oss 11%, Pc)	1.328 [1.316, 1.341]	1.25-1.9	held
Pc-P6-order-g11	both3p > fetch > foa (gpt-oss 11%, Pc)	1.000	ordering	held (point)
Pc-P9-g11	host plan at most 150 μ s per step (gpt-oss 11%, Pc)	202.4	$\leq 150 \mu$ s	failed
Pc-P7-g11	aa / foa below 0.95 on a < 32 GB/s link (gpt-oss 11%, Pc)	0.657 [0.654, 0.660]	< 0.95	held
Pc-P1-g25	engine reads within 4% of the replay (gpt-oss 25%, Pc)	0.016	≤ 0.04	held (point)
Pc-P3-w16-g25	w16 time share (gpt-oss 25%, Pc)	0.540 [0.533, 0.549]	0.45-0.85	held
Pc-P3-w4-g25	w4 time share (gpt-oss 25%, Pc)	0.228 [0.222, 0.235]	0.12-0.40	held
Pc-P3-w1-g25	w1 time share (gpt-oss 25%, Pc)	0.099 [0.094, 0.105]	≤ 0.30	held
Pc-P4-w8r5-g25	w8r5 time share (gpt-oss 25%, Pc)	0.149 [0.144, 0.155]	≤ 0.30	held
Pc-P4-allr5-g25	allr5 time share (gpt-oss 25%, Pc)	0.483 [0.474, 0.492]	0.30-0.60	held
Pc-P5a-g25	MIN's set worth more with one read than two (gpt-oss 25%, Pc)	0.974 [0.892, 1.065]	> 0 ms	held
Pc-P5b-g25	single read within 5% of deployed (gpt-oss 25%, Pc)	0.974 [0.969, 0.978]	0.95-1.05	held
Pc-P6-fetch-g25	fetch/base (gpt-oss 25%, Pc)	1.237 [1.224, 1.249]	1.1-1.55	held
Pc-P6-both3p-g25	both3p/base (gpt-oss 25%, Pc)	1.536 [1.521, 1.551]	1.25-1.9	held
Pc-P6-order-g25	both3p > fetch > foa (gpt-oss 25%, Pc)	1.000	ordering	held (point)
Pc-P9-g25	host plan at most 150 μ s per step (gpt-oss 25%, Pc)	218.0	$\leq 150 \mu$ s	failed
Pc-P7-g25	aa / foa below 0.95 on a < 32 GB/s link (gpt-oss 25%, Pc)	0.751 [0.745, 0.758]	< 0.95	held
Pc-P10-w2	Qwen3 w2 time share (Pc)	0.352 [0.343, 0.362]	0.30-0.70	held
Pc-P10-w8r5	Qwen3 w8r5 time share (Pc)	0.375 [0.367, 0.384]	0.25-0.60	held
Pc-P10-fetch	Qwen3 fetch/base (Pc)	1.211 [1.203, 1.218]	1.15-1.55	held
Pd-P1-g11	engine reads within 4% of the replay (gpt-oss 11%, Pd)	–	≤ 0.04	untested
Pd-P3-w16-g11	w16 time share (gpt-oss 11%, Pd)	0.803 [0.794, 0.812]	≥ 0.80	held (point)
Pd-P3-w4-g11	w4 time share (gpt-oss 11%, Pd)	0.342 [0.331, 0.354]	0.35-0.70	failed
Pd-P3-w1-g11	w1 time share (gpt-oss 11%, Pd)	0.147 [0.140, 0.154]	≤ 0.30	held
Pd-P4-w8r5-g11	w8r5 time share (gpt-oss 11%, Pd)	0.209 [0.199, 0.219]	0.15-0.45	held
Pd-P4-allr5-g11	allr5 time share (gpt-oss 11%, Pd)	0.521 [0.511, 0.532]	0.30-0.60	held
Pd-P5a-g11	MIN's set worth more with one read than two (gpt-oss 11%, Pd)	1.169 [1.031, 1.315]	> 0 ms	held
Pd-P5b-g11	single read within 5% of deployed (gpt-oss 11%, Pd)	0.981 [0.975, 0.987]	0.95-1.05	held
Pd-P6-fetch-g11	fetch/base (gpt-oss 11%, Pd)	1.051 [1.043, 1.058]	1.1-1.55	failed
Pd-P6-both3p-g11	both3p/base (gpt-oss 11%, Pd)	1.112 [1.105, 1.120]	1.25-1.9	failed
Pd-P6-order-g11	both3p > fetch > foa (gpt-oss 11%, Pd)	1.000	ordering	held (point)
Pd-P9-g11	host plan at most 150 μ s per step (gpt-oss 11%, Pd)	233.7	$\leq 150 \mu$ s	failed
Pd-P7-g11	aa / foa below 0.95 on a < 32 GB/s link (gpt-oss 11%, Pd)	0.566 [0.562, 0.570]	< 0.95	held
Pd-P1-g25	engine reads within 4% of the replay (gpt-oss 25%, Pd)	–	≤ 0.04	untested
Pd-P3-w16-g25	w16 time share (gpt-oss 25%, Pd)	0.522 [0.512, 0.532]	0.45-0.85	held
Pd-P3-w4-g25	w4 time share (gpt-oss 25%, Pd)	0.221 [0.213, 0.231]	0.12-0.40	held
Pd-P3-w1-g25	w1 time share (gpt-oss 25%, Pd)	0.099 [0.092, 0.107]	≤ 0.30	held
Pd-P4-w8r5-g25	w8r5 time share (gpt-oss 25%, Pd)	0.147 [0.140, 0.154]	≤ 0.30	held
Pd-P4-allr5-g25	allr5 time share (gpt-oss 25%, Pd)	0.435 [0.425, 0.445]	0.30-0.60	held
Pd-P5a-g25	MIN's set worth more with one read than two (gpt-oss 25%, Pd)	0.261 [0.101, 0.421]	> 0 ms	held
Pd-P5b-g25	single read within 5% of deployed (gpt-oss 25%, Pd)	0.922 [0.908, 0.936]	0.95-1.05	failed
Pd-P6-fetch-g25	fetch/base (gpt-oss 25%, Pd)	1.124 [1.108, 1.139]	1.1-1.55	held

continued

Clause	Statement	Measured	Predicted	Status
Pd-P6-both3p-g25	both3p/base (gpt-oss 25%, Pd)	1.315 [1.299, 1.331]	1.25-1.9	held
Pd-P6-order-g25	both3p > fetch > foa (gpt-oss 25%, Pd)	1.000	ordering	held (point)
Pd-P9-g25	host plan at most 150 μ s per step (gpt-oss 25%, Pd)	238.2	$\leq 150 \mu$ s	failed
Pd-P7-g25	aa / foa below 0.95 on a < 32 GB/s link (gpt-oss 25%, Pd)	0.692 [0.683, 0.701]	< 0.95	held
Pe-P1-g11	engine reads within 4% of the replay (gpt-oss 11%, Pe)	0.016	≤ 0.04	held (point)
Pe-P3-w16-g11	w16 time share (gpt-oss 11%, Pe)	0.935 [0.930, 0.939]	≥ 0.80	held
Pe-P3-w4-g11	w4 time share (gpt-oss 11%, Pe)	0.469 [0.457, 0.481]	0.35-0.70	held
Pe-P3-w1-g11	w1 time share (gpt-oss 11%, Pe)	0.164 [0.156, 0.173]	≤ 0.30	held
Pe-P4-w8r5-g11	w8r5 time share (gpt-oss 11%, Pe)	0.257 [0.246, 0.267]	0.15-0.45	held
Pe-P4-allr5-g11	allr5 time share (gpt-oss 11%, Pe)	0.415 [0.400, 0.430]	0.30-0.60	held
Pe-P5a-g11	MIN's set worth more with one read than two (gpt-oss 11%, Pe)	4.784 [4.621, 4.957]	> 0 ms	held
Pe-P5b-g11	single read within 5% of deployed (gpt-oss 11%, Pe)	1.008 [1.007, 1.009]	0.95-1.05	held
Pe-P6-fetch-g11	fetch/base (gpt-oss 11%, Pe)	1.348 [1.338, 1.357]	1.1-1.55	held
Pe-P6-both3p-g11	both3p/base (gpt-oss 11%, Pe)	1.461 [1.447, 1.474]	1.25-1.9	held
Pe-P6-order-g11	both3p > fetch > foa (gpt-oss 11%, Pe)	1.000	ordering	held (point)
Pe-P9-g11	host plan at most 150 μ s per step (gpt-oss 11%, Pe)	262.4	$\leq 150 \mu$ s	failed
Pe-P7-g11	aa / foa in [0.92, 1.04] on a ≥ 40 GB/s link (gpt-oss 11%, Pe)	0.966 [0.963, 0.969]	0.92-1.04	held
Pe-P1-g25	engine reads within 4% of the replay (gpt-oss 25%, Pe)	0.016	≤ 0.04	held (point)
Pe-P3-w16-g25	w16 time share (gpt-oss 25%, Pe)	0.631 [0.623, 0.640]	0.45-0.85	held
Pe-P3-w4-g25	w4 time share (gpt-oss 25%, Pe)	0.238 [0.232, 0.243]	0.12-0.40	held
Pe-P3-w1-g25	w1 time share (gpt-oss 25%, Pe)	0.071 [0.065, 0.076]	≤ 0.30	held
Pe-P4-w8r5-g25	w8r5 time share (gpt-oss 25%, Pe)	0.135 [0.131, 0.139]	≤ 0.30	held
Pe-P4-allr5-g25	allr5 time share (gpt-oss 25%, Pe)	0.405 [0.393, 0.416]	0.30-0.60	held
Pe-P5a-g25	MIN's set worth more with one read than two (gpt-oss 25%, Pe)	1.637 [1.529, 1.749]	> 0 ms	held
Pe-P5b-g25	single read within 5% of deployed (gpt-oss 25%, Pe)	1.014 [1.011, 1.017]	0.95-1.05	held
Pe-P6-fetch-g25	fetch/base (gpt-oss 25%, Pe)	1.403 [1.387, 1.419]	1.1-1.55	held
Pe-P6-both3p-g25	both3p/base (gpt-oss 25%, Pe)	1.710 [1.687, 1.734]	1.25-1.9	held
Pe-P6-order-g25	both3p > fetch > foa (gpt-oss 25%, Pe)	1.000	ordering	held (point)
Pe-P9-g25	host plan at most 150 μ s per step (gpt-oss 25%, Pe)	258.2	$\leq 150 \mu$ s	failed
Pe-P7-g25	aa / foa in [0.92, 1.04] on a ≥ 40 GB/s link (gpt-oss 25%, Pe)	0.988 [0.983, 0.993]	0.92-1.04	held
Pf-P1-g11	engine reads within 4% of the replay (gpt-oss 11%, Pf)	0.016	≤ 0.04	held (point)
Pf-P3-w16-g11	w16 time share (gpt-oss 11%, Pf)	0.781 [0.771, 0.791]	≥ 0.80	failed
Pf-P3-w4-g11	w4 time share (gpt-oss 11%, Pf)	0.320 [0.309, 0.331]	0.35-0.70	failed
Pf-P3-w1-g11	w1 time share (gpt-oss 11%, Pf)	0.141 [0.134, 0.148]	≤ 0.30	held
Pf-P4-w8r5-g11	w8r5 time share (gpt-oss 11%, Pf)	0.198 [0.188, 0.208]	0.15-0.45	held
Pf-P4-allr5-g11	allr5 time share (gpt-oss 11%, Pf)	0.540 [0.530, 0.549]	0.30-0.60	held
Pf-P5a-g11	MIN's set worth more with one read than two (gpt-oss 11%, Pf)	-0.484 [-1.055, -0.139]	> 0 ms	failed
Pf-P5b-g11	single read within 5% of deployed (gpt-oss 11%, Pf)	0.947 [0.928, 0.983]	0.95-1.05	failed
Pf-P6-fetch-g11	fetch/base (gpt-oss 11%, Pf)	0.875 [0.855, 0.911]	1.1-1.55	failed
Pf-P6-both3p-g11	both3p/base (gpt-oss 11%, Pf)	1.008 [0.975, 1.056]	1.25-1.9	failed
Pf-P6-order-g11	both3p > fetch > foa (gpt-oss 11%, Pf)	0.000	ordering	failed
Pf-P9-g11	host plan at most 150 μ s per step (gpt-oss 11%, Pf)	201.3	$\leq 150 \mu$ s	failed
Pf-P7-g11	aa / foa below 0.95 on a < 32 GB/s link (gpt-oss 11%, Pf)	0.448 [0.444, 0.452]	< 0.95	held
Pf-P1-g25	engine reads within 4% of the replay (gpt-oss 25%, Pf)	0.017	≤ 0.04	held (point)
Pf-P3-w16-g25	w16 time share (gpt-oss 25%, Pf)	0.493 [0.485, 0.502]	0.45-0.85	held
Pf-P3-w4-g25	w4 time share (gpt-oss 25%, Pf)	0.213 [0.207, 0.221]	0.12-0.40	held
Pf-P3-w1-g25	w1 time share (gpt-oss 25%, Pf)	0.095 [0.089, 0.102]	≤ 0.30	held
Pf-P4-w8r5-g25	w8r5 time share (gpt-oss 25%, Pf)	0.140 [0.134, 0.148]	≤ 0.30	held
Pf-P4-allr5-g25	allr5 time share (gpt-oss 25%, Pf)	0.434 [0.425, 0.444]	0.30-0.60	held
Pf-P5a-g25	MIN's set worth more with one read than two (gpt-oss 25%, Pf)	0.192 [0.143, 0.243]	> 0 ms	held
Pf-P5b-g25	single read within 5% of deployed (gpt-oss 25%, Pf)	0.853 [0.848, 0.859]	0.95-1.05	failed
Pf-P6-fetch-g25	fetch/base (gpt-oss 25%, Pf)	0.969 [0.964, 0.975]	1.1-1.55	failed
Pf-P6-both3p-g25	both3p/base (gpt-oss 25%, Pf)	1.202 [1.194, 1.211]	1.25-1.9	failed
Pf-P6-order-g25	both3p > fetch > foa (gpt-oss 25%, Pf)	1.000	ordering	held (point)
Pf-P9-g25	host plan at most 150 μ s per step (gpt-oss 25%, Pf)	198.6	$\leq 150 \mu$ s	failed

continued

Clause	Statement	Measured	Predicted	Status
Pf-P7-g25	aa / foa below 0.95 on a < 32 GB/s link (gpt-oss 25%, Pf)	0.612 [0.603, 0.622]	< 0.95	held
Pg-P1-g11	engine reads within 4% of the replay (gpt-oss 11%, Pg)	0.016	≤ 0.04	held (point)
Pg-P3-w16-g11	w16 time share (gpt-oss 11%, Pg)	0.835 [0.828, 0.842]	≥ 0.80	held
Pg-P3-w4-g11	w4 time share (gpt-oss 11%, Pg)	0.378 [0.367, 0.390]	0.35-0.70	held
Pg-P3-w1-g11	w1 time share (gpt-oss 11%, Pg)	0.155 [0.148, 0.162]	≤ 0.30	held
Pg-P4-w8r5-g11	w8r5 time share (gpt-oss 11%, Pg)	0.225 [0.214, 0.236]	0.15-0.45	held
Pg-P4-allr5-g11	allr5 time share (gpt-oss 11%, Pg)	0.510 [0.500, 0.521]	0.30-0.60	held
Pg-P5a-g11	MIN's set worth more with one read than two (gpt-oss 11%, Pg)	4.321 [4.105, 4.552]	> 0 ms	held
Pg-P5b-g11	single read within 5% of deployed (gpt-oss 11%, Pg)	1.012 [1.006, 1.018]	0.95-1.05	held
Pg-P6-fetch-g11	fetch/base (gpt-oss 11%, Pg)	1.245 [1.235, 1.256]	1.1-1.55	held
Pg-P6-both3p-g11	both3p/base (gpt-oss 11%, Pg)	1.381 [1.369, 1.393]	1.25-1.9	held
Pg-P6-order-g11	both3p > fetch > foa (gpt-oss 11%, Pg)	1.000	ordering	held (point)
Pg-P9-g11	host plan at most 150 μs per step (gpt-oss 11%, Pg)	277.3	≤ 150 μs	failed
Pg-P7-g11	aa / foa below 0.95 on a < 32 GB/s link (gpt-oss 11%, Pg)	0.722 [0.717, 0.726]	< 0.95	held
Pg-P1-g25	engine reads within 4% of the replay (gpt-oss 25%, Pg)	0.017	≤ 0.04	held (point)
Pg-P3-w16-g25	w16 time share (gpt-oss 25%, Pg)	0.544 [0.536, 0.553]	0.45-0.85	held
Pg-P3-w4-g25	w4 time share (gpt-oss 25%, Pg)	0.225 [0.219, 0.231]	0.12-0.40	held
Pg-P3-w1-g25	w1 time share (gpt-oss 25%, Pg)	0.090 [0.085, 0.095]	≤ 0.30	held
Pg-P4-w8r5-g25	w8r5 time share (gpt-oss 25%, Pg)	0.141 [0.136, 0.147]	≤ 0.30	held
Pg-P4-allr5-g25	allr5 time share (gpt-oss 25%, Pg)	0.438 [0.428, 0.448]	0.30-0.60	held
Pg-P5a-g25	MIN's set worth more with one read than two (gpt-oss 25%, Pg)	1.173 [1.090, 1.251]	> 0 ms	held
Pg-P5b-g25	single read within 5% of deployed (gpt-oss 25%, Pg)	0.997 [0.992, 1.002]	0.95-1.05	held
Pg-P6-fetch-g25	fetch/base (gpt-oss 25%, Pg)	1.320 [1.307, 1.332]	1.1-1.55	held
Pg-P6-both3p-g25	both3p/base (gpt-oss 25%, Pg)	1.630 [1.614, 1.648]	1.25-1.9	held
Pg-P6-order-g25	both3p > fetch > foa (gpt-oss 25%, Pg)	1.000	ordering	held (point)
Pg-P9-g25	host plan at most 150 μs per step (gpt-oss 25%, Pg)	242.3	≤ 150 μs	failed
Pg-P7-g25	aa / foa below 0.95 on a < 32 GB/s link (gpt-oss 25%, Pg)	0.794 [0.788, 0.801]	< 0.95	held
Ph-P1-g11	engine reads within 4% of the replay (gpt-oss 11%, Ph)	0.016	≤ 0.04	held (point)
Ph-P3-w16-g11	w16 time share (gpt-oss 11%, Ph)	0.887 [0.881, 0.892]	≥ 0.80	held
Ph-P3-w4-g11	w4 time share (gpt-oss 11%, Ph)	0.426 [0.415, 0.437]	0.35-0.70	held
Ph-P3-w1-g11	w1 time share (gpt-oss 11%, Ph)	0.172 [0.165, 0.179]	≤ 0.30	held
Ph-P4-w8r5-g11	w8r5 time share (gpt-oss 11%, Ph)	0.251 [0.241, 0.260]	0.15-0.45	held
Ph-P4-allr5-g11	allr5 time share (gpt-oss 11%, Ph)	0.499 [0.488, 0.510]	0.30-0.60	held
Ph-P5a-g11	MIN's set worth more with one read than two (gpt-oss 11%, Ph)	3.023 [2.907, 3.150]	> 0 ms	held
Ph-P5b-g11	single read within 5% of deployed (gpt-oss 11%, Ph)	0.991 [0.990, 0.993]	0.95-1.05	held
Ph-P6-fetch-g11	fetch/base (gpt-oss 11%, Ph)	1.246 [1.238, 1.254]	1.1-1.55	held
Ph-P6-both3p-g11	both3p/base (gpt-oss 11%, Ph)	1.424 [1.410, 1.437]	1.25-1.9	held
Ph-P6-order-g11	both3p > fetch > foa (gpt-oss 11%, Ph)	1.000	ordering	held (point)
Ph-P9-g11	host plan at most 150 μs per step (gpt-oss 11%, Ph)	216.2	≤ 150 μs	failed
Ph-P7-g11	aa / foa in [0.92, 1.04] on a ≥ 40 GB/s link (gpt-oss 11%, Ph)	0.845 [0.842, 0.848]	0.92-1.04	failed
Ph-P1-g25	engine reads within 4% of the replay (gpt-oss 25%, Ph)	0.016	≤ 0.04	held (point)
Ph-P3-w16-g25	w16 time share (gpt-oss 25%, Ph)	0.590 [0.581, 0.600]	0.45-0.85	held
Ph-P3-w4-g25	w4 time share (gpt-oss 25%, Ph)	0.242 [0.236, 0.247]	0.12-0.40	held
Ph-P3-w1-g25	w1 time share (gpt-oss 25%, Ph)	0.097 [0.092, 0.102]	≤ 0.30	held
Ph-P4-w8r5-g25	w8r5 time share (gpt-oss 25%, Ph)	0.144 [0.138, 0.151]	≤ 0.30	held
Ph-P4-allr5-g25	allr5 time share (gpt-oss 25%, Ph)	0.434 [0.425, 0.444]	0.30-0.60	held
Ph-P5a-g25	MIN's set worth more with one read than two (gpt-oss 25%, Ph)	1.095 [0.963, 1.224]	> 0 ms	held
Ph-P5b-g25	single read within 5% of deployed (gpt-oss 25%, Ph)	0.990 [0.986, 0.995]	0.95-1.05	held
Ph-P6-fetch-g25	fetch/base (gpt-oss 25%, Ph)	1.286 [1.274, 1.298]	1.1-1.55	held
Ph-P6-both3p-g25	both3p/base (gpt-oss 25%, Ph)	1.632 [1.609, 1.656]	1.25-1.9	held
Ph-P6-order-g25	both3p > fetch > foa (gpt-oss 25%, Ph)	1.000	ordering	held (point)
Ph-P9-g25	host plan at most 150 μs per step (gpt-oss 25%, Ph)	202.4	≤ 150 μs	failed
Ph-P7-g25	aa / foa in [0.92, 1.04] on a ≥ 40 GB/s link (gpt-oss 25%, Ph)	0.897 [0.893, 0.902]	0.92-1.04	failed
Pi-P1-g11	engine reads within 4% of the replay (gpt-oss 11%, Pi)	0.016	≤ 0.04	held (point)
Pi-P3-w16-g11	w16 time share (gpt-oss 11%, Pi)	0.844 [0.837, 0.851]	≥ 0.80	held
Pi-P3-w4-g11	w4 time share (gpt-oss 11%, Pi)	0.387 [0.375, 0.398]	0.35-0.70	held

continued

Clause	Statement	Measured	Predicted	Status
Pi-P3-w1-g11	w1 time share (gpt-oss 11%, Pi)	0.155 [0.147, 0.163]	≤ 0.30	held
Pi-P4-w8r5-g11	w8r5 time share (gpt-oss 11%, Pi)	0.226 [0.214, 0.237]	0.15-0.45	held
Pi-P4-allr5-g11	allr5 time share (gpt-oss 11%, Pi)	0.501 [0.489, 0.511]	0.30-0.60	held
Pi-P5a-g11	MIN's set worth more with one read than two (gpt-oss 11%, Pi)	8.898 [8.535, 9.306]	> 0 ms	held
Pi-P5b-g11	single read within 5% of deployed (gpt-oss 11%, Pi)	1.020 [1.019, 1.022]	0.95-1.05	held
Pi-P6-fetch-g11	fetch/base (gpt-oss 11%, Pi)	1.344 [1.335, 1.352]	1.1-1.55	held
Pi-P6-both3p-g11	both3p/base (gpt-oss 11%, Pi)	1.310 [1.300, 1.320]	1.25-1.9	held
Pi-P6-order-g11	both3p $>$ fetch $>$ foa (gpt-oss 11%, Pi)	0.000	ordering	failed
Pi-P9-g11	host plan at most 150 μ s per step (gpt-oss 11%, Pi)	207.8	$\leq 150 \mu$ s	failed
Pi-P7-g11	aa / foa below 0.95 on a < 32 GB/s link (gpt-oss 11%, Pi)	0.766 [0.763, 0.769]	< 0.95	held
Pi-P1-g25	engine reads within 4% of the replay (gpt-oss 25%, Pi)	0.017	≤ 0.04	held (point)
Pi-P3-w16-g25	w16 time share (gpt-oss 25%, Pi)	0.558 [0.550, 0.567]	0.45-0.85	held
Pi-P3-w4-g25	w4 time share (gpt-oss 25%, Pi)	0.226 [0.221, 0.232]	0.12-0.40	held
Pi-P3-w1-g25	w1 time share (gpt-oss 25%, Pi)	0.086 [0.082, 0.091]	≤ 0.30	held
Pi-P4-w8r5-g25	w8r5 time share (gpt-oss 25%, Pi)	0.139 [0.135, 0.143]	≤ 0.30	held
Pi-P4-allr5-g25	allr5 time share (gpt-oss 25%, Pi)	0.446 [0.437, 0.454]	0.30-0.60	held
Pi-P5a-g25	MIN's set worth more with one read than two (gpt-oss 25%, Pi)	2.382 [2.231, 2.539]	> 0 ms	held
Pi-P5b-g25	single read within 5% of deployed (gpt-oss 25%, Pi)	1.018 [1.014, 1.022]	0.95-1.05	held
Pi-P6-fetch-g25	fetch/base (gpt-oss 25%, Pi)	1.444 [1.428, 1.459]	1.1-1.55	held
Pi-P6-both3p-g25	both3p/base (gpt-oss 25%, Pi)	1.578 [1.559, 1.597]	1.25-1.9	held
Pi-P6-order-g25	both3p $>$ fetch $>$ foa (gpt-oss 25%, Pi)	1.000	ordering	held (point)
Pi-P9-g25	host plan at most 150 μ s per step (gpt-oss 25%, Pi)	201.3	$\leq 150 \mu$ s	failed
Pi-P7-g25	aa / foa below 0.95 on a < 32 GB/s link (gpt-oss 25%, Pi)	0.820 [0.813, 0.827]	< 0.95	held
Pj-P1-g11	engine reads within 4% of the replay (gpt-oss 11%, Pj)	0.016	≤ 0.04	held (point)
Pj-P3-w16-g11	w16 time share (gpt-oss 11%, Pj)	0.905 [0.900, 0.910]	≥ 0.80	held
Pj-P3-w4-g11	w4 time share (gpt-oss 11%, Pj)	0.441 [0.429, 0.452]	0.35-0.70	held
Pj-P3-w1-g11	w1 time share (gpt-oss 11%, Pj)	0.173 [0.166, 0.181]	≤ 0.30	held
Pj-P4-w8r5-g11	w8r5 time share (gpt-oss 11%, Pj)	0.257 [0.247, 0.266]	0.15-0.45	held
Pj-P4-allr5-g11	allr5 time share (gpt-oss 11%, Pj)	0.482 [0.471, 0.494]	0.30-0.60	held
Pj-P5a-g11	MIN's set worth more with one read than two (gpt-oss 11%, Pj)	3.093 [2.977, 3.214]	> 0 ms	held
Pj-P5b-g11	single read within 5% of deployed (gpt-oss 11%, Pj)	0.998 [0.997, 0.999]	0.95-1.05	held
Pj-P6-fetch-g11	fetch/base (gpt-oss 11%, Pj)	1.265 [1.257, 1.273]	1.1-1.55	held
Pj-P6-both3p-g11	both3p/base (gpt-oss 11%, Pj)	1.428 [1.414, 1.442]	1.25-1.9	held
Pj-P6-order-g11	both3p $>$ fetch $>$ foa (gpt-oss 11%, Pj)	1.000	ordering	held (point)
Pj-P9-g11	host plan at most 150 μ s per step (gpt-oss 11%, Pj)	198.2	$\leq 150 \mu$ s	failed
Pj-P7-g11	aa / foa in [0.92, 1.04] on a ≥ 40 GB/s link (gpt-oss 11%, Pj)	0.880 [0.877, 0.882]	0.92-1.04	failed
Pj-P1-g25	engine reads within 4% of the replay (gpt-oss 25%, Pj)	0.016	≤ 0.04	held (point)
Pj-P3-w16-g25	w16 time share (gpt-oss 25%, Pj)	0.607 [0.600, 0.615]	0.45-0.85	held
Pj-P3-w4-g25	w4 time share (gpt-oss 25%, Pj)	0.244 [0.239, 0.250]	0.12-0.40	held
Pj-P3-w1-g25	w1 time share (gpt-oss 25%, Pj)	0.090 [0.077, 0.099]	≤ 0.30	held
Pj-P4-w8r5-g25	w8r5 time share (gpt-oss 25%, Pj)	0.151 [0.147, 0.156]	≤ 0.30	held
Pj-P4-allr5-g25	allr5 time share (gpt-oss 25%, Pj)	0.430 [0.420, 0.441]	0.30-0.60	held
Pj-P5a-g25	MIN's set worth more with one read than two (gpt-oss 25%, Pj)	0.971 [0.899, 1.040]	> 0 ms	held
Pj-P5b-g25	single read within 5% of deployed (gpt-oss 25%, Pj)	1.002 [0.999, 1.005]	0.95-1.05	held
Pj-P6-fetch-g25	fetch/base (gpt-oss 25%, Pj)	1.309 [1.296, 1.322]	1.1-1.55	held
Pj-P6-both3p-g25	both3p/base (gpt-oss 25%, Pj)	1.658 [1.634, 1.681]	1.25-1.9	held
Pj-P6-order-g25	both3p $>$ fetch $>$ foa (gpt-oss 25%, Pj)	1.000	ordering	held (point)
Pj-P9-g25	host plan at most 150 μ s per step (gpt-oss 25%, Pj)	193.3	$\leq 150 \mu$ s	failed
Pj-P7-g25	aa / foa in [0.92, 1.04] on a ≥ 40 GB/s link (gpt-oss 25%, Pj)	0.921 [0.917, 0.925]	0.92-1.04	held (point)
P2a	time share within 0.15 of read share at $\geq 80\%$ of hybrid host-cells	0.940	≥ 0.80	held (point)
P2b	w1 $<$ w4 $<$ w16 at $\geq 80\%$ of host-cells	1.000	≥ 0.80	held (point)
P8-g11	spread of fetch/base across hosts > 3 x the median within-host half-width (gpt-oss 11%)	55.5	> 3	held (point)
P8-g25	spread of fetch/base across hosts > 3 x the median within-host half-width (gpt-oss 25%)	36.5	> 3	held (point)

Table S6: Job 100 (foresight on the deployed path; the model from the probe alone), scored by machine by `scripts/job100.py` under the same rule. Totals: 216 clauses, 41 held, 122 held (point), 27 failed, 26 untested. Hosts a (Pd again), b (5700X3D), c (Ph again), f (9950X, x8 link), e (13900KF, never ran); cells gpt-oss 11% (g11) and 25% (g25). Measured: the point estimate and, where one exists, the 95% interval (bootstrap over problems within the host; pooled clauses have none).

Clause	Statement	Measured	Predicted	Status
a-P1-aa-g11	aa counters equal job 099's (gpt-oss 11%, Pd again)	0.000	$\leq 0.5\%$	held (point)
a-P1-w4-g11	w4 counters equal job 099's (gpt-oss 11%, Pd again)	0.000	$\leq 0.5\%$	held (point)
a-P1-w16-g11	w16 counters equal job 099's (gpt-oss 11%, Pd again)	0.000	$\leq 0.5\%$	held (point)
a-P1-fetch-g11	fetch counters equal job 099's (gpt-oss 11%, Pd again)	0.001	$\leq 0.5\%$	held (point)
a-P2-aa-g11	probe-only aa time within 12% (gpt-oss 11%, Pd again)	0.009	$\leq 12\%$	held (point)
a-P2-w4-g11	probe-only w4 time within 12% (gpt-oss 11%, Pd again)	0.014	$\leq 12\%$	held (point)
a-P2-fetchsign-g11	probe-only fetch time under-predicted (gpt-oss 11%, Pd again)	-0.186	< 0	held (point)
a-P3-fetch-g11	model sign of fetch/base, predicted 1.17 (gpt-oss 11%, Pd again)	1.048 [1.040, 1.056]	≥ 1	held
a-P3-aa-g11	model sign of aa/base, predicted 0.51 (gpt-oss 11%, Pd again)	0.554 [0.550, 0.557]	≥ 1	held
a-P4-b16-g11	b16/base in [1.0, 1.25] (gpt-oss 11%, Pd again)	1.026 [1.018, 1.033]	1.0-1.25	held
a-P4-b4-g11	b4/base in [0.98, 1.12] (gpt-oss 11%, Pd again)	1.018 [1.014, 1.022]	0.98-1.12	held
a-P4-b8r5-g11	b8r5/base in [0.95, 1.08] (gpt-oss 11%, Pd again)	0.943 [0.939, 0.947]	0.95-1.08	failed
a-P4-b16bypass-g11	b16/base $\leq$ bypass/base + 0.03 (gpt-oss 11%, Pd again)	0.026	≤ 0.03	held (point)
a-P4-mono-g11	b16 at least as fast as b4 (gpt-oss 11%, Pd again)	1.008 [1.002, 1.014]	≥ 1	held
a-P5-g11	b16/base $>$ w16/base at link/CPU 0.41 (gpt-oss 11%, Pd again)	0.135	> 0	held (point)
a-P7-w4-g11	host plan of w4 at most 150 μ s per step (gpt-oss 11%, Pd again)	13.3	$\leq 150 \mu$ s	held (point)
a-P7-w16-g11	host plan of w16 at most 150 μ s per step (gpt-oss 11%, Pd again)	19.7	$\leq 150 \mu$ s	held (point)
a-P7-b4-g11	host plan of b4 at most 150 μ s per step (gpt-oss 11%, Pd again)	–	$\leq 150 \mu$ s	untested
a-P7-b16-g11	host plan of b16 at most 150 μ s per step (gpt-oss 11%, Pd again)	–	$\leq 150 \mu$ s	untested
a-P7-b8r5-g11	host plan of b8r5 at most 150 μ s per step (gpt-oss 11%, Pd again)	–	$\leq 150 \mu$ s	untested
a-P7-fetch-g11	host plan of fetch at most 150 μ s per step (gpt-oss 11%, Pd again)	64.8	$\leq 150 \mu$ s	held (point)
a-P8-b4-g11	engine misses of b4 within 10% of the replay (gpt-oss 11%, Pd again)	0.142	$\leq 10\%$	failed
a-P8-b16-g11	engine misses of b16 within 10% of the replay (gpt-oss 11%, Pd again)	0.178	$\leq 10\%$	failed
a-P8-b8r5-g11	engine misses of b8r5 within 10% of the replay (gpt-oss 11%, Pd again)	0.091	$\leq 10\%$	held (point)
a-P1-aa-g25	aa counters equal job 099's (gpt-oss 25%, Pd again)	0.000	$\leq 0.5\%$	held (point)
a-P1-w4-g25	w4 counters equal job 099's (gpt-oss 25%, Pd again)	0.000	$\leq 0.5\%$	held (point)
a-P1-w16-g25	w16 counters equal job 099's (gpt-oss 25%, Pd again)	0.000	$\leq 0.5\%$	held (point)
a-P1-fetch-g25	fetch counters equal job 099's (gpt-oss 25%, Pd again)	0.001	$\leq 0.5\%$	held (point)
a-P2-aa-g25	probe-only aa time within 12% (gpt-oss 25%, Pd again)	0.004	$\leq 12\%$	held (point)
a-P2-w4-g25	probe-only w4 time within 12% (gpt-oss 25%, Pd again)	0.000	$\leq 12\%$	held (point)
a-P2-fetchsign-g25	probe-only fetch time under-predicted (gpt-oss 25%, Pd again)	-0.089	< 0	held (point)
a-P3-fetch-g25	model sign of fetch/base, predicted 1.14 (gpt-oss 25%, Pd again)	1.111 [1.103, 1.118]	≥ 1	held
a-P3-aa-g25	model sign of aa/base, predicted 0.58 (gpt-oss 25%, Pd again)	0.629 [0.621, 0.638]	≥ 1	held
a-P4-b16-g25	b16/base in [1.0, 1.25] (gpt-oss 25%, Pd again)	1.123 [1.112, 1.134]	1.0-1.25	held
a-P4-b4-g25	b4/base in [0.98, 1.12] (gpt-oss 25%, Pd again)	1.034 [1.024, 1.043]	0.98-1.12	held
a-P4-b8r5-g25	b8r5/base in [0.95, 1.08] (gpt-oss 25%, Pd again)	1.009 [1.001, 1.015]	0.95-1.08	held
a-P4-b16bypass-g25	b16/base $\leq$ bypass/base + 0.03 (gpt-oss 25%, Pd again)	-0.069	≤ 0.03	held (point)
a-P4-mono-g25	b16 at least as fast as b4 (gpt-oss 25%, Pd again)	1.086 [1.073, 1.101]	≥ 1	held
a-P7-w4-g25	host plan of w4 at most 150 μ s per step (gpt-oss 25%, Pd again)	12.6	$\leq 150 \mu$ s	held (point)
a-P7-w16-g25	host plan of w16 at most 150 μ s per step (gpt-oss 25%, Pd again)	20.0	$\leq 150 \mu$ s	held (point)

continued

Clause	Statement	Measured	Predicted	Status
a-P7-b4-g25	host plan of b4 at most 150 μ s per step (gpt-oss 25%, Pd again)	–	$\leq 150 \mu$ s	untested
a-P7-b16-g25	host plan of b16 at most 150 μ s per step (gpt-oss 25%, Pd again)	–	$\leq 150 \mu$ s	untested
a-P7-b8r5-g25	host plan of b8r5 at most 150 μ s per step (gpt-oss 25%, Pd again)	–	$\leq 150 \mu$ s	untested
a-P7-fetch-g25	host plan of fetch at most 150 μ s per step (gpt-oss 25%, Pd again)	63.5	$\leq 150 \mu$ s	held (point)
a-P8-b4-g25	engine misses of b4 within 10% of the replay (gpt-oss 25%, Pd again)	0.097	$\leq 10\%$	held (point)
a-P8-b16-g25	engine misses of b16 within 10% of the replay (gpt-oss 25%, Pd again)	0.134	$\leq 10\%$	failed
a-P8-b8r5-g25	engine misses of b8r5 within 10% of the replay (gpt-oss 25%, Pd again)	0.066	$\leq 10\%$	held (point)
a-P6-base-g11	base time within 3% of job 099d (gpt-oss 11%, Pd again)	0.003	$\leq 3\%$	held (point)
a-P6-foa-g11	foa/base within 0.03 of job 099d (gpt-oss 11%, Pd again)	0.001	≤ 0.03	held (point)
a-P6-aa-g11	aa/base within 0.03 of job 099d (gpt-oss 11%, Pd again)	0.001	≤ 0.03	held (point)
a-P6-fetch-g11	fetch/base within 0.03 of job 099d (gpt-oss 11%, Pd again)	0.003	≤ 0.03	held (point)
a-P6-both3p-g11	both3p/base within 0.03 of job 099d (gpt-oss 11%, Pd again)	0.002	≤ 0.03	held (point)
a-P6-w4-g11	w4/base within 0.03 of job 099d (gpt-oss 11%, Pd again)	0.001	≤ 0.03	held (point)
a-P6-w16-g11	w16/base within 0.03 of job 099d (gpt-oss 11%, Pd again)	0.002	≤ 0.03	held (point)
a-P6-base-g25	base time within 3% of job 099d (gpt-oss 25%, Pd again)	0.018	$\leq 3\%$	held (point)
a-P6-foa-g25	foa/base within 0.03 of job 099d (gpt-oss 25%, Pd again)	0.018	≤ 0.03	held (point)
a-P6-aa-g25	aa/base within 0.03 of job 099d (gpt-oss 25%, Pd again)	0.009	≤ 0.03	held (point)
a-P6-fetch-g25	fetch/base within 0.03 of job 099d (gpt-oss 25%, Pd again)	0.013	≤ 0.03	held (point)
a-P6-both3p-g25	both3p/base within 0.03 of job 099d (gpt-oss 25%, Pd again)	0.029	≤ 0.03	held (point)
a-P6-w4-g25	w4/base within 0.03 of job 099d (gpt-oss 25%, Pd again)	0.011	≤ 0.03	held (point)
a-P6-w16-g25	w16/base within 0.03 of job 099d (gpt-oss 25%, Pd again)	0.014	≤ 0.03	held (point)
b-P1-aa-g11	aa counters equal job 099's (gpt-oss 11%, 5700X3D)	0.000	$\leq 0.5\%$	held (point)
b-P1-w4-g11	w4 counters equal job 099's (gpt-oss 11%, 5700X3D)	0.000	$\leq 0.5\%$	held (point)
b-P1-w16-g11	w16 counters equal job 099's (gpt-oss 11%, 5700X3D)	0.000	$\leq 0.5\%$	held (point)
b-P1-fetch-g11	fetch counters equal job 099's (gpt-oss 11%, 5700X3D)	0.001	$\leq 0.5\%$	held (point)
b-P2-aa-g11	probe-only aa time within 12% (gpt-oss 11%, 5700X3D)	0.073	$\leq 12\%$	held (point)
b-P2-w4-g11	probe-only w4 time within 12% (gpt-oss 11%, 5700X3D)	0.048	$\leq 12\%$	held (point)
b-P3-fetch-g11	model sign of fetch/base, predicted 1.40 (gpt-oss 11%, 5700X3D)	1.367 [1.356, 1.377]	≥ 1	held
b-P3-aa-g11	model sign of aa/base, predicted 0.87 (gpt-oss 11%, 5700X3D)	0.928 [0.924, 0.931]	≥ 1	held
b-P4-b16-g11	b16/base in [1.0, 1.25] (gpt-oss 11%, 5700X3D)	1.000 [0.995, 1.004]	1.0-1.25	failed
b-P4-b4-g11	b4/base in [0.98, 1.12] (gpt-oss 11%, 5700X3D)	0.992 [0.989, 0.995]	0.98-1.12	held
b-P4-b8r5-g11	b8r5/base in [0.95, 1.08] (gpt-oss 11%, 5700X3D)	0.952 [0.949, 0.954]	0.95-1.08	held (point)
b-P4-b16bypass-g11	b16/base $\leq$ bypass/base + 0.03 (gpt-oss 11%, 5700X3D)	0.011	≤ 0.03	held (point)
b-P4-mono-g11	b16 at least as fast as b4 (gpt-oss 11%, 5700X3D)	1.008 [1.005, 1.011]	≥ 1	held
b-P5-g11	w16/base $>$ b16/base at link/CPU 0.92 (gpt-oss 11%, 5700X3D)	0.321	> 0	held (point)
b-P7-w4-g11	host plan of w4 at most 150 μ s per step (gpt-oss 11%, 5700X3D)	15.8	$\leq 150 \mu$ s	held (point)
b-P7-w16-g11	host plan of w16 at most 150 μ s per step (gpt-oss 11%, 5700X3D)	26.7	$\leq 150 \mu$ s	held (point)
b-P7-b4-g11	host plan of b4 at most 150 μ s per step (gpt-oss 11%, 5700X3D)	–	$\leq 150 \mu$ s	untested
b-P7-b16-g11	host plan of b16 at most 150 μ s per step (gpt-oss 11%, 5700X3D)	–	$\leq 150 \mu$ s	untested
b-P7-b8r5-g11	host plan of b8r5 at most 150 μ s per step (gpt-oss 11%, 5700X3D)	–	$\leq 150 \mu$ s	untested
b-P7-fetch-g11	host plan of fetch at most 150 μ s per step (gpt-oss 11%, 5700X3D)	106.4	$\leq 150 \mu$ s	held (point)
b-P8-b4-g11	engine misses of b4 within 10% of the replay (gpt-oss 11%, 5700X3D)	0.196	$\leq 10\%$	failed
b-P8-b16-g11	engine misses of b16 within 10% of the replay (gpt-oss 11%, 5700X3D)	0.329	$\leq 10\%$	failed

continued

Clause	Statement	Measured	Predicted	Status
b-P8-b8r5-g11	engine misses of b8r5 within 10% of the replay (gpt-oss 11%, 5700X3D)	0.123	$\leq 10\%$	failed
b-P1-aa-g25	aa counters equal job 099's (gpt-oss 25%, 5700X3D)	0.000	$\leq 0.5\%$	held (point)
b-P1-w4-g25	w4 counters equal job 099's (gpt-oss 25%, 5700X3D)	0.000	$\leq 0.5\%$	held (point)
b-P1-w16-g25	w16 counters equal job 099's (gpt-oss 25%, 5700X3D)	0.000	$\leq 0.5\%$	held (point)
b-P1-fetch-g25	fetch counters equal job 099's (gpt-oss 25%, 5700X3D)	0.001	$\leq 0.5\%$	held (point)
b-P2-aa-g25	probe-only aa time within 12% (gpt-oss 25%, 5700X3D)	0.063	$\leq 12\%$	held (point)
b-P2-w4-g25	probe-only w4 time within 12% (gpt-oss 25%, 5700X3D)	0.037	$\leq 12\%$	held (point)
b-P3-fetch-g25	model sign of fetch/base, predicted 1.50 (gpt-oss 25%, 5700X3D)	1.484 [1.465, 1.501]	≥ 1	held
b-P3-aa-g25	model sign of aa/base, predicted 0.90 (gpt-oss 25%, 5700X3D)	0.975 [0.968, 0.981]	≥ 1	held
b-P4-b16-g25	b16/base in [1.0, 1.25] (gpt-oss 25%, 5700X3D)	1.072 [1.068, 1.077]	1.0-1.25	held
b-P4-b4-g25	b4/base in [0.98, 1.12] (gpt-oss 25%, 5700X3D)	1.026 [1.023, 1.028]	0.98-1.12	held
b-P4-b8r5-g25	b8r5/base in [0.95, 1.08] (gpt-oss 25%, 5700X3D)	1.000 [0.997, 1.004]	0.95-1.08	held
b-P4-b16bypass-g25	b16/base $\leq$ bypass/base + 0.03 (gpt-oss 25%, 5700X3D)	-0.092	≤ 0.03	held (point)
b-P4-mono-g25	b16 at least as fast as b4 (gpt-oss 25%, 5700X3D)	1.045 [1.042, 1.048]	≥ 1	held
b-P7-w4-g25	host plan of w4 at most 150 μ s per step (gpt-oss 25%, 5700X3D)	15.6	$\leq 150 \mu$ s	held (point)
b-P7-w16-g25	host plan of w16 at most 150 μ s per step (gpt-oss 25%, 5700X3D)	23.2	$\leq 150 \mu$ s	held (point)
b-P7-b4-g25	host plan of b4 at most 150 μ s per step (gpt-oss 25%, 5700X3D)	–	$\leq 150 \mu$ s	untested
b-P7-b16-g25	host plan of b16 at most 150 μ s per step (gpt-oss 25%, 5700X3D)	–	$\leq 150 \mu$ s	untested
b-P7-b8r5-g25	host plan of b8r5 at most 150 μ s per step (gpt-oss 25%, 5700X3D)	–	$\leq 150 \mu$ s	untested
b-P7-fetch-g25	host plan of fetch at most 150 μ s per step (gpt-oss 25%, 5700X3D)	81.9	$\leq 150 \mu$ s	held (point)
b-P8-b4-g25	engine misses of b4 within 10% of the replay (gpt-oss 25%, 5700X3D)	0.108	$\leq 10\%$	failed
b-P8-b16-g25	engine misses of b16 within 10% of the replay (gpt-oss 25%, 5700X3D)	0.256	$\leq 10\%$	failed
b-P8-b8r5-g25	engine misses of b8r5 within 10% of the replay (gpt-oss 25%, 5700X3D)	0.074	$\leq 10\%$	held (point)
c-P1-aa-g11	aa counters equal job 099's (gpt-oss 11%, Ph again)	0.000	$\leq 0.5\%$	held (point)
c-P1-w4-g11	w4 counters equal job 099's (gpt-oss 11%, Ph again)	0.000	$\leq 0.5\%$	held (point)
c-P1-w16-g11	w16 counters equal job 099's (gpt-oss 11%, Ph again)	0.000	$\leq 0.5\%$	held (point)
c-P1-fetch-g11	fetch counters equal job 099's (gpt-oss 11%, Ph again)	0.001	$\leq 0.5\%$	held (point)
c-P2-aa-g11	probe-only aa time within 12% (gpt-oss 11%, Ph again)	0.003	$\leq 12\%$	held (point)
c-P2-w4-g11	probe-only w4 time within 12% (gpt-oss 11%, Ph again)	0.009	$\leq 12\%$	held (point)
c-P3-fetch-g11	model sign of fetch/base, predicted 1.33 (gpt-oss 11%, Ph again)	1.270 [1.262, 1.278]	≥ 1	held
c-P3-aa-g11	model sign of aa/base, predicted 0.85 (gpt-oss 11%, Ph again)	0.855 [0.852, 0.858]	≥ 1	held
c-P4-b16-g11	b16/base in [1.0, 1.25] (gpt-oss 11%, Ph again)	1.035 [1.030, 1.040]	1.0-1.25	held
c-P4-b4-g11	b4/base in [0.98, 1.12] (gpt-oss 11%, Ph again)	1.011 [1.008, 1.013]	0.98-1.12	held
c-P4-b8r5-g11	b8r5/base in [0.95, 1.08] (gpt-oss 11%, Ph again)	0.980 [0.978, 0.983]	0.95-1.08	held
c-P4-b16bypass-g11	b16/base $\leq$ bypass/base + 0.03 (gpt-oss 11%, Ph again)	0.007	≤ 0.03	held (point)
c-P4-mono-g11	b16 at least as fast as b4 (gpt-oss 11%, Ph again)	1.024 [1.019, 1.028]	≥ 1	held
c-P5-g11	w16/base > b16/base at link/CPU 0.96 (gpt-oss 11%, Ph again)	0.170	> 0	held (point)
c-P7-w4-g11	host plan of w4 at most 150 μ s per step (gpt-oss 11%, Ph again)	11.0	$\leq 150 \mu$ s	held (point)
c-P7-w16-g11	host plan of w16 at most 150 μ s per step (gpt-oss 11%, Ph again)	19.6	$\leq 150 \mu$ s	held (point)
c-P7-b4-g11	host plan of b4 at most 150 μ s per step (gpt-oss 11%, Ph again)	–	$\leq 150 \mu$ s	untested
c-P7-b16-g11	host plan of b16 at most 150 μ s per step (gpt-oss 11%, Ph again)	–	$\leq 150 \mu$ s	untested
c-P7-b8r5-g11	host plan of b8r5 at most 150 μ s per step (gpt-oss 11%, Ph again)	–	$\leq 150 \mu$ s	untested
c-P7-fetch-g11	host plan of fetch at most 150 μ s per step (gpt-oss 11%, Ph again)	63.0	$\leq 150 \mu$ s	held (point)
c-P8-b4-g11	engine misses of b4 within 10% of the replay (gpt-oss 11%, Ph again)	0.195	$\leq 10\%$	failed

continued

Clause	Statement	Measured	Predicted	Status
c-P8-b16-g11	engine misses of b16 within 10% of the replay (gpt-oss 11%, Ph again)	0.330	$\leq 10\%$	failed
c-P8-b8r5-g11	engine misses of b8r5 within 10% of the replay (gpt-oss 11%, Ph again)	0.123	$\leq 10\%$	failed
c-P1-aa-g25	aa counters equal job 099's (gpt-oss 25%, Ph again)	0.000	$\leq 0.5\%$	held (point)
c-P1-w4-g25	w4 counters equal job 099's (gpt-oss 25%, Ph again)	0.000	$\leq 0.5\%$	held (point)
c-P1-w16-g25	w16 counters equal job 099's (gpt-oss 25%, Ph again)	0.000	$\leq 0.5\%$	held (point)
c-P1-fetch-g25	fetch counters equal job 099's (gpt-oss 25%, Ph again)	0.001	$\leq 0.5\%$	held (point)
c-P2-aa-g25	probe-only aa time within 12% (gpt-oss 25%, Ph again)	0.006	$\leq 12\%$	held (point)
c-P2-w4-g25	probe-only w4 time within 12% (gpt-oss 25%, Ph again)	0.001	$\leq 12\%$	held (point)
c-P3-fetch-g25	model sign of fetch/base, predicted 1.35 (gpt-oss 25%, Ph again)	1.287 [1.272, 1.301]	≥ 1	held
c-P3-aa-g25	model sign of aa/base, predicted 0.89 (gpt-oss 25%, Ph again)	0.891 [0.887, 0.896]	≥ 1	held
c-P4-b16-g25	b16/base in [1.0, 1.25] (gpt-oss 25%, Ph again)	1.072 [1.069, 1.075]	1.0-1.25	held
c-P4-b4-g25	b4/base in [0.98, 1.12] (gpt-oss 25%, Ph again)	1.026 [1.020, 1.030]	0.98-1.12	held
c-P4-b8r5-g25	b8r5/base in [0.95, 1.08] (gpt-oss 25%, Ph again)	1.010 [1.006, 1.014]	0.95-1.08	held
c-P4-b16bypass-g25	b16/base $\leq$ bypass/base + 0.03 (gpt-oss 25%, Ph again)	-0.087	≤ 0.03	held (point)
c-P4-mono-g25	b16 at least as fast as b4 (gpt-oss 25%, Ph again)	1.045 [1.041, 1.051]	≥ 1	held
c-P7-w4-g25	host plan of w4 at most 150 μ s per step (gpt-oss 25%, Ph again)	10.6	$\leq 150 \mu$ s	held (point)
c-P7-w16-g25	host plan of w16 at most 150 μ s per step (gpt-oss 25%, Ph again)	16.1	$\leq 150 \mu$ s	held (point)
c-P7-b4-g25	host plan of b4 at most 150 μ s per step (gpt-oss 25%, Ph again)	–	$\leq 150 \mu$ s	untested
c-P7-b16-g25	host plan of b16 at most 150 μ s per step (gpt-oss 25%, Ph again)	–	$\leq 150 \mu$ s	untested
c-P7-b8r5-g25	host plan of b8r5 at most 150 μ s per step (gpt-oss 25%, Ph again)	–	$\leq 150 \mu$ s	untested
c-P7-fetch-g25	host plan of fetch at most 150 μ s per step (gpt-oss 25%, Ph again)	56.8	$\leq 150 \mu$ s	held (point)
c-P8-b4-g25	engine misses of b4 within 10% of the replay (gpt-oss 25%, Ph again)	0.109	$\leq 10\%$	failed
c-P8-b16-g25	engine misses of b16 within 10% of the replay (gpt-oss 25%, Ph again)	0.256	$\leq 10\%$	failed
c-P8-b8r5-g25	engine misses of b8r5 within 10% of the replay (gpt-oss 25%, Ph again)	0.074	$\leq 10\%$	held (point)
c-P6-base-g11	base time within 3% of job 099h (gpt-oss 11%, Ph again)	0.019	$\leq 3\%$	held (point)
c-P6-foa-g11	foa/base within 0.03 of job 099h (gpt-oss 11%, Ph again)	0.001	≤ 0.03	held (point)
c-P6-aa-g11	aa/base within 0.03 of job 099h (gpt-oss 11%, Ph again)	0.017	≤ 0.03	held (point)
c-P6-fetch-g11	fetch/base within 0.03 of job 099h (gpt-oss 11%, Ph again)	0.024	≤ 0.03	held (point)
c-P6-both3p-g11	both3p/base within 0.03 of job 099h (gpt-oss 11%, Ph again)	0.030	≤ 0.03	held (point)
c-P6-w4-g11	w4/base within 0.03 of job 099h (gpt-oss 11%, Ph again)	0.019	≤ 0.03	held (point)
c-P6-w16-g11	w16/base within 0.03 of job 099h (gpt-oss 11%, Ph again)	0.024	≤ 0.03	held (point)
c-P6-base-g25	base time within 3% of job 099h (gpt-oss 25%, Ph again)	0.003	$\leq 3\%$	held (point)
c-P6-foa-g25	foa/base within 0.03 of job 099h (gpt-oss 25%, Ph again)	0.007	≤ 0.03	held (point)
c-P6-aa-g25	aa/base within 0.03 of job 099h (gpt-oss 25%, Ph again)	0.003	≤ 0.03	held (point)
c-P6-fetch-g25	fetch/base within 0.03 of job 099h (gpt-oss 25%, Ph again)	0.001	≤ 0.03	held (point)
c-P6-both3p-g25	both3p/base within 0.03 of job 099h (gpt-oss 25%, Ph again)	0.009	≤ 0.03	held (point)
c-P6-w4-g25	w4/base within 0.03 of job 099h (gpt-oss 25%, Ph again)	0.005	≤ 0.03	held (point)
c-P6-w16-g25	w16/base within 0.03 of job 099h (gpt-oss 25%, Ph again)	0.005	≤ 0.03	held (point)
f-P1-aa-g11	aa counters equal job 099's (gpt-oss 11%, 9950X, slower link)	0.000	$\leq 0.5\%$	held (point)
f-P1-w4-g11	w4 counters equal job 099's (gpt-oss 11%, 9950X, slower link)	0.000	$\leq 0.5\%$	held (point)
f-P1-w16-g11	w16 counters equal job 099's (gpt-oss 11%, 9950X, slower link)	0.000	$\leq 0.5\%$	held (point)
f-P1-fetch-g11	fetch counters equal job 099's (gpt-oss 11%, 9950X, slower link)	0.001	$\leq 0.5\%$	held (point)
f-P2-aa-g11	probe-only aa time within 12% (gpt-oss 11%, 9950X, slower link)	0.201	$\leq 12\%$	failed

continued

Clause	Statement	Measured	Predicted	Status
f-P2-w4-g11	probe-only w4 time within 12% (gpt-oss 11%, 9950X, slower link)	0.191	$\leq 12\%$	failed
f-P2-fetchsign-g11	probe-only fetch time under-predicted (gpt-oss 11%, 9950X, slower link)	-0.020	< 0	held (point)
f-P3-fetch-g11	model sign of fetch/base, predicted 1.25 (gpt-oss 11%, 9950X, slower link)	1.067 [1.054, 1.081]	$\downarrow 1$	held
f-P3-aa-g11	model sign of aa/base, predicted 0.52 (gpt-oss 11%, 9950X, slower link)	0.542 [0.536, 0.549]	$\downarrow 1$	held
f-P4-b16-g11	b16/base in [1.0, 1.25] (gpt-oss 11%, 9950X, slower link)	1.008 [0.996, 1.022]	1.0-1.25	held (point)
f-P4-b4-g11	b4/base in [0.98, 1.12] (gpt-oss 11%, 9950X, slower link)	1.009 [0.998, 1.022]	0.98-1.12	held
f-P4-b8r5-g11	b8r5/base in [0.95, 1.08] (gpt-oss 11%, 9950X, slower link)	0.960 [0.950, 0.973]	0.95-1.08	held (point)
f-P4-b16bypass-g11	b16/base $\leq$ bypass/base + 0.03 (gpt-oss 11%, 9950X, slower link)	0.022	≤ 0.03	held (point)
f-P4-mono-g11	b16 at least as fast as b4 (gpt-oss 11%, 9950X, slower link)	0.999 [0.995, 1.004]	≥ 1	failed
f-P7-w4-g11	host plan of w4 at most 150 μ s per step (gpt-oss 11%, 9950X, slower link)	10.1	$\leq 150 \mu$ s	held (point)
f-P7-w16-g11	host plan of w16 at most 150 μ s per step (gpt-oss 11%, 9950X, slower link)	15.9	$\leq 150 \mu$ s	held (point)
f-P7-b4-g11	host plan of b4 at most 150 μ s per step (gpt-oss 11%, 9950X, slower link)	–	$\leq 150 \mu$ s	untested
f-P7-b16-g11	host plan of b16 at most 150 μ s per step (gpt-oss 11%, 9950X, slower link)	–	$\leq 150 \mu$ s	untested
f-P7-b8r5-g11	host plan of b8r5 at most 150 μ s per step (gpt-oss 11%, 9950X, slower link)	–	$\leq 150 \mu$ s	untested
f-P7-fetch-g11	host plan of fetch at most 150 μ s per step (gpt-oss 11%, 9950X, slower link)	49.0	$\leq 150 \mu$ s	held (point)
f-P8-b4-g11	engine misses of b4 within 10% of the replay (gpt-oss 11%, 9950X, slower link)	0.171	$\leq 10\%$	failed
f-P8-b16-g11	engine misses of b16 within 10% of the replay (gpt-oss 11%, 9950X, slower link)	0.259	$\leq 10\%$	failed
f-P8-b8r5-g11	engine misses of b8r5 within 10% of the replay (gpt-oss 11%, 9950X, slower link)	0.106	$\leq 10\%$	failed
f-P1-aa-g25	aa counters equal job 099's (gpt-oss 25%, 9950X, slower link)	0.000	$\leq 0.5\%$	held (point)
f-P1-w4-g25	w4 counters equal job 099's (gpt-oss 25%, 9950X, slower link)	0.000	$\leq 0.5\%$	held (point)
f-P1-w16-g25	w16 counters equal job 099's (gpt-oss 25%, 9950X, slower link)	0.000	$\leq 0.5\%$	held (point)
f-P1-fetch-g25	fetch counters equal job 099's (gpt-oss 25%, 9950X, slower link)	0.001	$\leq 0.5\%$	held (point)
f-P2-aa-g25	probe-only aa time within 12% (gpt-oss 25%, 9950X, slower link)	0.178	$\leq 12\%$	failed
f-P2-w4-g25	probe-only w4 time within 12% (gpt-oss 25%, 9950X, slower link)	0.167	$\leq 12\%$	failed
f-P2-fetchsign-g25	probe-only fetch time under-predicted (gpt-oss 25%, 9950X, slower link)	0.019	< 0	failed
f-P3-fetch-g25	model sign of fetch/base, predicted 1.20 (gpt-oss 25%, 9950X, slower link)	1.111 [1.103, 1.118]	$\downarrow 1$	held
f-P3-aa-g25	model sign of aa/base, predicted 0.58 (gpt-oss 25%, 9950X, slower link)	0.620 [0.614, 0.626]	$\downarrow 1$	held
f-P4-b16-g25	b16/base in [1.0, 1.25] (gpt-oss 25%, 9950X, slower link)	1.064 [1.049, 1.073]	1.0-1.25	held
f-P4-b4-g25	b4/base in [0.98, 1.12] (gpt-oss 25%, 9950X, slower link)	0.960 [0.948, 0.971]	0.98-1.12	failed
f-P4-b8r5-g25	b8r5/base in [0.95, 1.08] (gpt-oss 25%, 9950X, slower link)	1.003 [0.999, 1.007]	0.95-1.08	held
f-P4-b16bypass-g25	b16/base $\leq$ bypass/base + 0.03 (gpt-oss 25%, 9950X, slower link)	-0.032	≤ 0.03	held (point)
f-P4-mono-g25	b16 at least as fast as b4 (gpt-oss 25%, 9950X, slower link)	1.108 [1.092, 1.124]	≥ 1	held
f-P7-w4-g25	host plan of w4 at most 150 μ s per step (gpt-oss 25%, 9950X, slower link)	10.3	$\leq 150 \mu$ s	held (point)
f-P7-w16-g25	host plan of w16 at most 150 μ s per step (gpt-oss 25%, 9950X, slower link)	15.2	$\leq 150 \mu$ s	held (point)
f-P7-b4-g25	host plan of b4 at most 150 μ s per step (gpt-oss 25%, 9950X, slower link)	–	$\leq 150 \mu$ s	untested
f-P7-b16-g25	host plan of b16 at most 150 μ s per step (gpt-oss 25%, 9950X, slower link)	–	$\leq 150 \mu$ s	untested

continued

Clause	Statement	Measured	Predicted	Status
f-P7-b8r5-g25	host plan of b8r5 at most 150 μ s per step (gpt-oss 25%, 9950X, slower link)	–	$\leq 150 \mu$ s	untested
f-P7-fetch-g25	host plan of fetch at most 150 μ s per step (gpt-oss 25%, 9950X, slower link)	46.4	$\leq 150 \mu$ s	held (point)
f-P8-b4-g25	engine misses of b4 within 10% of the replay (gpt-oss 25%, 9950X, slower link)	0.105	$\leq 10\%$	failed
f-P8-b16-g25	engine misses of b16 within 10% of the replay (gpt-oss 25%, 9950X, slower link)	0.228	$\leq 10\%$	failed
f-P8-b8r5-g25	engine misses of b8r5 within 10% of the replay (gpt-oss 25%, 9950X, slower link)	0.071	$\leq 10\%$	held (point)
e-P9-099d	fetch/base at 11%: 13900KF, faster link above 099d	–	> 0	untested
e-P9-100a	fetch/base at 11%: 13900KF, faster link above 100a	–	> 0	untested
f-P9-099e	fetch/base at 11%: 9950X, slower link below 099e	0.281	> 0	held (point)
f-P9-099j	fetch/base at 11%: 9950X, slower link below 099j	0.198	> 0	held (point)
P2-median	median —error— of the probe-only predictions over all new host-cells	0.038	$\leq 5\%$	held (point)

Table S7: Job 101 (the crossover machine relaunched, and a second 9800X3D), scored by machine by `scripts/job101.py` under the same rule. Totals: 25 clauses, 6 held, 19 held (point), 0 failed, 0 untested. Hosts a (Pf again), b (9800X3D, slower link); cells gpt-oss 11% (g11) and 25% (g25). Measured: the point estimate and, where one exists, the 95% interval (bootstrap over problems within the host; pooled clauses have none).

Clause	Statement	Measured	Predicted	Status
a-P1-loss-g11	fetch/base below 1 at 11% (Pf again)	0.860 [0.855, 0.866]	< 1	held
a-P1-fetch-g11	fetch/base within 0.03 of job 099f (g11)	0.015	≤ 0.03	held (point)
a-P1-foa-g11	foa/base within 0.03 of job 099f (g11)	0.026	≤ 0.03	held (point)
a-P1-aa-g11	aa/base within 0.03 of job 099f (g11)	0.007	≤ 0.03	held (point)
a-P1-both3p-g11	both3p/base within 0.03 of job 099f (g11)	0.011	≤ 0.03	held (point)
a-P1-bypass-g11	bypass/base within 0.03 of job 099f (g11)	0.016	≤ 0.03	held (point)
a-P1-base-g11	base time within 3% of job 099f (g11)	0.017	$\leq 3\%$	held (point)
a-P1-fetch-g25	fetch/base within 0.03 of job 099f (g25)	0.003	≤ 0.03	held (point)
a-P1-foa-g25	foa/base within 0.03 of job 099f (g25)	0.003	≤ 0.03	held (point)
a-P1-aa-g25	aa/base within 0.03 of job 099f (g25)	0.000	≤ 0.03	held (point)
a-P1-both3p-g25	both3p/base within 0.03 of job 099f (g25)	0.007	≤ 0.03	held (point)
a-P1-bypass-g25	bypass/base within 0.03 of job 099f (g25)	0.002	≤ 0.03	held (point)
a-P1-base-g25	base time within 3% of job 099f (g25)	0.000	$\leq 3\%$	held (point)
b-P2-belowPb	fetch/base at 11% below Pb's (job 099b)	0.206	> 0	held (point)
b-P2-aboveOne	fetch/base at 11% above 1	1.137 [1.132, 1.142]	> 1	held
a-P3-sign-g11	model sign of fetch/base, predicted 0.92 (Pf again, g11)	0.860 [0.855, 0.866]	$\downarrow 1$	held
a-P3-aa-g11	probe-only aa within 12% (Pf again, g11)	0.016	$\leq 12\%$	held (point)
a-P3-fetchunder-g11	probe-only fetch under-predicted (Pf again, g11)	-0.102	< 0	held (point)
a-P3-sign-g25	model sign of fetch/base, predicted 0.95 (Pf again, g25)	0.966 [0.958, 0.974]	$\downarrow 1$	held
a-P3-aa-g25	probe-only aa within 12% (Pf again, g25)	0.022	$\leq 12\%$	held (point)
a-P3-fetchunder-g25	probe-only fetch under-predicted (Pf again, g25)	-0.045	< 0	held (point)
b-P3-sign-g11	model sign of fetch/base, predicted 1.36 (9800X3D, slow, g11)	1.137 [1.132, 1.142]	$\downarrow 1$	held
b-P3-aa-g11	probe-only aa within 12% (9800X3D, slow, g11)	0.001	$\leq 12\%$	held (point)
b-P3-sign-g25	model sign of fetch/base, predicted 1.39 (9800X3D, slow, g25)	1.223 [1.212, 1.234]	$\downarrow 1$	held
b-P3-aa-g25	probe-only aa within 12% (9800X3D, slow, g25)	0.005	$\leq 12\%$	held (point)

Table S8: Job 102 (can the bound’s host term be reached), scored by machine by `scripts/job102.py` under the same rule. Totals: 33 clauses, 0 held, 28 held (point), 5 failed, 0 untested. Hosts a (Pd again), b (9950X), c (9800X3D, link 0.55 of its CPU rate); cells gpt-oss $C=14$ (11%) and $C=32$ (25%). Clause P2below’s measured value is the analytic per-layer fraction minus the measured one. Measured: the point estimate and, where one exists, the 95% interval (bootstrap over problems within the host; pooled clauses have none).

Clause	Statement	Measured	Predicted	Status
a-P1-C14	token mode reaches ≥ 0.80 of the bound’s host term (Pd again, $C = 14$)	0.933	≥ 0.80	held (point)
a-P2floor-C14	layer mode reaches ≥ 0.50 of it (Pd again, $C = 14$)	0.861	≥ 0.50	held (point)
a-P2below-C14	layer mode at least 0.03 below the analytic per-layer fraction 0.98 (Pd again, $C = 14$)	0.123	≥ 0.03	held (point)
a-P4-C14	layer.link $> 10\%$ slower than layer (CPU path 2.49x the link; Pd again, $C = 14$)	1.429	> 0.10	held (point)
a-P5-C14	two repetitions agree within 3%, worst mode (Pd again, $C = 14$)	0.009	≤ 0.03	held (point)
a-P1-C32	token mode reaches ≥ 0.80 of the bound’s host term (Pd again, $C = 32$)	0.929	≥ 0.80	held (point)
a-P2floor-C32	layer mode reaches ≥ 0.50 of it (Pd again, $C = 32$)	0.889	≥ 0.50	held (point)
a-P2below-C32	layer mode at least 0.03 below the analytic per-layer fraction 0.98 (Pd again, $C = 32$)	0.091	≥ 0.03	held (point)
a-P4-C32	layer.link $> 10\%$ slower than layer (CPU path 2.49x the link; Pd again, $C = 32$)	1.507	> 0.10	held (point)
a-P5-C32	two repetitions agree within 3%, worst mode (Pd again, $C = 32$)	0.001	≤ 0.03	held (point)
a-P3	layer mode’s fraction lower at $C = 32$ than at 14 (Pd again)	-0.029	> 0	failed
b-P1-C14	token mode reaches ≥ 0.80 of the bound’s host term (9950X, $C = 14$)	0.974	≥ 0.80	held (point)
b-P2floor-C14	layer mode reaches ≥ 0.50 of it (9950X, $C = 14$)	0.963	≥ 0.50	held (point)
b-P2below-C14	layer mode at least 0.03 below the analytic per-layer fraction 0.94 (9950X, $C = 14$)	-0.022	≥ 0.03	failed
b-P4-C14	layer.link within 10% of layer (paths match, link/CPU 1.02; 9950X, $C = 14$)	0.018	≤ 0.10	held (point)
b-P5-C14	two repetitions agree within 3%, worst mode (9950X, $C = 14$)	0.001	≤ 0.03	held (point)
b-P1-C32	token mode reaches ≥ 0.80 of the bound’s host term (9950X, $C = 32$)	0.966	≥ 0.80	held (point)
b-P2floor-C32	layer mode reaches ≥ 0.50 of it (9950X, $C = 32$)	0.955	≥ 0.50	held (point)
b-P2below-C32	layer mode at least 0.03 below the analytic per-layer fraction 0.92 (9950X, $C = 32$)	-0.035	≥ 0.03	failed
b-P4-C32	layer.link within 10% of layer (paths match, link/CPU 1.02; 9950X, $C = 32$)	0.014	≤ 0.10	held (point)
b-P5-C32	two repetitions agree within 3%, worst mode (9950X, $C = 32$)	0.001	≤ 0.03	held (point)
b-P3	layer mode’s fraction lower at $C = 32$ than at 14 (9950X)	0.008	> 0	held (point)
c-P1-C14	token mode reaches ≥ 0.80 of the bound’s host term (9800X3D, $C = 14$)	0.955	≥ 0.80	held (point)
c-P2floor-C14	layer mode reaches ≥ 0.50 of it (9800X3D, $C = 14$)	0.950	≥ 0.50	held (point)
c-P2below-C14	layer mode at least 0.03 below the analytic per-layer fraction 0.95 (9800X3D, $C = 14$)	0.001	≥ 0.03	failed
c-P4-C14	layer.link $> 10\%$ slower than layer (CPU path 1.82x the link; 9800X3D, $C = 14$)	0.788	> 0.10	held (point)
c-P5-C14	two repetitions agree within 3%, worst mode (9800X3D, $C = 14$)	0.001	≤ 0.03	held (point)
c-P1-C32	token mode reaches ≥ 0.80 of the bound’s host term (9800X3D, $C = 32$)	0.939	≥ 0.80	held (point)
c-P2floor-C32	layer mode reaches ≥ 0.50 of it (9800X3D, $C = 32$)	0.933	≥ 0.50	held (point)
c-P2below-C32	layer mode at least 0.03 below the analytic per-layer fraction 0.94 (9800X3D, $C = 32$)	0.002	≥ 0.03	failed
c-P4-C32	layer.link $> 10\%$ slower than layer (CPU path 1.82x the link; 9800X3D, $C = 32$)	0.763	> 0.10	held (point)
c-P5-C32	two repetitions agree within 3%, worst mode (9800X3D, $C = 32$)	0.000	≤ 0.03	held (point)
c-P3	layer mode’s fraction lower at $C = 32$ than at 14 (9800X3D)	0.017	> 0	held (point)

Table S9: Job 103 (the fewest-admission MIN set, first attempt), scored by machine by `scripts/job103.py` under the same rule. Totals: 45 clauses, 0 held, 4 held (point), 0 failed, 41 untested. Hosts a (Pf again), b (285K, 40 GB/s link), d (5950X), e (3970X); c never started. The plan could not be computed (scipy did not install), so the engine ran greedy MIN in the plan configurations: every plan clause is untested. Measured: the point estimate and, where one exists, the 95% interval (bootstrap over problems within the host; pooled clauses have none).

Clause	Statement	Measured	Predicted	Status
a-P1-copies-g11	P1-copies (Pf again, g11): no plan on this host	–		untested
a-P1-misses-g11	P1-misses (Pf again, g11): no plan on this host	–		untested
a-P2-reads-g11	P2-reads (Pf again, g11): no plan on this host	–		untested
a-P6-g11	P6 (Pf again, g11): no plan on this host	–		untested
a-P3-g11	P3 (Pf again, g11): no plan on this host	–		untested
a-P4-g11	P4 (Pf again, g11): no plan on this host	–		untested
a-P7-g11	P7 (Pf again, g11): no plan on this host	–		untested
a-P1-copies-g25	P1-copies (Pf again, g25): no plan on this host	–		untested
a-P1-misses-g25	P1-misses (Pf again, g25): no plan on this host	–		untested
a-P2-reads-g25	P2-reads (Pf again, g25): no plan on this host	–		untested
a-P6-g25	P6 (Pf again, g25): no plan on this host	–		untested
a-P8-base-g11	base time within 3% of job 101a (g11)	0.002	≤ 0.03	held (point)
a-P8-fetch-g11	fetch/base within 0.03 of job 101a (g11)	0.001	≤ 0.03	held (point)
a-P8-base-g25	base time within 3% of job 101a (g25)	0.003	≤ 0.03	held (point)
a-P8-fetch-g25	fetch/base within 0.03 of job 101a (g25)	0.005	≤ 0.03	held (point)
b-P1-copies-g11	P1-copies (285K, second, g11): no plan on this host	–		untested
b-P1-misses-g11	P1-misses (285K, second, g11): no plan on this host	–		untested
b-P2-reads-g11	P2-reads (285K, second, g11): no plan on this host	–		untested
b-P6-g11	P6 (285K, second, g11): no plan on this host	–		untested
b-P4-g11	P4 (285K, second, g11): no plan on this host	–		untested
b-P7-g11	P7 (285K, second, g11): no plan on this host	–		untested
b-P1-copies-g25	P1-copies (285K, second, g25): no plan on this host	–		untested
b-P1-misses-g25	P1-misses (285K, second, g25): no plan on this host	–		untested
b-P2-reads-g25	P2-reads (285K, second, g25): no plan on this host	–		untested
b-P6-g25	P6 (285K, second, g25): no plan on this host	–		untested
d-P1-copies-g11	P1-copies (Pg again (5950X), g11): no plan on this host	–		untested
d-P1-misses-g11	P1-misses (Pg again (5950X), g11): no plan on this host	–		untested
d-P2-reads-g11	P2-reads (Pg again (5950X), g11): no plan on this host	–		untested
d-P6-g11	P6 (Pg again (5950X), g11): no plan on this host	–		untested
d-P4-g11	P4 (Pg again (5950X), g11): no plan on this host	–		untested
d-P7-g11	P7 (Pg again (5950X), g11): no plan on this host	–		untested
d-P1-copies-g25	P1-copies (Pg again (5950X), g25): no plan on this host	–		untested
d-P1-misses-g25	P1-misses (Pg again (5950X), g25): no plan on this host	–		untested
d-P2-reads-g25	P2-reads (Pg again (5950X), g25): no plan on this host	–		untested
d-P6-g25	P6 (Pg again (5950X), g25): no plan on this host	–		untested
e-P1-copies-g11	P1-copies (3970X, g11): no plan on this host	–		untested
e-P1-misses-g11	P1-misses (3970X, g11): no plan on this host	–		untested
e-P2-reads-g11	P2-reads (3970X, g11): no plan on this host	–		untested
e-P6-g11	P6 (3970X, g11): no plan on this host	–		untested
e-P4-g11	P4 (3970X, g11): no plan on this host	–		untested
e-P7-g11	P7 (3970X, g11): no plan on this host	–		untested
e-P1-copies-g25	P1-copies (3970X, g25): no plan on this host	–		untested
e-P1-misses-g25	P1-misses (3970X, g25): no plan on this host	–		untested
e-P2-reads-g25	P2-reads (3970X, g25): no plan on this host	–		untested
e-P6-g25	P6 (3970X, g25): no plan on this host	–		untested

Table S10: Job 104 (the fewest-admission MIN set), scored by machine by `scripts/job103.py` under the same rule. Totals: 47 clauses, 1 held, 22 held (point), 24 failed, 0 untested. Hosts a (Pf again), b (the 285K of 103b), c (the 5950X of 103d); cells gpt-oss 11% (g11) and 25% (g25). Measured: the point estimate and, where one exists, the 95% interval (bootstrap over problems within the host; pooled clauses have none).

Clause	Statement	Measured	Predicted	Status
a-P1-copies-g11	fetchplan's in-step copies $\leq 0.70x$ fetch's (Pf again, g11)	0.597	≤ 0.70	held (point)
a-P1-misses-g11	fetchplan's misses within 3% of fetch's (Pf again, g11)	0.025	≤ 0.03	held (point)
a-P1-replay-fetch-g11	fetch: engine misses within 2% of the host's replay (Pf again, g11)	0.016	≤ 0.02	held (point)
a-P1-replay-fetchplan-g11	fetchplan: engine misses within 2% of the host's replay (Pf again, g11)	0.038	≤ 0.02	failed
a-P2-reads-g11	bypassplan's host reads $\leq 0.90x$ bypass's (Pf again, g11)	0.896	≤ 0.90	held (point)
a-P3-g11	fetchplan/base above 1 (Pf again, g11)	1.035 [1.023, 1.052]	> 1	held
a-P4-g11	fetchplan/base exceeds fetch/base by ≥ 0.03 (Pf again, ratio 0.29, g11)	0.172	≥ 0.03	held (point)
a-P6-g11	bypassplan/base $\geq$ bypass/base - 0.01 (Pf again, g11)	0.172	≥ -0.01	held (point)
a-P7-g11	interaction share smaller with the plan's set (Pf again, g11)	-0.045	> 0	failed
a-P1-copies-g25	fetchplan's in-step copies $\leq 0.70x$ fetch's (Pf again, g25)	0.726	≤ 0.70	failed
a-P1-misses-g25	fetchplan's misses within 3% of fetch's (Pf again, g25)	0.053	≤ 0.03	failed
a-P1-replay-fetch-g25	fetch: engine misses within 2% of the host's replay (Pf again, g25)	0.024	≤ 0.02	failed
a-P1-replay-fetchplan-g25	fetchplan: engine misses within 2% of the host's replay (Pf again, g25)	0.068	≤ 0.02	failed
a-P2-reads-g25	bypassplan's host reads $\leq 0.90x$ bypass's (Pf again, g25)	0.911	≤ 0.90	failed
a-P6-g25	bypassplan/base $\geq$ bypass/base - 0.01 (Pf again, g25)	0.084	≥ -0.01	held (point)
a-P8-base-g11	base time within 3% of job 101a (g11)	0.003	≤ 0.03	held (point)
a-P8-fetch-g11	fetch/base within 0.03 of job 101a (g11)	0.003	≤ 0.03	held (point)
a-P8-base-g25	base time within 3% of job 101a (g25)	0.002	≤ 0.03	held (point)
a-P8-fetch-g25	fetch/base within 0.03 of job 101a (g25)	0.002	≤ 0.03	held (point)
b-P1-copies-g11	fetchplan's in-step copies $\leq 0.70x$ fetch's (285K, second, g11)	0.597	≤ 0.70	held (point)
b-P1-misses-g11	fetchplan's misses within 3% of fetch's (285K, second, g11)	0.025	≤ 0.03	held (point)
b-P1-replay-fetch-g11	fetch: engine misses within 2% of the host's replay (285K, second, g11)	0.016	≤ 0.02	held (point)
b-P1-replay-fetchplan-g11	fetchplan: engine misses within 2% of the host's replay (285K, second, g11)	0.038	≤ 0.02	failed
b-P2-reads-g11	bypassplan's host reads $\leq 0.90x$ bypass's (285K, second, g11)	0.940	≤ 0.90	failed
b-P4-g11	fetchplan/base exceeds fetch/base by ≥ 0.03 (285K, second, ratio 0.49, g11)	0.127	≥ 0.03	held (point)
b-P6-g11	bypassplan/base $\geq$ bypass/base - 0.01 (285K, second, g11)	0.053	≥ -0.01	held (point)
b-P7-g11	interaction share smaller with the plan's set (285K, second, g11)	-0.097	> 0	failed
b-P1-copies-g25	fetchplan's in-step copies $\leq 0.70x$ fetch's (285K, second, g25)	0.726	≤ 0.70	failed
b-P1-misses-g25	fetchplan's misses within 3% of fetch's (285K, second, g25)	0.053	≤ 0.03	failed
b-P1-replay-fetch-g25	fetch: engine misses within 2% of the host's replay (285K, second, g25)	0.024	≤ 0.02	failed
b-P1-replay-fetchplan-g25	fetchplan: engine misses within 2% of the host's replay (285K, second, g25)	0.068	≤ 0.02	failed
b-P2-reads-g25	bypassplan's host reads $\leq 0.90x$ bypass's (285K, second, g25)	0.955	≤ 0.90	failed
b-P6-g25	bypassplan/base $\geq$ bypass/base - 0.01 (285K, second, g25)	0.009	≥ -0.01	held (point)
c-P1-copies-g11	fetchplan's in-step copies $\leq 0.70x$ fetch's (Pg again (5950X), g11)	0.597	≤ 0.70	held (point)
c-P1-misses-g11	fetchplan's misses within 3% of fetch's (Pg again (5950X), g11)	0.025	≤ 0.03	held (point)
c-P1-replay-fetch-g11	fetch: engine misses within 2% of the host's replay (Pg again (5950X), g11)	0.016	≤ 0.02	held (point)
c-P1-replay-fetchplan-g11	fetchplan: engine misses within 2% of the host's replay (Pg again (5950X), g11)	0.038	≤ 0.02	failed

continued

Clause	Statement	Measured	Predicted	Status
c-P2-reads-g11	bypassplan's host reads $\leq 0.90x$ bypass's (Pg again (5950X), g11)	0.940	≤ 0.90	failed
c-P4-g11	fetchplan/base exceeds fetch/base by ≥ 0.03 (Pg again (5950X), ratio 0.63, g11)	0.082	≥ 0.03	held (point)
c-P6-g11	bypassplan/base $\geq$ bypass/base - 0.01 (Pg again (5950X), g11)	0.038	≥ -0.01	held (point)
c-P7-g11	interaction share smaller with the plan's set (Pg again (5950X), g11)	-0.027	> 0	failed
c-P1-copies-g25	fetchplan's in-step copies $\leq 0.70x$ fetch's (Pg again (5950X), g25)	0.726	≤ 0.70	failed
c-P1-misses-g25	fetchplan's misses within 3% of fetch's (Pg again (5950X), g25)	0.053	≤ 0.03	failed
c-P1-replay-fetch-g25	fetch: engine misses within 2% of the host's replay (Pg again (5950X), g25)	0.024	≤ 0.02	failed
c-P1-replay-fetchplan-g25	fetchplan: engine misses within 2% of the host's replay (Pg again (5950X), g25)	0.068	≤ 0.02	failed
c-P2-reads-g25	bypassplan's host reads $\leq 0.90x$ bypass's (Pg again (5950X), g25)	1.007	≤ 0.90	failed
c-P6-g25	bypassplan/base $\geq$ bypass/base - 0.01 (Pg again (5950X), g25)	-0.047	≥ -0.01	failed

Table S11: Job 105 (the sum law with profiled GPU compute; layer-ahead copies across machines), scored by machine by `scripts/job105.py` under the same rule. Totals: 31 clauses, 7 held, 20 held (point), 4 failed, 0 untested. Hosts a (EPYC 7402P), b (Threadripper 9960X), e (Pf again), f (O4 again); c never started, and d and g had throttled cards that the gate stopped. Cells gpt-oss 11% (g11) and 25% (g25). Measured: the point estimate and, where one exists, the 95% interval (bootstrap over problems within the host; pooled clauses have none).

Clause	Statement	Measured	Predicted	Status
a-P1-g11	profiled GPU compute 4.0-5.0 ms (EPYC 7402P, g11)	4.301	4.0 to 5.0	held (point)
a-P2-g11	deployed time within 6% of G_prof + R S / B_host (EPYC 7402P, g11)	0.047	≤ 0.06	held (point)
a-P4-g11	fewest-admission set copied in the step beats the deployed cache (EPYC 7402P, g11)	1.149 [1.141, 1.156]	> 1	held
a-P4g-g11	fewest-admission set beats the greedy set by ≥ 0.03 (ratio 0.45; EPYC 7402P, g11)	0.140	≥ 0.03	held (point)
a-P5-g11	fewest-admission copies $\leq 0.65x$ the greedy set's (EPYC 7402P, g11)	0.597	≤ 0.65	held (point)
a-P1-g25	profiled GPU compute 4.0-5.0 ms (EPYC 7402P, g25)	4.498	4.0 to 5.0	held (point)
a-P2-g25	deployed time within 6% of G_prof + R S / B_host (EPYC 7402P, g25)	0.093	≤ 0.06	failed
b-P1-g11	profiled GPU compute 4.0-5.0 ms (Threadripper 9960X, g11)	4.143	4.0 to 5.0	held (point)
b-P2-g11	deployed time within 6% of G_prof + R S / B_host (Threadripper 9960X, g11)	0.020	≤ 0.06	held (point)
b-P3-g11	PREFETCH loses (ratio 0.32; Threadripper 9960X, g11)	0.909 [0.903, 0.915]	< 1	held
b-P4-g11	fewest-admission set copied in the step beats the deployed cache (Threadripper 9960X, g11)	1.039 [1.034, 1.043]	> 1	held
b-P4g-g11	fewest-admission set beats the greedy set by ≥ 0.03 (ratio 0.32; Threadripper 9960X, g11)	0.126	≥ 0.03	held (point)
b-P5-g11	fewest-admission copies $\leq 0.65x$ the greedy set's (Threadripper 9960X, g11)	0.599	≤ 0.65	held (point)
b-P1-g25	profiled GPU compute 4.0-5.0 ms (Threadripper 9960X, g25)	4.379	4.0 to 5.0	held (point)
b-P2-g25	deployed time within 6% of G_prof + R S / B_host (Threadripper 9960X, g25)	0.070	≤ 0.06	failed
e-P1-g11	profiled GPU compute 4.0-5.0 ms (Pf again, g11)	4.073	4.0 to 5.0	held (point)
e-P2-g11	deployed time within 6% of G_prof + R S / B_host (Pf again, g11)	0.015	≤ 0.06	held (point)
e-P3-g11	PREFETCH loses (ratio 0.28; Pf again, g11)	0.722 [0.713, 0.731]	< 1	held
e-P4-g11	fewest-admission set copied in the step beats the deployed cache (Pf again, g11)	1.028 [1.015, 1.042]	> 1	held
e-P4g-g11	fewest-admission set beats the greedy set by ≥ 0.03 (ratio 0.28; Pf again, g11)	0.168	≥ 0.03	held (point)
e-P5-g11	fewest-admission copies $\leq 0.65x$ the greedy set's (Pf again, g11)	0.597	≤ 0.65	held (point)
e-P1-g25	profiled GPU compute 4.0-5.0 ms (Pf again, g25)	4.401	4.0 to 5.0	held (point)
e-P2-g25	deployed time within 6% of G_prof + R S / B_host (Pf again, g25)	0.038	≤ 0.06	held (point)
f-P1-g11	profiled GPU compute 4.0-5.0 ms (O4 again (9950X), g11)	4.404	4.0 to 5.0	held (point)
f-P2-g11	deployed time within 6% of G_prof + R S / B_host (O4 again (9950X), g11)	0.014	≤ 0.06	held (point)
f-P3-g11	PREFETCH gains $\geq 3\%$ (ratio 1.04; O4 again (9950X), g11)	1.042 [1.034, 1.049]	≥ 1.03	held
f-P4-g11	fewest-admission set copied in the step beats the deployed cache (O4 again (9950X), g11)	1.341 [1.330, 1.353]	> 1	held
f-P5-g11	fewest-admission copies $\leq 0.65x$ the greedy set's (O4 again (9950X), g11)	0.599	≤ 0.65	held (point)
f-P1-g25	profiled GPU compute 4.0-5.0 ms (O4 again (9950X), g25)	4.440	4.0 to 5.0	held (point)
f-P2-g25	deployed time within 6% of G_prof + R S / B_host (O4 again (9950X), g25)	0.069	≤ 0.06	failed
P2-median	median —error— of the sum law at most 4%	0.042	≤ 0.04	failed

Table S12: Job 106 (the law with its overlap term; online admission rules; slow links launch by launch), scored by machine by `scripts/job106.py` under the same rule. Totals: 181 clauses, 21 held, 112 held (point), 48 failed, 0 untested. Hosts a (the Threadripper 9960X of 105b again), b (Pf again), c (EPYC 7302, a 13 GB/s link), d (Xeon Platinum 8347C), e (Ryzen 9 9950X). Cells gpt-oss 11% (g11, three rounds) and 25% (g25, two), Qwen3 12.5% (q12) and 25% (q25), one round each. On hosts c and d the deployed cache's rounds were 12% and 52% apart (slowest over fastest). Measured: the point estimate and, where one exists, the 95% interval (bootstrap over problems within the host; pooled clauses have none).

Clause	Statement	Measured	Predicted	Status
a-P1-g11	profiled GPU compute 4.0-5.0 ms (Threadripper 9960X again, g11)	4.146	4.0 to 5.0	held (point)
a-P2-g11	overlap law within 6% of base (Threadripper 9960X again, g11)	0.033	≤ 0.06	held (point)
a-P3-reads-g11	dk reads fewer host experts than base (Threadripper 9960X again, g11)	0.947	< 1	held (point)
a-P3-speed-g11	dk/base ≥ 1.01 (Threadripper 9960X again, g11)	1.003 [0.991, 1.017]	≥ 1.01	failed
a-P4-reads-g11	lrn reads at most dk's (Threadripper 9960X again, g11)	0.983	≤ 1	held (point)
a-P4-speed-g11	lrn/base ≥ 1.00 (Threadripper 9960X again, g11)	0.982 [0.968, 0.995]	≥ 1.00	failed
a-P5-dk-g11	law-predicted dk/base within 0.03 (Threadripper 9960X again, g11)	0.006	≤ 0.03	held (point)
a-P5-lrn-g11	law-predicted lrn/base within 0.03 (Threadripper 9960X again, g11)	0.003	≤ 0.03	held (point)
a-P4-host-g11	learned host time $\leq 400 \mu\text{s}$ per step (Threadripper 9960X again, g11)	320.6	≤ 400	held (point)
a-P6-fetch-r1	greedy MIN in the step loses in round 1 (Threadripper 9960X again)	0.931	< 1	held (point)
a-P6-plan-r1	fewest-admission set gains in round 1 (Threadripper 9960X again)	1.052	> 1	held (point)
a-P6-fetch-r2	greedy MIN in the step loses in round 2 (Threadripper 9960X again)	0.933	< 1	held (point)
a-P6-plan-r2	fewest-admission set gains in round 2 (Threadripper 9960X again)	1.061	> 1	held (point)
a-P6-fetch-r3	greedy MIN in the step loses in round 3 (Threadripper 9960X again)	0.936	< 1	held (point)
a-P6-plan-r3	fewest-admission set gains in round 3 (Threadripper 9960X again)	1.053	> 1	held (point)
a-P6-launch	launch-level interval of fetchplan/base above 1 (Threadripper 9960X again)	1.056 [1.048, 1.064]	> 1	held
a-P7-speed	fetchplan/base > 1 (Threadripper 9960X again)	1.056 [1.048, 1.064]	> 1	held
a-P7-copies	fetchplan copies $\leq 0.65\text{x}$ fetch's (Threadripper 9960X again)	0.599	≤ 0.65	held (point)
a-P8-g11	base varies $\leq 2\%$ between rounds (Threadripper 9960X again, g11)	0.001	≤ 0.02	held (point)
a-P1-g25	profiled GPU compute 4.0-5.0 ms (Threadripper 9960X again, g25)	4.363	4.0 to 5.0	held (point)
a-P2-g25	overlap law within 6% of base (Threadripper 9960X again, g25)	0.013	≤ 0.06	held (point)
a-P3-reads-g25	dk reads fewer host experts than base (Threadripper 9960X again, g25)	0.897	< 1	held (point)
a-P3-speed-g25	dk/base ≥ 1.01 (Threadripper 9960X again, g25)	1.019 [1.003, 1.037]	≥ 1.01	held (point)
a-P4-reads-g25	lrn reads at most dk's (Threadripper 9960X again, g25)	0.959	≤ 1	held (point)
a-P4-speed-g25	lrn/base ≥ 1.00 (Threadripper 9960X again, g25)	0.988 [0.972, 1.003]	≥ 1.00	failed
a-P5-dk-g25	law-predicted dk/base within 0.03 (Threadripper 9960X again, g25)	0.008	≤ 0.03	held (point)
a-P5-lrn-g25	law-predicted lrn/base within 0.03 (Threadripper 9960X again, g25)	0.013	≤ 0.03	held (point)
a-P4-host-g25	learned host time $\leq 400 \mu\text{s}$ per step (Threadripper 9960X again, g25)	300.2	≤ 400	held (point)
a-P8-g25	base varies $\leq 2\%$ between rounds (Threadripper 9960X again, g25)	0.009	≤ 0.02	held (point)
a-P1-q12	profiled GPU compute 4.5-6.5 ms (Threadripper 9960X again, q12)	5.417	4.5 to 6.5	held (point)
a-P2-q12	overlap law within 6% of base (Threadripper 9960X again, q12)	0.049	≤ 0.06	held (point)
a-P3-speed-q12	dk/base ≥ 1.00 (Threadripper 9960X again, q12)	1.016 [1.015, 1.018]	≥ 1.00	held
a-P4-host-q12	learned host time $\leq 700 \mu\text{s}$ per step (Threadripper 9960X again, q12)	587.4	≤ 700	held (point)

continued

Clause	Statement	Measured	Predicted	Status
a-P1-q25	profiled GPU compute 4.5-6.5 ms (Threadripper 9960X again, q25)	5.689	4.5 to 6.5	held (point)
a-P2-q25	overlap law within 6% of base (Threadripper 9960X again, q25)	0.060	≤ 0.06	held (point)
a-P3-speed-q25	dk/base ≥ 1.00 (Threadripper 9960X again, q25)	1.012 [1.009, 1.014]	≥ 1.00	held
a-P4-host-q25	learned host time $\leq 700 \mu s$ per step (Threadripper 9960X again, q25)	593.2	≤ 700	held (point)
b-P1-g11	profiled GPU compute 4.0-5.0 ms (Pf again (285K), g11)	4.283	4.0 to 5.0	held (point)
b-P2-g11	overlap law within 6% of base (Pf again (285K), g11)	0.009	≤ 0.06	held (point)
b-P3-reads-g11	dk reads fewer host experts than base (Pf again (285K), g11)	0.942	< 1	held (point)
b-P3-speed-g11	dk/base ≥ 1.01 (Pf again (285K), g11)	1.024 [1.005, 1.037]	≥ 1.01	held (point)
b-P4-reads-g11	lrn reads at most dk's (Pf again (285K), g11)	0.987	≤ 1	held (point)
b-P4-speed-g11	lrn/base ≥ 1.00 (Pf again (285K), g11)	1.028 [1.022, 1.035]	≥ 1.00	held
b-P5-dk-g11	law-predicted dk/base within 0.03 (Pf again (285K), g11)	0.001	≤ 0.03	held (point)
b-P5-lrn-g11	law-predicted lrn/base within 0.03 (Pf again (285K), g11)	0.014	≤ 0.03	held (point)
b-P4-host-g11	learned host time $\leq 400 \mu s$ per step (Pf again (285K), g11)	266.7	≤ 400	held (point)
b-P6-fetch-r1	greedy MIN in the step loses in round 1 (Pf again (285K))	0.850	< 1	held (point)
b-P6-plan-r1	fewest-admission set gains in round 1 (Pf again (285K))	1.018	> 1	held (point)
b-P6-fetch-r2	greedy MIN in the step loses in round 2 (Pf again (285K))	0.855	< 1	held (point)
b-P6-plan-r2	fewest-admission set gains in round 2 (Pf again (285K))	1.022	> 1	held (point)
b-P6-fetch-r3	greedy MIN in the step loses in round 3 (Pf again (285K))	0.853	< 1	held (point)
b-P6-plan-r3	fewest-admission set gains in round 3 (Pf again (285K))	1.018	> 1	held (point)
b-P6-launch	launch-level interval of fetchplan/base above 1 (Pf again (285K))	1.019 [1.013, 1.026]	> 1	held
b-P7-speed	fetchplan/base > 1 (Pf again (285K))	1.019 [1.013, 1.026]	> 1	held
b-P7-copies	fetchplan copies $\leq 0.65x$ fetch's (Pf again (285K))	0.597	≤ 0.65	held (point)
b-P8-g11	base varies $\leq 2\%$ between rounds (Pf again (285K), g11)	0.003	≤ 0.02	held (point)
b-P1-g25	profiled GPU compute 4.0-5.0 ms (Pf again (285K), g25)	4.652	4.0 to 5.0	held (point)
b-P2-g25	overlap law within 6% of base (Pf again (285K), g25)	0.016	≤ 0.06	held (point)
b-P3-reads-g25	dk reads fewer host experts than base (Pf again (285K), g25)	0.895	< 1	held (point)
b-P3-speed-g25	dk/base ≥ 1.01 (Pf again (285K), g25)	1.057 [1.043, 1.066]	≥ 1.01	held
b-P4-reads-g25	lrn reads at most dk's (Pf again (285K), g25)	0.958	≤ 1	held (point)
b-P4-speed-g25	lrn/base ≥ 1.00 (Pf again (285K), g25)	1.055 [1.043, 1.066]	≥ 1.00	held
b-P5-dk-g25	law-predicted dk/base within 0.03 (Pf again (285K), g25)	0.030	≤ 0.03	failed
b-P5-lrn-g25	law-predicted lrn/base within 0.03 (Pf again (285K), g25)	0.045	≤ 0.03	failed
b-P4-host-g25	learned host time $\leq 400 \mu s$ per step (Pf again (285K), g25)	272.1	≤ 400	held (point)
b-P8-g25	base varies $\leq 2\%$ between rounds (Pf again (285K), g25)	0.003	≤ 0.02	held (point)
b-P1-q12	profiled GPU compute 4.5-6.5 ms (Pf again (285K), q12)	5.426	4.5 to 6.5	held (point)
b-P2-q12	overlap law within 6% of base (Pf again (285K), q12)	0.018	≤ 0.06	held (point)
b-P3-speed-q12	dk/base ≥ 1.00 (Pf again (285K), q12)	1.018 [1.002, 1.032]	≥ 1.00	held
b-P4-host-q12	learned host time $\leq 700 \mu s$ per step (Pf again (285K), q12)	624.1	≤ 700	held (point)
b-P1-q25	profiled GPU compute 4.5-6.5 ms (Pf again (285K), q25)	5.930	4.5 to 6.5	held (point)
b-P2-q25	overlap law within 6% of base (Pf again (285K), q25)	0.045	≤ 0.06	held (point)
b-P3-speed-q25	dk/base ≥ 1.00 (Pf again (285K), q25)	1.038 [1.033, 1.044]	≥ 1.00	held
b-P4-host-q25	learned host time $\leq 700 \mu s$ per step (Pf again (285K), q25)	619.3	≤ 700	held (point)
c-P1-g11	profiled GPU compute 4.0-5.0 ms (EPYC 7302, g11)	4.271	4.0 to 5.0	held (point)
c-P2-g11	overlap law within 6% of base (EPYC 7302, g11)	0.309	≤ 0.06	failed
c-P3-reads-g11	dk reads fewer host experts than base (EPYC 7302, g11)	0.943	< 1	held (point)
c-P3-speed-g11	dk/base ≥ 1.01 (EPYC 7302, g11)	1.061 [1.014, 1.099]	≥ 1.01	held
c-P4-reads-g11	lrn reads at most dk's (EPYC 7302, g11)	0.988	≤ 1	held (point)
c-P4-speed-g11	lrn/base ≥ 1.00 (EPYC 7302, g11)	0.965 [0.919, 1.012]	≥ 1.00	failed

continued

Clause	Statement	Measured	Predicted	Status
c-P5-dk-g11	law-predicted dk/base within 0.03 (EPYC 7302, g11)	0.040	≤ 0.03	failed
c-P5-lrn-g11	law-predicted lrn/base within 0.03 (EPYC 7302, g11)	0.010	≤ 0.03	held (point)
c-P4-host-g11	learned host time $\leq 400 \mu s$ per step (EPYC 7302, g11)	652.8	≤ 400	failed
c-P6-fetch-r1	greedy MIN in the step loses in round 1 (EPYC 7302)	0.650	< 1	held (point)
c-P6-plan-r1	fewest-admission set gains in round 1 (EPYC 7302)	0.834	> 1	failed
c-P6-fetch-r2	greedy MIN in the step loses in round 2 (EPYC 7302)	0.693	< 1	held (point)
c-P6-plan-r2	fewest-admission set gains in round 2 (EPYC 7302)	0.878	> 1	failed
c-P6-fetch-r3	greedy MIN in the step loses in round 3 (EPYC 7302)	0.708	< 1	held (point)
c-P6-plan-r3	fewest-admission set gains in round 3 (EPYC 7302)	0.824	> 1	failed
c-P6-launch	launch-level interval of fetchplan/base above 1 (EPYC 7302)	0.846 [0.820, 0.877]	> 1	failed
c-P7-speed	fetchplan/base > 1 (EPYC 7302)	0.846 [0.820, 0.877]	> 1	failed
c-P7-copies	fetchplan copies $\leq 0.65x$ fetch's (EPYC 7302)	0.574	≤ 0.65	held (point)
c-P8-g11	base varies $\leq 2\%$ between rounds (EPYC 7302, g11)	0.110	≤ 0.02	failed
c-P1-g25	profiled GPU compute 4.0-5.0 ms (EPYC 7302, g25)	4.668	4.0 to 5.0	held (point)
c-P2-g25	overlap law within 6% of base (EPYC 7302, g25)	0.286	≤ 0.06	failed
c-P3-reads-g25	dk reads fewer host experts than base (EPYC 7302, g25)	0.896	< 1	held (point)
c-P3-speed-g25	dk/base ≥ 1.01 (EPYC 7302, g25)	1.083 [0.997, 1.169]	≥ 1.01	held (point)
c-P4-reads-g25	lrn reads at most dk's (EPYC 7302, g25)	0.971	≤ 1	held (point)
c-P4-speed-g25	lrn/base ≥ 1.00 (EPYC 7302, g25)	1.101 [1.060, 1.139]	≥ 1.00	held
c-P5-dk-g25	law-predicted dk/base within 0.03 (EPYC 7302, g25)	0.063	≤ 0.03	failed
c-P5-lrn-g25	law-predicted lrn/base within 0.03 (EPYC 7302, g25)	0.139	≤ 0.03	failed
c-P4-host-g25	learned host time $\leq 400 \mu s$ per step (EPYC 7302, g25)	540.9	≤ 400	failed
c-P8-g25	base varies $\leq 2\%$ between rounds (EPYC 7302, g25)	0.016	≤ 0.02	held (point)
c-P1-q12	profiled GPU compute 4.5-6.5 ms (EPYC 7302, q12)	5.817	4.5 to 6.5	held (point)
c-P2-q12	overlap law within 6% of base (EPYC 7302, q12)	0.198	≤ 0.06	failed
c-P3-speed-q12	dk/base ≥ 1.00 (EPYC 7302, q12)	1.074 [1.017, 1.133]	≥ 1.00	held
c-P4-host-q12	learned host time $\leq 700 \mu s$ per step (EPYC 7302, q12)	1151.4	≤ 700	failed
c-P1-q25	profiled GPU compute 4.5-6.5 ms (EPYC 7302, q25)	6.333	4.5 to 6.5	held (point)
c-P2-q25	overlap law within 6% of base (EPYC 7302, q25)	0.169	≤ 0.06	failed
c-P3-speed-q25	dk/base ≥ 1.00 (EPYC 7302, q25)	1.014 [0.925, 1.076]	≥ 1.00	held (point)
c-P4-host-q25	learned host time $\leq 700 \mu s$ per step (EPYC 7302, q25)	995.1	≤ 700	failed
d-P1-g11	profiled GPU compute 4.0-5.0 ms (Xeon 8347C, g11)	4.101	4.0 to 5.0	held (point)
d-P2-g11	overlap law within 6% of base (Xeon 8347C, g11)	0.384	≤ 0.06	failed
d-P3-reads-g11	dk reads fewer host experts than base (Xeon 8347C, g11)	0.942	< 1	held (point)
d-P3-speed-g11	dk/base ≥ 1.01 (Xeon 8347C, g11)	1.278 [0.989, 1.559]	≥ 1.01	held (point)
d-P4-reads-g11	lrn reads at most dk's (Xeon 8347C, g11)	0.987	≤ 1	held (point)
d-P4-speed-g11	lrn/base ≥ 1.00 (Xeon 8347C, g11)	1.078 [0.795, 1.447]	≥ 1.00	held (point)
d-P5-dk-g11	law-predicted dk/base within 0.03 (Xeon 8347C, g11)	0.255	≤ 0.03	failed
d-P5-lrn-g11	law-predicted lrn/base within 0.03 (Xeon 8347C, g11)	0.105	≤ 0.03	failed
d-P4-host-g11	learned host time $\leq 400 \mu s$ per step (Xeon 8347C, g11)	693.2	≤ 400	failed
d-P6-fetch-r1	greedy MIN in the step loses in round 1 (Xeon 8347C)	0.850	< 1	held (point)
d-P6-plan-r1	fewest-admission set gains in round 1 (Xeon 8347C)	0.996	> 1	failed
d-P6-fetch-r2	greedy MIN in the step loses in round 2 (Xeon 8347C)	1.270	< 1	failed
d-P6-plan-r2	fewest-admission set gains in round 2 (Xeon 8347C)	1.489	> 1	held (point)
d-P6-fetch-r3	greedy MIN in the step loses in round 3 (Xeon 8347C)	1.265	< 1	failed
d-P6-plan-r3	fewest-admission set gains in round 3 (Xeon 8347C)	1.487	> 1	held (point)
d-P6-launch	launch-level interval of fetchplan/base above 1 (Xeon 8347C)	1.326 [0.997, 1.613]	> 1	held (point)
d-P7-speed	fetchplan/base > 1 (Xeon 8347C)	1.326 [0.997, 1.613]	> 1	held (point)
d-P7-copies	fetchplan copies $\leq 0.65x$ fetch's (Xeon 8347C)	0.599	≤ 0.65	held (point)
d-P8-g11	base varies $\leq 2\%$ between rounds (Xeon 8347C, g11)	0.388	≤ 0.02	failed
d-P1-g25	profiled GPU compute 4.0-5.0 ms (Xeon 8347C, g25)	4.488	4.0 to 5.0	held (point)
d-P2-g25	overlap law within 6% of base (Xeon 8347C, g25)	0.148	≤ 0.06	failed
d-P3-reads-g25	dk reads fewer host experts than base (Xeon 8347C, g25)	0.895	< 1	held (point)
d-P3-speed-g25	dk/base ≥ 1.01 (Xeon 8347C, g25)	1.063 [1.050, 1.081]	≥ 1.01	held
d-P4-reads-g25	lrn reads at most dk's (Xeon 8347C, g25)	0.960	≤ 1	held (point)
d-P4-speed-g25	lrn/base ≥ 1.00 (Xeon 8347C, g25)	1.016 [0.990, 1.042]	≥ 1.00	held (point)
d-P5-dk-g25	law-predicted dk/base within 0.03 (Xeon 8347C, g25)	0.041	≤ 0.03	failed
d-P5-lrn-g25	law-predicted lrn/base within 0.03 (Xeon 8347C, g25)	0.050	≤ 0.03	failed
d-P4-host-g25	learned host time $\leq 400 \mu s$ per step (Xeon 8347C, g25)	575.2	≤ 400	failed
d-P8-g25	base varies $\leq 2\%$ between rounds (Xeon 8347C, g25)	0.027	≤ 0.02	failed
d-P1-q12	profiled GPU compute 4.5-6.5 ms (Xeon 8347C, q12)	5.295	4.5 to 6.5	held (point)
d-P2-q12	overlap law within 6% of base (Xeon 8347C, q12)	0.524	≤ 0.06	failed
d-P3-speed-q12	dk/base ≥ 1.00 (Xeon 8347C, q12)	1.958 [1.670, 2.241]	≥ 1.00	held
d-P4-host-q12	learned host time $\leq 700 \mu s$ per step (Xeon 8347C, q12)	1225.0	≤ 700	failed

continued

Clause	Statement	Measured	Predicted	Status
d-P1-q25	profiled GPU compute 4.5-6.5 ms (Xeon 8347C, q25)	5.825	4.5 to 6.5	held (point)
d-P2-q25	overlap law within 6% of base (Xeon 8347C, q25)	0.102	≤ 0.06	failed
d-P3-speed-q25	dk/base ≥ 1.00 (Xeon 8347C, q25)	0.863 [0.710, 1.021]	≥ 1.00	failed
d-P4-host-q25	learned host time $\leq 700 \mu s$ per step (Xeon 8347C, q25)	1234.9	≤ 700	failed
e-P1-g11	profiled GPU compute 4.0-5.0 ms (Pe again (9950X), g11)	4.453	4.0 to 5.0	held (point)
e-P2-g11	overlap law within 6% of base (Pe again (9950X), g11)	0.005	≤ 0.06	held (point)
e-P3-reads-g11	dk reads fewer host experts than base (Pe again (9950X), g11)	0.956	< 1	held (point)
e-P3-speed-g11	dk/base ≥ 1.01 (Pe again (9950X), g11)	1.010 [1.009, 1.011]	≥ 1.01	failed
e-P4-reads-g11	lrn reads at most dk's (Pe again (9950X), g11)	0.977	≤ 1	held (point)
e-P4-speed-g11	lrn/base ≥ 1.00 (Pe again (9950X), g11)	1.014 [1.011, 1.017]	≥ 1.00	held
e-P5-dk-g11	law-predicted dk/base within 0.03 (Pe again (9950X), g11)	0.017	≤ 0.03	held (point)
e-P5-lrn-g11	law-predicted lrn/base within 0.03 (Pe again (9950X), g11)	0.011	≤ 0.03	held (point)
e-P4-host-g11	learned host time $\leq 400 \mu s$ per step (Pe again (9950X), g11)	434.7	≤ 400	failed
e-P7-speed	fetchplan/base > 1 (Pe again (9950X))	1.339 [1.333, 1.346]	> 1	held
e-P7-copies	fetchplan copies $\leq 0.65x$ fetch's (Pe again (9950X))	0.599	≤ 0.65	held (point)
e-P8-g11	base varies $\leq 2\%$ between rounds (Pe again (9950X), g11)	0.002	≤ 0.02	held (point)
e-P1-g25	profiled GPU compute 4.0-5.0 ms (Pe again (9950X), g25)	4.500	4.0 to 5.0	held (point)
e-P2-g25	overlap law within 6% of base (Pe again (9950X), g25)	0.029	≤ 0.06	held (point)
e-P3-reads-g25	dk reads fewer host experts than base (Pe again (9950X), g25)	0.937	< 1	held (point)
e-P3-speed-g25	dk/base ≥ 1.01 (Pe again (9950X), g25)	1.008 [1.006, 1.010]	≥ 1.01	failed
e-P4-reads-g25	lrn reads at most dk's (Pe again (9950X), g25)	0.941	≤ 1	held (point)
e-P4-speed-g25	lrn/base ≥ 1.00 (Pe again (9950X), g25)	1.010 [1.002, 1.017]	≥ 1.00	held
e-P5-dk-g25	law-predicted dk/base within 0.03 (Pe again (9950X), g25)	0.015	≤ 0.03	held (point)
e-P5-lrn-g25	law-predicted lrn/base within 0.03 (Pe again (9950X), g25)	0.015	≤ 0.03	held (point)
e-P4-host-g25	learned host time $\leq 400 \mu s$ per step (Pe again (9950X), g25)	439.0	≤ 400	failed
e-P8-g25	base varies $\leq 2\%$ between rounds (Pe again (9950X), g25)	0.000	≤ 0.02	held (point)
e-P1-q12	profiled GPU compute 4.5-6.5 ms (Pe again (9950X), q12)	5.818	4.5 to 6.5	held (point)
e-P2-q12	overlap law within 6% of base (Pe again (9950X), q12)	0.045	≤ 0.06	held (point)
e-P3-speed-q12	dk/base ≥ 1.00 (Pe again (9950X), q12)	1.011 [1.010, 1.012]	≥ 1.00	held
e-P4-host-q12	learned host time $\leq 700 \mu s$ per step (Pe again (9950X), q12)	739.5	≤ 700	failed
e-P1-q25	profiled GPU compute 4.5-6.5 ms (Pe again (9950X), q25)	5.933	4.5 to 6.5	held (point)
e-P2-q25	overlap law within 6% of base (Pe again (9950X), q25)	0.010	≤ 0.06	held (point)
e-P3-speed-q25	dk/base ≥ 1.00 (Pe again (9950X), q25)	1.010 [1.009, 1.012]	≥ 1.00	held
e-P4-host-q25	learned host time $\leq 700 \mu s$ per step (Pe again (9950X), q25)	1029.7	≤ 700	failed
P2-median	median —error— of the overlap law at most 4%	0.047	≤ 0.04	failed
P2-g25	at gpt-oss 25% the overlap law's median —error— at most half the plain law's	0.355	≤ 0.5	held (point)
P3-pooled	pooled median dk/base at gpt-oss ≥ 1.02	1.040	≥ 1.02	held (point)

Table S13: Job 107 (the time relation on machines never rented before), scored by machine by `scripts/job107.py` under the same rule. Totals: 45 clauses, 11 held, 22 held (point), 12 failed, 0 untested. Validity condition (not a prediction): not met. Valid hosts b (Ryzen 9 5900XT) and d (a dual-socket AMD engineering sample), both never rented before. Hosts c (EPYC 9754) and e (EPYC 7663) failed the registered 2% check between rounds and are not scored; a, f and g stopped at the gate (host memory in use). Cells gpt-oss 11% (g11) and 25% (g25), two rounds each, Qwen3 12.5% (q12) and 25% (q25), one round each. Measured: the point estimate and, where one exists, the 95% interval (bootstrap over problems within the host; pooled clauses have none).

Clause	Statement	Measured	Predicted	Status
b-P1-g11	profiled GPU compute 4.0-5.0 ms (107b Ryzen 9 5900XT, g11)	4.439	4.0 to 5.0	held (point)
b-P2-g11	overlap form within 6% of base (107b Ryzen 9 5900XT, g11)	0.071	≤ 0.06	failed
b-P3-reads-g11	dk reads fewer host experts than base (107b Ryzen 9 5900XT, g11)	0.957	< 1	held (point)
b-P3-speed-g11	dk/base ≥ 0.995 (107b Ryzen 9 5900XT, g11)	1.021 [1.018, 1.024]	≥ 0.995	held
b-P4-g11	law-predicted dk/base within 0.03 (107b Ryzen 9 5900XT, g11)	0.009	≤ 0.03	held (point)
b-P5-fetchplan	fetchplan/base > 1 at gpt-oss 11% (107b Ryzen 9 5900XT)	1.419 [1.406, 1.432]	> 1	held
b-P5-bypassplan	bypassplan/base > 1 at gpt-oss 11% (107b Ryzen 9 5900XT)	1.038 [1.035, 1.040]	> 1	held
b-P6	fetchplan beats bypassplan at ratio 0.75 (107b Ryzen 9 5900XT)	0.381	> 0	held (point)
b-P1-g25	profiled GPU compute 4.0-5.0 ms (107b Ryzen 9 5900XT, g25)	4.514	4.0 to 5.0	held (point)
b-P2-g25	overlap form within 6% of base (107b Ryzen 9 5900XT, g25)	0.023	≤ 0.06	held (point)
b-P3-reads-g25	dk reads fewer host experts than base (107b Ryzen 9 5900XT, g25)	0.934	< 1	held (point)
b-P3-speed-g25	dk/base ≥ 0.995 (107b Ryzen 9 5900XT, g25)	1.019 [1.013, 1.023]	≥ 0.995	held
b-P4-g25	law-predicted dk/base within 0.03 (107b Ryzen 9 5900XT, g25)	0.014	≤ 0.03	held (point)
b-P1-q12	profiled GPU compute 4.5-6.5 ms (107b Ryzen 9 5900XT, q12)	5.752	4.5 to 6.5	held (point)
b-P2-q12	overlap form within 6% of base (107b Ryzen 9 5900XT, q12)	0.113	≤ 0.06	failed
b-P3-speed-q12	dk/base ≥ 0.995 (107b Ryzen 9 5900XT, q12)	1.021 [1.018, 1.024]	≥ 0.995	held
b-P4-q12	law-predicted dk/base within 0.03 (107b Ryzen 9 5900XT, q12)	0.001	≤ 0.03	held (point)
b-P1-q25	profiled GPU compute 4.5-6.5 ms (107b Ryzen 9 5900XT, q25)	5.948	4.5 to 6.5	held (point)
b-P2-q25	overlap form within 6% of base (107b Ryzen 9 5900XT, q25)	0.064	≤ 0.06	failed
b-P3-speed-q25	dk/base ≥ 0.995 (107b Ryzen 9 5900XT, q25)	1.018 [1.016, 1.021]	≥ 0.995	held
b-P4-q25	law-predicted dk/base within 0.03 (107b Ryzen 9 5900XT, q25)	0.005	≤ 0.03	held (point)
d-P1-g11	profiled GPU compute 4.0-5.0 ms (107d AMD engineering sample, g11)	4.051	4.0 to 5.0	held (point)
d-P2-g11	overlap form within 6% of base (107d AMD engineering sample, g11)	0.590	≤ 0.06	failed
d-P3-reads-g11	dk reads fewer host experts than base (107d AMD engineering sample, g11)	0.948	< 1	held (point)
d-P3-speed-g11	dk/base ≥ 0.995 (107d AMD engineering sample, g11)	1.049 [1.036, 1.062]	≥ 0.995	held
d-P4-g11	law-predicted dk/base within 0.03 (107d AMD engineering sample, g11)	0.041	≤ 0.03	failed
d-P5-fetchplan	fetchplan/base > 1 at gpt-oss 11% (107d AMD engineering sample)	1.157 [1.143, 1.172]	> 1	held
d-P5-bypassplan	bypassplan/base > 1 at gpt-oss 11% (107d AMD engineering sample)	1.126 [1.110, 1.142]	> 1	held
d-P6	fetchplan beats bypassplan at ratio 0.44 (107d AMD engineering sample)	0.032	> 0	held (point)
d-P1-g25	profiled GPU compute 4.0-5.0 ms (107d AMD engineering sample, g25)	4.245	4.0 to 5.0	held (point)
d-P2-g25	overlap form within 6% of base (107d AMD engineering sample, g25)	0.522	≤ 0.06	failed

continued

Clause	Statement	Measured	Predicted	Status
d-P3-reads-g25	dk reads fewer host experts than base (107d AMD engineering sample, g25)	0.907	< 1	held (point)
d-P3-speed-g25	dk/base ≥ 0.995 (107d AMD engineering sample, g25)	0.987 [0.978, 1.000]	≥ 0.995	failed
d-P4-g25	law-predicted dk/base within 0.03 (107d AMD engineering sample, g25)	0.018	≤ 0.03	held (point)
d-P1-q12	profiled GPU compute 4.5-6.5 ms (107d AMD engineering sample, q12)	5.603	4.5 to 6.5	held (point)
d-P2-q12	overlap form within 6% of base (107d AMD engineering sample, q12)	0.583	≤ 0.06	failed
d-P3-speed-q12	dk/base ≥ 0.995 (107d AMD engineering sample, q12)	1.033 [1.024, 1.043]	≥ 0.995	held
d-P4-q12	law-predicted dk/base within 0.03 (107d AMD engineering sample, q12)	0.021	≤ 0.03	held (point)
d-P1-q25	profiled GPU compute 4.5-6.5 ms (107d AMD engineering sample, q25)	5.854	4.5 to 6.5	held (point)
d-P2-q25	overlap form within 6% of base (107d AMD engineering sample, q25)	0.563	≤ 0.06	failed
d-P3-speed-q25	dk/base ≥ 0.995 (107d AMD engineering sample, q25)	1.074 [1.061, 1.087]	≥ 0.995	held
d-P4-q25	law-predicted dk/base within 0.03 (107d AMD engineering sample, q25)	0.067	≤ 0.03	failed
P2-median	median —error— of the overlap form at most 4%	0.318	≤ 0.04	failed
P2-plain	overlap form's median —error— below the plain form's	-0.014	> 0	failed
P3-median	median dk/base over all cells ≥ 1.01	1.021	≥ 1.01	held (point)
valid	at least 3 valid hosts	2.000	≥ 3	failed

Table S14: Job 108 (the time relation on new machines of the class it was found on), scored by machine by `scripts/job108.py` under the same rule. Totals: 20 clauses, 0 held, 12 held (point), 6 failed, 2 untested. Validity condition (not a prediction): met. Valid hosts a (EPYC 7543; G not profiled, the median of the profiles of jobs 069c and 105 used), d (Ryzen 9 9950X3D) and f (Ryzen 9 5950X). Host b (EPYC 7K62) failed the round check; c and e stopped at the gate. Cells gpt-oss 11% (g11) and 25% (g25), two rounds each. Measured: the point estimate and, where one exists, the 95% interval (bootstrap over problems within the host; pooled clauses have none).

Clause	Statement	Measured	Predicted	Status
a-P1-g11	profiled GPU compute 4.0-5.0 ms (108a EPYC 7543, g11)	–	4.0 to 5.0	untested
a-P2-g11	relation within 8% of base (108a EPYC 7543, g11)	0.343	≤ 0.08	failed
a-P3-g11	implied read rate 0.85-1.25 of B_host (108a EPYC 7543, g11)	0.584	0.85 to 1.25	failed
a-P1-g25	profiled GPU compute 4.0-5.0 ms (108a EPYC 7543, g25)	–	4.0 to 5.0	untested
a-P2-g25	relation within 8% of base (108a EPYC 7543, g25)	0.277	≤ 0.08	failed
a-P3-g25	implied read rate 0.85-1.25 of B_host (108a EPYC 7543, g25)	0.610	0.85 to 1.25	failed
d-P1-g11	profiled GPU compute 4.0-5.0 ms (108d Ryzen 9 9950X3D, g11)	4.186	4.0 to 5.0	held (point)
d-P2-g11	relation within 8% of base (108d Ryzen 9 9950X3D, g11)	0.069	≤ 0.08	held (point)
d-P3-g11	implied read rate 0.85-1.25 of B_host (108d Ryzen 9 9950X3D, g11)	0.911	0.85 to 1.25	held (point)
d-P1-g25	profiled GPU compute 4.0-5.0 ms (108d Ryzen 9 9950X3D, g25)	4.241	4.0 to 5.0	held (point)
d-P2-g25	relation within 8% of base (108d Ryzen 9 9950X3D, g25)	0.041	≤ 0.08	held (point)
d-P3-g25	implied read rate 0.85-1.25 of B_host (108d Ryzen 9 9950X3D, g25)	0.937	0.85 to 1.25	held (point)
f-P1-g11	profiled GPU compute 4.0-5.0 ms (108f Ryzen 9 5950X, g11)	4.488	4.0 to 5.0	held (point)
f-P2-g11	relation within 8% of base (108f Ryzen 9 5950X, g11)	0.065	≤ 0.08	held (point)
f-P3-g11	implied read rate 0.85-1.25 of B_host (108f Ryzen 9 5950X, g11)	0.924	0.85 to 1.25	held (point)
f-P1-g25	profiled GPU compute 4.0-5.0 ms (108f Ryzen 9 5950X, g25)	4.568	4.0 to 5.0	held (point)
f-P2-g25	relation within 8% of base (108f Ryzen 9 5950X, g25)	0.009	≤ 0.08	held (point)
f-P3-g25	implied read rate 0.85-1.25 of B_host (108f Ryzen 9 5950X, g25)	0.987	0.85 to 1.25	held (point)
P2-six	at most one cell beyond 6%	4.000	≤ 1	failed
P2-median	median —error— at most 4%	0.067	≤ 0.04	failed
valid	at least 3 valid machines	3.000	≥ 3	held (point)

Table S15: Job 109 (the 2x2 and online policies on new machines), scored by machine by `scripts/job109.py` under the same rule. Totals: 90 clauses, 61 held, 29 held (point), 0 failed, 0 untested. Validity condition (not a prediction): met. Hosts a, c, d, e, f (new desktop-class machines; offer b was no longer listed); cells gpt-oss 11% (g11, two rounds) and 25% (g25, one). Intervals are over problems, paired across configurations and rounds. Measured: the point estimate and, where one exists, the 95% interval (bootstrap over problems within the host; pooled clauses have none).

Clause	Statement	Measured	Predicted	Status
a-P1-g11	MIN read once beats both single changes by 0.04 (109a Ryzen 7 9800X3D)	0.115 [0.111, 0.120]	≥ 0.04	held
a-P1-int-g11	interaction positive (109a Ryzen 7 9800X3D, g11)	1.525 [1.455, 1.599]	> 0 ms	held
a-P1-int-g25	interaction positive (109a Ryzen 7 9800X3D, g25)	0.502 [0.442, 0.557]	> 0 ms	held
a-P2-foa	foa/base within [0.96, 1.05] at 11% (109a Ryzen 7 9800X3D)	0.989 [0.986, 0.991]	0.96 to 1.05	held
a-P2-bypass	bypass/base within [0.96, 1.05] at 11% (109a Ryzen 7 9800X3D)	1.039 [1.033, 1.044]	0.96 to 1.05	held
a-P4-g25	both3p beats fetch at 25% (109a Ryzen 7 9800X3D)	0.330	> 0	held (point)
a-P5	bypass/base ≥ 1.08 at 25% (109a Ryzen 7 9800X3D)	1.165 [1.157, 1.173]	≥ 1.08	held
a-P7-best-g11	best online policy ≤ 1.10 (109a Ryzen 7 9800X3D, g11)	1.014 [1.008, 1.020]	≤ 1.10	held
a-P7-cap-g11	capture ≤ 0.35 (109a Ryzen 7 9800X3D, g11)	0.041 [0.022, 0.058]	≤ 0.35	held
a-P7-best-g25	best online policy ≤ 1.10 (109a Ryzen 7 9800X3D, g25)	1.011 [1.007, 1.015]	≤ 1.10	held
a-P7-cap-g25	capture ≤ 0.35 (109a Ryzen 7 9800X3D, g25)	0.021 [0.014, 0.027]	≤ 0.35	held
a-P9	—R1 - pf— ≤ 0.04 at 11% (109a Ryzen 7 9800X3D)	0.010	≤ 0.04	held (point)
a-P10-dk-g11	dk/base within [0.98, 1.06] (109a Ryzen 7 9800X3D, g11)	1.004 [1.001, 1.008]	0.98 to 1.06	held
a-P10-r2-g11	R2/R1 within [0.97, 1.06] (109a Ryzen 7 9800X3D, g11)	1.018	0.97 to 1.06	held (point)
a-P11-g11	R1 and R2 read ≥ 1.3 R* (109a Ryzen 7 9800X3D, g11)	1.667	≥ 1.3	held (point)
a-P10-dk-g25	dk/base within [0.98, 1.06] (109a Ryzen 7 9800X3D, g25)	1.011 [1.007, 1.015]	0.98 to 1.06	held
a-P10-r2-g25	R2/R1 within [0.97, 1.06] (109a Ryzen 7 9800X3D, g25)	1.024	0.97 to 1.06	held (point)
a-P11-g25	R1 and R2 read ≥ 1.3 R* (109a Ryzen 7 9800X3D, g25)	2.199	≥ 1.3	held (point)
c-P1-g11	MIN read once beats both single changes by 0.04 (109c Ryzen 9 7945HX)	0.212 [0.205, 0.219]	≥ 0.04	held
c-P1-int-g11	interaction positive (109c Ryzen 9 7945HX, g11)	2.966 [2.839, 3.098]	> 0 ms	held
c-P1-int-g25	interaction positive (109c Ryzen 9 7945HX, g25)	1.022 [0.956, 1.087]	> 0 ms	held
c-P2-foa	foa/base within [0.96, 1.05] at 11% (109c Ryzen 9 7945HX)	0.992 [0.991, 0.993]	0.96 to 1.05	held
c-P2-bypass	bypass/base within [0.96, 1.05] at 11% (109c Ryzen 9 7945HX)	1.028 [1.023, 1.034]	0.96 to 1.05	held
c-P4-g25	both3p beats fetch at 25% (109c Ryzen 9 7945HX)	0.344	> 0	held (point)
c-P5	bypass/base ≥ 1.08 at 25% (109c Ryzen 9 7945HX)	1.159 [1.150, 1.167]	≥ 1.08	held
c-P7-best-g11	best online policy ≤ 1.10 (109c Ryzen 9 7945HX, g11)	1.024 [1.018, 1.029]	≤ 1.10	held
c-P7-cap-g11	capture ≤ 0.35 (109c Ryzen 9 7945HX, g11)	0.057 [0.044, 0.069]	≤ 0.35	held
c-P7-best-g25	best online policy ≤ 1.10 (109c Ryzen 9 7945HX, g25)	0.995 [0.993, 1.003]	≤ 1.10	held
c-P7-cap-g25	capture ≤ 0.35 (109c Ryzen 9 7945HX, g25)	-0.008 [-0.012, 0.004]	≤ 0.35	held
c-P9	—R1 - pf— ≤ 0.04 at 11% (109c Ryzen 9 7945HX)	0.010	≤ 0.04	held (point)
c-P10-dk-g11	dk/base within [0.98, 1.06] (109c Ryzen 9 7945HX, g11)	1.003 [1.002, 1.004]	0.98 to 1.06	held
c-P10-r2-g11	R2/R1 within [0.97, 1.06] (109c Ryzen 9 7945HX, g11)	1.010	0.97 to 1.06	held (point)
c-P11-g11	R1 and R2 read ≥ 1.3 R* (109c Ryzen 9 7945HX, g11)	1.665	≥ 1.3	held (point)
c-P10-dk-g25	dk/base within [0.98, 1.06] (109c Ryzen 9 7945HX, g25)	0.994 [0.992, 0.997]	0.98 to 1.06	held
c-P10-r2-g25	R2/R1 within [0.97, 1.06] (109c Ryzen 9 7945HX, g25)	1.030	0.97 to 1.06	held (point)
c-P11-g25	R1 and R2 read ≥ 1.3 R* (109c Ryzen 9 7945HX, g25)	2.195	≥ 1.3	held (point)
d-P1-g11	MIN read once beats both single changes by 0.04 (109d Core i9-13900KF)	0.210 [0.204, 0.217]	≥ 0.04	held
d-P1-int-g11	interaction positive (109d Core i9-13900KF, g11)	2.904 [2.781, 3.045]	> 0 ms	held
d-P1-int-g25	interaction positive (109d Core i9-13900KF, g25)	0.780 [0.698, 0.852]	> 0 ms	held
d-P2-foa	foa/base within [0.96, 1.05] at 11% (109d Core i9-13900KF)	0.998 [0.996, 0.999]	0.96 to 1.05	held
d-P2-bypass	bypass/base within [0.96, 1.05] at 11% (109d Core i9-13900KF)	1.036 [1.030, 1.042]	0.96 to 1.05	held
d-P4-g25	both3p beats fetch at 25% (109d Core i9-13900KF)	0.357	> 0	held (point)
d-P5	bypass/base ≥ 1.08 at 25% (109d Core i9-13900KF)	1.179 [1.173, 1.186]	≥ 1.08	held
d-P7-best-g11	best online policy ≤ 1.10 (109d Core i9-13900KF, g11)	1.007 [1.006, 1.008]	≤ 1.10	held
d-P7-cap-g11	capture ≤ 0.35 (109d Core i9-13900KF, g11)	0.015 [0.012, 0.018]	≤ 0.35	held
d-P7-best-g25	best online policy ≤ 1.10 (109d Core i9-13900KF, g25)	1.010 [1.006, 1.016]	≤ 1.10	held
d-P7-cap-g25	capture ≤ 0.35 (109d Core i9-13900KF, g25)	0.017 [0.009, 0.027]	≤ 0.35	held
d-P8	pf/base below 1 at ratio < 0.75 (109d Core i9-13900KF)	0.968 [0.961, 0.974]	< 1.00	held
d-P9	—R1 - pf— ≤ 0.04 at 11% (109d Core i9-13900KF)	0.008	≤ 0.04	held (point)

continued

Clause	Statement	Measured	Predicted	Status
d-P10-dk-g11	dk/base within [0.98, 1.06] (109d Core i9-13900KF, g11)	1.007 [1.006, 1.008]	0.98 to 1.06	held
d-P10-r2-g11	R2/R1 within [0.97, 1.06] (109d Core i9-13900KF, g11)	1.017	0.97 to 1.06	held (point)
d-P11-g11	R1 and R2 read ≥ 1.3 R* (109d Core i9-13900KF, g11)	1.667	≥ 1.3	held (point)
d-P10-dk-g25	dk/base within [0.98, 1.06] (109d Core i9-13900KF, g25)	1.010 [1.006, 1.016]	0.98 to 1.06	held
d-P10-r2-g25	R2/R1 within [0.97, 1.06] (109d Core i9-13900KF, g25)	1.018	0.97 to 1.06	held (point)
d-P11-g25	R1 and R2 read ≥ 1.3 R* (109d Core i9-13900KF, g25)	2.198	≥ 1.3	held (point)
e-P1-g11	MIN read once beats both single changes by 0.04 (109e Core Ultra 9 285K)	0.211 [0.206, 0.216]	≥ 0.04	held
e-P1-int-g11	interaction positive (109e Core Ultra 9 285K, g11)	3.551 [3.379, 3.736]	> 0 ms	held
e-P1-int-g25	interaction positive (109e Core Ultra 9 285K, g25)	0.809 [0.750, 0.865]	> 0 ms	held
e-P2-foa	foa/base within [0.96, 1.05] at 11% (109e Core Ultra 9 285K)	1.004 [1.003, 1.005]	0.96 to 1.05	held
e-P2-bypass	bypass/base within [0.96, 1.05] at 11% (109e Core Ultra 9 285K)	0.991 [0.985, 0.997]	0.96 to 1.05	held
e-P4-g25	both3p beats fetch at 25% (109e Core Ultra 9 285K)	0.399	> 0	held (point)
e-P5	bypass/base ≥ 1.08 at 25% (109e Core Ultra 9 285K)	1.172 [1.164, 1.181]	≥ 1.08	held
e-P7-best-g11	best online policy ≤ 1.10 (109e Core Ultra 9 285K, g11)	1.021 [1.019, 1.022]	≤ 1.10	held
e-P7-cap-g11	capture ≤ 0.35 (109e Core Ultra 9 285K, g11)	0.045 [0.041, 0.049]	≤ 0.35	held
e-P7-best-g25	best online policy ≤ 1.10 (109e Core Ultra 9 285K, g25)	1.027 [1.024, 1.030]	≤ 1.10	held
e-P7-cap-g25	capture ≤ 0.35 (109e Core Ultra 9 285K, g25)	0.042 [0.038, 0.046]	≤ 0.35	held
e-P8	pf/base below 1 at ratio < 0.75 (109e Core Ultra 9 285K)	0.852 [0.845, 0.859]	< 1.00	held
e-P9	—R1 - pf— ≤ 0.04 at 11% (109e Core Ultra 9 285K)	0.001	≤ 0.04	held (point)
e-P10-dk-g11	dk/base within [0.98, 1.06] (109e Core Ultra 9 285K, g11)	1.021 [1.019, 1.022]	0.98 to 1.06	held
e-P10-r2-g11	R2/R1 within [0.97, 1.06] (109e Core Ultra 9 285K, g11)	1.028	0.97 to 1.06	held (point)
e-P11-g11	R1 and R2 read ≥ 1.3 R* (109e Core Ultra 9 285K, g11)	1.667	≥ 1.3	held (point)
e-P10-dk-g25	dk/base within [0.98, 1.06] (109e Core Ultra 9 285K, g25)	1.027 [1.024, 1.030]	0.98 to 1.06	held
e-P10-r2-g25	R2/R1 within [0.97, 1.06] (109e Core Ultra 9 285K, g25)	1.028	0.97 to 1.06	held (point)
e-P11-g25	R1 and R2 read ≥ 1.3 R* (109e Core Ultra 9 285K, g25)	2.198	≥ 1.3	held (point)
f-P1-g11	MIN read once beats both single changes by 0.04 (109f Ryzen 9 5950X)	0.358 [0.350, 0.364]	≥ 0.04	held
f-P1-int-g11	interaction positive (109f Ryzen 9 5950X, g11)	7.487 [7.200, 7.814]	> 0 ms	held
f-P2-foa	foa/base within [0.96, 1.05] at 11% (109f Ryzen 9 5950X)	1.007 [1.005, 1.009]	0.96 to 1.05	held
f-P2-bypass	bypass/base within [0.96, 1.05] at 11% (109f Ryzen 9 5950X)	1.006 [1.000, 1.012]	0.96 to 1.05	held
f-P7-best-g11	best online policy ≤ 1.10 (109f Ryzen 9 5950X, g11)	1.010 [1.008, 1.012]	≤ 1.10	held
f-P7-cap-g11	capture ≤ 0.35 (109f Ryzen 9 5950X, g11)	0.019 [0.015, 0.024]	≤ 0.35	held
f-P9	—R1 - pf— ≤ 0.04 at 11% (109f Ryzen 9 5950X)	0.001	≤ 0.04	held (point)
f-P10-dk-g11	dk/base within [0.98, 1.06] (109f Ryzen 9 5950X, g11)	1.010 [1.008, 1.012]	0.98 to 1.06	held
f-P10-r2-g11	R2/R1 within [0.97, 1.06] (109f Ryzen 9 5950X, g11)	1.014	0.97 to 1.06	held (point)
f-P11-g11	R1 and R2 read ≥ 1.3 R* (109f Ryzen 9 5950X, g11)	1.665	≥ 1.3	held (point)
P4-g11	both3p beats fetch at 11% on all but at most one	0.000	≤ 1 miss	held (point)
P3-g11	mean together share in [0.25, 0.55]	0.347 [0.219, 0.475]	0.25 to 0.55	held (point)
P3-g25	mean together share in [0.2, 0.5]	0.273 [0.230, 0.315]	0.2 to 0.5	held
P6-g11	mean residual within ± 0.1	0.005 [-0.031, 0.042]	—x— ≤ 0.1	held
P6-g25	mean residual within ± 0.15	-0.051 [-0.074, -0.029]	—x— ≤ 0.15	held
P7-mean	mean capture at 11% at most 0.25	0.035 [0.013, 0.057]	≤ 0.25	held
valid	at least 4 valid machines	5.000	≥ 4	held (point)

Table S16: Job 110 (the RTX 5090 trend on RTX 4090s, as registered), scored by machine by `scripts/job109.py` under the same rule. Totals: 1 clauses, 0 held, 0 held (point), 0 failed, 1 untested. Validity condition (not a prediction): not met. RTX 4090 hosts a–e; a stopped at the ratio gate, the others at the output check after one round (the RTX 5090 reference loss): no host is valid, as registered. Measured: the point estimate and, where one exists, the 95% interval (bootstrap over problems within the host; pooled clauses have none).

Clause	Statement	Measured	Predicted	Status
Q1-g25	fetch and both3p within 0.10 (log) of the trend at 25%	–	≤ 1 miss	untested
valid	on all but one host-configuration at least 3 valid hosts	0.000	≥ 3	failed

Table S17: Job 111 (job 110’s predictions on the relaunched RTX 4090s), scored by machine by `scripts/job109.py` under the same rule. Totals: 37 clauses, 24 held, 13 held (point), 0 failed, 0 untested. Validity condition (not a prediction): met. The RTX 4090 machines of job 110 relaunched with the card’s reference loss; job 110’s clauses scored on its valid hosts. Intervals are over problems. Measured: the point estimate and, where one exists, the 95% interval (bootstrap over problems within the host; pooled clauses have none).

Clause	Statement	Measured	Predicted	Status
d-Q1-fetch-g11	fetch/base within 0.10 (log) of the 5090 trend at 11% (111d Core i9-14900KF)	-0.027 [-0.032, -0.023]	—x— 0.10	≤ held
d-Q1-both3p-g11	both3p/base within 0.10 (log) of the 5090 trend at 11% (111d Core i9-14900KF)	0.015 [0.007, 0.023]	—x— 0.10	≤ held
d-Q1-foa-g11	foa/base within 0.06 (log) of the 5090 trend at 11% (111d Core i9-14900KF)	-0.006 [-0.009, -0.004]	—x— 0.06	≤ held
d-Q1-bypass-g11	bypass/base within 0.06 (log) of the 5090 trend at 11% (111d Core i9-14900KF)	0.009 [0.003, 0.014]	—x— 0.06	≤ held
d-Q2-pf	pf/base below 1 at 11% (111d Core i9-14900KF)	0.682 [0.676, 0.687]	< 1.00	held
d-Q2-R1	R1/base below 1 at 11% (111d Core i9-14900KF)	0.644 [0.638, 0.650]	< 1.00	held
d-Q3-g11	best online policy ≤ 1.10 (111d Core i9-14900KF, g11)	1.033 [1.030, 1.036]	≤ 1.10	held
d-Q4-g11	base / Eq. (1) within [1.8, 3.5] (111d Core i9-14900KF, g11)	2.488	1.8 to 3.5	held (point)
d-Q5-g11	R1 and R2 read ≥ 1.3 R* (111d Core i9-14900KF, g11)	1.671	≥ 1.3	held (point)
d-Q3-g25	best online policy ≤ 1.10 (111d Core i9-14900KF, g25)	1.039 [1.035, 1.044]	≤ 1.10	held
d-Q4-g25	base / Eq. (1) within [2.5, 6.5] (111d Core i9-14900KF, g25)	4.117	2.5 to 6.5	held (point)
d-Q5-g25	R1 and R2 read ≥ 1.3 R* (111d Core i9-14900KF, g25)	2.204	≥ 1.3	held (point)
f-Q1-fetch-g11	fetch/base within 0.10 (log) of the 5090 trend at 11% (111f Core Ultra 9 285K)	-0.004 [-0.008, 0.001]	—x— 0.10	≤ held
f-Q1-both3p-g11	both3p/base within 0.10 (log) of the 5090 trend at 11% (111f Core Ultra 9 285K)	0.056 [0.050, 0.063]	—x— 0.10	≤ held
f-Q1-foa-g11	foa/base within 0.06 (log) of the 5090 trend at 11% (111f Core Ultra 9 285K)	0.007 [0.005, 0.009]	—x— 0.06	≤ held
f-Q1-bypass-g11	bypass/base within 0.06 (log) of the 5090 trend at 11% (111f Core Ultra 9 285K)	0.034 [0.029, 0.040]	—x— 0.06	≤ held
f-Q2-pf	pf/base below 1 at 11% (111f Core Ultra 9 285K)	0.709 [0.703, 0.715]	< 1.00	held
f-Q2-R1	R1/base below 1 at 11% (111f Core Ultra 9 285K)	0.681 [0.674, 0.686]	< 1.00	held
f-Q3-g11	best online policy ≤ 1.10 (111f Core Ultra 9 285K, g11)	1.041 [1.038, 1.044]	≤ 1.10	held
f-Q4-g11	base / Eq. (1) within [1.8, 3.5] (111f Core Ultra 9 285K, g11)	2.539	1.8 to 3.5	held (point)
f-Q5-g11	R1 and R2 read ≥ 1.3 R* (111f Core Ultra 9 285K, g11)	1.671	≥ 1.3	held (point)
f-Q3-g25	best online policy ≤ 1.10 (111f Core Ultra 9 285K, g25)	1.045 [1.041, 1.049]	≤ 1.10	held
f-Q4-g25	base / Eq. (1) within [2.5, 6.5] (111f Core Ultra 9 285K, g25)	4.121	2.5 to 6.5	held (point)
f-Q5-g25	R1 and R2 read ≥ 1.3 R* (111f Core Ultra 9 285K, g25)	2.204	≥ 1.3	held (point)
g-Q1-fetch-g11	fetch/base within 0.10 (log) of the 5090 trend at 11% (111g Core Ultra 9 285K)	-0.002 [-0.007, 0.003]	—x— 0.10	≤ held
g-Q1-both3p-g11	both3p/base within 0.10 (log) of the 5090 trend at 11% (111g Core Ultra 9 285K)	0.058 [0.051, 0.065]	—x— 0.10	≤ held
g-Q1-foa-g11	foa/base within 0.06 (log) of the 5090 trend at 11% (111g Core Ultra 9 285K)	0.006 [0.004, 0.009]	—x— 0.06	≤ held
g-Q1-bypass-g11	bypass/base within 0.06 (log) of the 5090 trend at 11% (111g Core Ultra 9 285K)	0.034 [0.028, 0.040]	—x— 0.06	≤ held
g-Q2-pf	pf/base below 1 at 11% (111g Core Ultra 9 285K)	0.708 [0.701, 0.713]	< 1.00	held
g-Q2-R1	R1/base below 1 at 11% (111g Core Ultra 9 285K)	0.679 [0.673, 0.685]	< 1.00	held
g-Q3-g11	best online policy ≤ 1.10 (111g Core Ultra 9 285K, g11)	1.038 [1.035, 1.042]	≤ 1.10	held
g-Q4-g11	base / Eq. (1) within [1.8, 3.5] (111g Core Ultra 9 285K, g11)	2.535	1.8 to 3.5	held (point)
g-Q5-g11	R1 and R2 read ≥ 1.3 R* (111g Core Ultra 9 285K, g11)	1.671	≥ 1.3	held (point)
g-Q3-g25	best online policy ≤ 1.10 (111g Core Ultra 9 285K, g25)	1.043 [1.039, 1.047]	≤ 1.10	held
g-Q4-g25	base / Eq. (1) within [2.5, 6.5] (111g Core Ultra 9 285K, g25)	4.106	2.5 to 6.5	held (point)
g-Q5-g25	R1 and R2 read ≥ 1.3 R* (111g Core Ultra 9 285K, g25)	2.204	≥ 1.3	held (point)
Q1-g25	fetch and both3p within 0.10 (log) of the trend at 25% on all but one host-configuration	0.000	≤ 1 miss	held (point)
valid	at least 3 valid hosts	3.000	≥ 3	held (point)

Table S18: Job 112 (the last measurements: a second card at higher ratios, the timing control, a second probe), scored by machine by `scripts/job112.py` under the same rule. Totals: 57 clauses, 24 held, 21 held (point), 12 failed, 0 untested. Validity condition (not a prediction): met. Job 110's clauses (Q, ids 112-Q) on new RTX 4090 machines at higher ratios; the timing control (T1–T4), the second probe (T5) and the relaunched of job 109's RTX 5090 machines (T6, T7). Intervals are over problems. Measured: the point estimate and, where one exists, the 95% interval (bootstrap over problems within the host; pooled clauses have none).

Clause	Statement	Measured	Predicted	Status
a-Q1-fetch-g11	fetch/base within 0.10 (log) of the 5090 trend at 11% (112a Core i5-12400)	0.049 [0.043, 0.056]	—x— 0.10	≤ held
a-Q1-both3p-g11	both3p/base within 0.10 (log) of the 5090 trend at 11% (112a Core i5-12400)	0.064 [0.054, 0.074]	—x— 0.10	≤ held
a-Q1-foa-g11	foa/base within 0.06 (log) of the 5090 trend at 11% (112a Core i5-12400)	0.008 [0.006, 0.010]	—x— 0.06	≤ held
a-Q1-bypass-g11	bypass/base within 0.06 (log) of the 5090 trend at 11% (112a Core i5-12400)	0.001 [-0.006, 0.007]	—x— 0.06	≤ held
a-Q3-g11	best online policy ≤ 1.10 (112a Core i5-12400, g11)	1.014 [1.010, 1.017]	≤ 1.10	held
a-Q4-g11	base / Eq. (1) within [1.8, 3.5] (112a Core i5-12400, g11)	2.196	1.8 to 3.5	held (point)
a-Q5-g11	R1 and R2 read ≥ 1.3 R* (112a Core i5-12400, g11)	1.664	≥ 1.3	held (point)
a-Q3-g25	best online policy ≤ 1.10 (112a Core i5-12400, g25)	1.021 [1.018, 1.024]	≤ 1.10	held
a-Q4-g25	base / Eq. (1) within [2.5, 6.5] (112a Core i5-12400, g25)	3.284	2.5 to 6.5	held (point)
a-Q5-g25	R1 and R2 read ≥ 1.3 R* (112a Core i5-12400, g25)	2.196	≥ 1.3	held (point)
a-Q6	interaction positive at 11% (112a Core i5-12400)	6.604 [6.345, 6.890]	> 0 ms	held
b-Q1-fetch-g11	fetch/base within 0.10 (log) of the 5090 trend at 11% (112b Ryzen 7 7800X3D)	-0.003 [-0.007, 0.002]	—x— 0.10	≤ held
b-Q1-both3p-g11	both3p/base within 0.10 (log) of the 5090 trend at 11% (112b Ryzen 7 7800X3D)	0.030 [0.019, 0.040]	—x— 0.10	≤ held
b-Q1-foa-g11	foa/base within 0.06 (log) of the 5090 trend at 11% (112b Ryzen 7 7800X3D)	-0.005 [-0.008, -0.002]	—x— 0.06	≤ held
b-Q1-bypass-g11	bypass/base within 0.06 (log) of the 5090 trend at 11% (112b Ryzen 7 7800X3D)	0.007 [0.002, 0.011]	—x— 0.06	≤ held
b-Q2-pf	pf/base below 1 at 11% (112b Ryzen 7 7800X3D)	0.793 [0.789, 0.798]	< 1.00	held
b-Q2-R1	R1/base below 1 at 11% (112b Ryzen 7 7800X3D)	0.763 [0.757, 0.767]	< 1.00	held
b-Q3-g11	best online policy ≤ 1.10 (112b Ryzen 7 7800X3D, g11)	1.012 [1.007, 1.018]	≤ 1.10	held
b-Q4-g11	base / Eq. (1) within [1.8, 3.5] (112b Ryzen 7 7800X3D, g11)	2.179	1.8 to 3.5	held (point)
b-Q5-g11	R1 and R2 read ≥ 1.3 R* (112b Ryzen 7 7800X3D, g11)	1.666	≥ 1.3	held (point)
b-Q3-g25	best online policy ≤ 1.10 (112b Ryzen 7 7800X3D, g25)	1.023 [1.018, 1.028]	≤ 1.10	held
b-Q4-g25	base / Eq. (1) within [2.5, 6.5] (112b Ryzen 7 7800X3D, g25)	3.449	2.5 to 6.5	held (point)
b-Q5-g25	R1 and R2 read ≥ 1.3 R* (112b Ryzen 7 7800X3D, g25)	2.196	≥ 1.3	held (point)
b-Q6	interaction positive at 11% (112b Ryzen 7 7800X3D)	2.978 [2.820, 3.141]	> 0 ms	held
Q1-g25	fetch and both3p within 0.10 (log) of the trend at 25% on all but one host-configuration	0.000	≤ 1 miss	held (point)
a-T1-g11	Few-2R-early misses ≤ 0.95x Few-2R's (112a Core i5-12400, g11)	0.979	≤ 0.95	failed
a-T2-g11	Few-2R-early/base ≥ Few-2R/base - 0.01 (112a Core i5-12400, g11)	-0.047 [-0.050, -0.044]	≥ -0.01	failed
a-T1-g25	Few-2R-early misses ≤ 0.95x Few-2R's (112a Core i5-12400, g25)	0.949	≤ 0.95	held (point)
a-T2-g25	Few-2R-early/base ≥ Few-2R/base - 0.01 (112a Core i5-12400, g25)	-0.019 [-0.028, -0.012]	≥ -0.01	failed
a-T3	second read costs more than the late landing at 11% (112a Core i5-12400)	0.413 [0.401, 0.424]	> 0	held
a-T4	Few-2R-early/base ≤ 1.10 at 11% (112a Core i5-12400)	0.980 [0.975, 0.985]	≤ 1.10	held
a-T5	second probe within 10% of the first (112a Core i5-12400)	0.008	—x— 0.10	≤ held (point)
b-T1-g11	Few-2R-early misses ≤ 0.95x Few-2R's (112b Ryzen 7 7800X3D, g11)	0.968	≤ 0.95	failed
b-T2-g11	Few-2R-early/base ≥ Few-2R/base - 0.01 (112b Ryzen 7 7800X3D, g11)	-0.032 [-0.036, -0.028]	≥ -0.01	failed
b-T1-g25	Few-2R-early misses ≤ 0.95x Few-2R's (112b Ryzen 7 7800X3D, g25)	0.953	≤ 0.95	failed
b-T2-g25	Few-2R-early/base ≥ Few-2R/base - 0.01 (112b Ryzen 7 7800X3D, g25)	-0.020 [-0.029, -0.013]	≥ -0.01	failed

continued

Clause	Statement	Measured	Predicted	Status
b-T3	second read costs more than the late landing at 11% (112b Ryzen 7 7800X3D)	0.245 [0.233, 0.256]	> 0	held
b-T4	Few-2R-early/base ≤ 1.10 at 11% (112b Ryzen 7 7800X3D)	0.995 [0.990, 1.000]	≤ 1.10	held
b-T5	second probe within 10% of the first (112b Ryzen 7 7800X3D)	0.049	---x--- ≤ 0.10	held (point)
d-T1-g11	Few-2R-early misses $\leq 0.95x$ Few-2R's (112d Ryzen 9 7945HX, g11)	0.980	≤ 0.95	failed
d-T2-g11	Few-2R-early/base $\geq$ Few-2R/base - 0.01 (112d Ryzen 9 7945HX, g11)	-0.031 [-0.034, -0.029]	≥ -0.01	failed
d-T1-g25	Few-2R-early misses $\leq 0.95x$ Few-2R's (112d Ryzen 9 7945HX, g25)	0.949	≤ 0.95	held (point)
d-T2-g25	Few-2R-early/base $\geq$ Few-2R/base - 0.01 (112d Ryzen 9 7945HX, g25)	-0.006 [-0.014, 0.001]	≥ -0.01	held (point)
d-T3	second read costs more than the late landing at 11% (112d Ryzen 9 7945HX)	0.281 [0.272, 0.291]	> 0	held
d-T4	Few-2R-early/base ≤ 1.10 at 11% (112d Ryzen 9 7945HX)	0.997 [0.993, 1.000]	≤ 1.10	held
d-T5	second probe within 10% of the first (112d Ryzen 9 7945HX)	-0.004	---x--- ≤ 0.10	held (point)
g-T1-g11	Few-2R-early misses $\leq 0.95x$ Few-2R's (112g Ryzen 9 5950X, g11)	0.980	≤ 0.95	failed
g-T2-g11	Few-2R-early/base $\geq$ Few-2R/base - 0.01 (112g Ryzen 9 5950X, g11)	-0.053 [-0.057, -0.049]	≥ -0.01	failed
g-T1-g25	Few-2R-early misses $\leq 0.95x$ Few-2R's (112g Ryzen 9 5950X, g25)	0.950	≤ 0.95	held (point)
g-T2-g25	Few-2R-early/base $\geq$ Few-2R/base - 0.01 (112g Ryzen 9 5950X, g25)	-0.027 [-0.037, -0.019]	≥ -0.01	failed
g-T3	second read costs more than the late landing at 11% (112g Ryzen 9 5950X)	0.481 [0.468, 0.494]	> 0	held
g-T4	Few-2R-early/base ≤ 1.10 at 11% (112g Ryzen 9 5950X)	0.978 [0.972, 0.982]	≤ 1.10	held
g-T5	second probe within 10% of the first (112g Ryzen 9 5950X)	-0.003	---x--- ≤ 0.10	held (point)
d-T6	base within 5% of job 109's on this machine (112d Ryzen 9 7945HX)	0.004	---x--- ≤ 0.05	held (point)
d-T7	MIN-1R beats both single changes by 0.04 at 11% (112d Ryzen 9 7945HX)	0.210	≥ 0.04	held (point)
g-T6	base within 5% of job 109's on this machine (112g Ryzen 9 5950X)	-0.007	---x--- ≤ 0.05	held (point)
g-T7	MIN-1R beats both single changes by 0.04 at 11% (112g Ryzen 9 5950X)	0.351	≥ 0.04	held (point)
valid	at least three valid hosts over the two cards for T1-T5	4.000	≥ 3	held (point)

Table S19: Job 113 (the two-read path with the in-step fetches off), scored by machine by `scripts/job113.py` under the same rule. Totals: 14 clauses, 6 held, 6 held (point), 2 failed, 0 untested. Validity condition (not a prediction): met. Hosts b (Ryzen 9 5950X) and d (Core Ultra 9 285K); offers a and c were not listed at launch (results/113_offers_at_launch.txt). Cells gpt-oss 11% (g11, two rounds) and 25% (g25, one). Intervals are over problems; the counter clauses (H1, H2) are deterministic counts. Measured: the point estimate and, where one exists, the 95% interval (bootstrap over problems within the host; pooled clauses have none).

Clause	Statement	Measured	Predicted	Status
b-H1-g11	bypassplan0 misses $\leq 0.88x$ bypassplan's (113b Ryzen 9 5950X, g11)	0.773	≤ 0.88	held (point)
b-H1-g25	bypassplan0 misses $\leq 0.88x$ bypassplan's (113b Ryzen 9 5950X, g25)	0.718	≤ 0.88	held (point)
b-H2	bypassplan0 host reads $\leq 0.90x$ base0's at 11% (113b Ryzen 9 5950X)	0.784	≤ 0.90	held (point)
b-H3	base0/base within [0.80, 1.00] at 11% (113b Ryzen 9 5950X)	0.981 [0.974, 0.987]	0.80 to 1.00	held
b-H4	bypassplan0/base0 ≥ 1.10 at 11% (113b Ryzen 9 5950X)	1.259 [1.247, 1.272]	≥ 1.10	held
b-H5	bypass0/base0 ≥ 1.03 at 11% (113b Ryzen 9 5950X)	1.122 [1.110, 1.134]	≥ 1.03	held
b-H6	fetchplan/base $\geq$ bypassplan0/base + 0.05 at 11% (113b Ryzen 9 5950X)	0.162 [0.153, 0.171]	≥ 0.05	held
d-H1-g11	bypassplan0 misses $\leq 0.88x$ bypassplan's (113d Core Ultra 9 285K, g11)	0.786	≤ 0.88	held (point)
d-H1-g25	bypassplan0 misses $\leq 0.88x$ bypassplan's (113d Core Ultra 9 285K, g25)	0.742	≤ 0.88	held (point)
d-H2	bypassplan0 host reads $\leq 0.90x$ base0's at 11% (113d Core Ultra 9 285K)	0.784	≤ 0.90	held (point)
d-H3	base0/base within [0.80, 1.00] at 11% (113d Core Ultra 9 285K)	1.073 [1.062, 1.082]	0.80 to 1.00	failed
d-H4	bypassplan0/base0 ≥ 1.10 at 11% (113d Core Ultra 9 285K)	1.231 [1.217, 1.246]	≥ 1.10	held
d-H5	bypass0/base0 ≥ 1.03 at 11% (113d Core Ultra 9 285K)	1.095 [1.083, 1.108]	≥ 1.03	held
d-H6	fetchplan/base $\geq$ bypassplan0/base + 0.05 at 11% (113d Core Ultra 9 285K)	0.024 [0.017, 0.030]	≥ 0.05	failed
valid	at least two valid hosts (else single-machine results)	2.000	≥ 2	met

Table S20: Job 114 (the 2×2 with the in-step fetches off), scored by machine by `scripts/job114.py` under the same rule. Totals: 33 clauses, 17 held, 14 held (point), 2 failed, 0 untested. Validity condition (not a prediction): met. Valid hosts b (Ryzen 9 5950X; its probe gave an all-zero fetch table), c (Core Ultra 7 265K) and f (Ryzen 9 5950X, job 109f’s machine); a, d and g stopped at the first gate (GPUs rented before), and e (Ryzen 9 9950X3D2) failed the round check and is not scored (results/114_offers_at_*.txt). Cells gpt-oss 11% (g11, two rounds) and 25% (g25, one). Intervals are over problems; the counter clauses (H1) are deterministic counts; H5’s interval is a paired bootstrap of the interaction in ms. Measured: the point estimate and, where one exists, the 95% interval (bootstrap over problems within the host; pooled clauses have none).

Clause	Statement	Measured	Predicted	Status
b-H1a-g11	MIN-1R misses and fetches within 1% of jobs 109/112 (114b Ryzen 9 5950X, g11)	0.001	≤ 0.01 (largest relative deviation)	held (point)
b-H1a-g25	MIN-1R misses and fetches within 1% of jobs 109/112 (114b Ryzen 9 5950X, g25)	0.002	≤ 0.01 (largest relative deviation)	held (point)
b-H1b-base0	base0: no in-step fetch; misses and admissions within 1% of job 113’s at 11% (114b Ryzen 9 5950X)	0.000	≤ 0.01 and no fetch	held (point)
b-H1b-bypass0	bypass0: no in-step fetch; misses and admissions within 1% of job 113’s at 11% (114b Ryzen 9 5950X)	0.001	≤ 0.01 and no fetch	held (point)
b-H1c	foa0’s in-step fetches ≤ 0.5x foa’s at 11% (114b Ryzen 9 5950X)	1.000	≤ 0.5	failed
b-H2	bypass0/bypass ≥ 1.03 at 11% (114b Ryzen 9 5950X)	1.002 [1.001, 1.003]	≥ 1.03	failed
b-H3-g11	bypass0/base0 ≥ 1.05 at g11 (114b Ryzen 9 5950X)	1.123 [1.112, 1.135]	≥ 1.05	held
b-H3-g25	bypass0/base0 ≥ 1.15 at g25 (114b Ryzen 9 5950X)	1.350 [1.331, 1.369]	≥ 1.15	held
b-H4	foa0/base0 within [0.95, 1.10] at 11% (114b Ryzen 9 5950X)	1.068 [1.064, 1.073]	0.95 to 1.10	held
b-H5	interaction > 0 on the CPU-only path at 11% (114b Ryzen 9 5950X; ms per token)	3.175 [3.008, 3.349]	> 0	held
b-H6	base0/base within [0.92, 1.10] at 11% (114b Ryzen 9 5950X)	1.000 [0.998, 1.002]	0.92 to 1.10	held
c-H1a-g11	MIN-1R misses and fetches within 1% of jobs 109/112 (114c Core Ultra 7 265K, g11)	0.001	≤ 0.01 (largest relative deviation)	held (point)
c-H1a-g25	MIN-1R misses and fetches within 1% of jobs 109/112 (114c Core Ultra 7 265K, g25)	0.002	≤ 0.01 (largest relative deviation)	held (point)
c-H1b-base0	base0: no in-step fetch; misses and admissions within 1% of job 113’s at 11% (114c Core Ultra 7 265K)	0.000	≤ 0.01 and no fetch	held (point)
c-H1b-bypass0	bypass0: no in-step fetch; misses and admissions within 1% of job 113’s at 11% (114c Core Ultra 7 265K)	0.001	≤ 0.01 and no fetch	held (point)
c-H1c	foa0’s in-step fetches ≤ 0.5x foa’s at 11% (114c Core Ultra 7 265K)	0.350	≤ 0.5	held (point)
c-H2	bypass0/bypass ≥ 1.03 at 11% (114c Core Ultra 7 265K)	1.165 [1.159, 1.172]	≥ 1.03	held
c-H3-g11	bypass0/base0 ≥ 1.05 at g11 (114c Core Ultra 7 265K)	1.102 [1.092, 1.112]	≥ 1.05	held
c-H3-g25	bypass0/base0 ≥ 1.15 at g25 (114c Core Ultra 7 265K)	1.276 [1.260, 1.294]	≥ 1.15	held
c-H4	foa0/base0 within [0.95, 1.10] at 11% (114c Core Ultra 7 265K)	1.026 [1.021, 1.031]	0.95 to 1.10	held
c-H5	interaction > 0 on the CPU-only path at 11% (114c Core Ultra 7 265K; ms per token)	0.360 [0.257, 0.467]	> 0	held
c-H6	base0/base within [0.92, 1.10] at 11% (114c Core Ultra 7 265K)	1.056 [1.049, 1.063]	0.92 to 1.10	held
f-H1a-g11	MIN-1R misses and fetches within 1% of jobs 109/112 (114f Ryzen 9 5950X, g11)	0.001	≤ 0.01 (largest relative deviation)	held (point)
f-H1a-g25	MIN-1R misses and fetches within 1% of jobs 109/112 (114f Ryzen 9 5950X, g25)	0.002	≤ 0.01 (largest relative deviation)	held (point)
f-H1b-base0	base0: no in-step fetch; misses and admissions within 1% of job 113’s at 11% (114f Ryzen 9 5950X)	0.000	≤ 0.01 and no fetch	held (point)

continued

Clause	Statement	Measured	Predicted	Status
f-H1b-bypass0	bypass0: no in-step fetch; misses and admissions within 1% of job 113's at 11% (114f Ryzen 9 5950X)	0.001	≤ 0.01 and no fetch	held (point)
f-H1c	foa0's in-step fetches $\leq 0.5x$ foa's at 11% (114f Ryzen 9 5950X)	0.256	≤ 0.5	held (point)
f-H2	bypass0/bypass ≥ 1.03 at 11% (114f Ryzen 9 5950X)	1.115 [1.111, 1.120]	≥ 1.03	held
f-H3-g11	bypass0/base0 ≥ 1.05 at g11 (114f Ryzen 9 5950X)	1.124 [1.114, 1.134]	≥ 1.05	held
f-H3-g25	bypass0/base0 ≥ 1.15 at g25 (114f Ryzen 9 5950X)	1.357 [1.340, 1.374]	≥ 1.15	held
f-H4	foa0/base0 within [0.95, 1.10] at 11% (114f Ryzen 9 5950X)	1.069 [1.064, 1.075]	0.95 to 1.10	held
f-H5	interaction > 0 on the CPU-only path at 11% (114f Ryzen 9 5950X; ms per token)	3.101 [2.928, 3.282]	> 0	held
f-H6	base0/base within [0.92, 1.10] at 11% (114f Ryzen 9 5950X)	1.000 [0.993, 1.008]	0.92 to 1.10	held
valid	at least two valid hosts (else single-machine results)	3.000	≥ 2	met

Table S21: Job 115 (the 2×2 with the in-step fetches off on more machines, with idle probe runs), scored by machine by `scripts/job114.py` under the same rule. Totals: 28 clauses, 12 held, 15 held (point), 1 failed, 0 untested. Validity condition (not a prediction): met. Hosts and gates as listed in Appendix D (job 115). Cells gpt-oss 11% (g11, two rounds) and 25% (g25, one). Intervals are over problems; H1 and H7 are deterministic counts and probe readings; H8 is pooled over the valid machines of jobs 114 and 115. Measured: the point estimate and, where one exists, the 95% interval (bootstrap over problems within the host; pooled clauses have none).

Clause	Statement	Measured	Predicted	Status
b-H1a-g11	MIN-1R misses and fetches within 1% of jobs 109-114 (115b Core i9-14900K, g11)	0.001	≤ 0.01 (largest relative deviation)	held (point)
b-H1a-g25	MIN-1R misses and fetches within 1% of jobs 109-114 (115b Core i9-14900K, g25)	0.002	≤ 0.01 (largest relative deviation)	held (point)
b-H1b-base0	base0: no in-step fetch; misses and admissions within 1% of job 114's at 11% (115b Core i9-14900K)	0.000	≤ 0.01 and no fetch	held (point)
b-H1b-bypass0	bypass0: no in-step fetch; misses and admissions within 1% of job 114's at 11% (115b Core i9-14900K)	0.001	≤ 0.01 and no fetch	held (point)
b-H1c	foa0's in-step fetches ≤ 0.5x foa's at 11% (115b Core i9-14900K)	0.394	≤ 0.5	held (point)
b-H2	bypass0/bypass ≥ 1.05 at 11% (115b Core i9-14900K)	1.100 [1.089, 1.110]	≥ 1.05	held
b-H3-g11	bypass0/base0 ≥ 1.07 at g11 (115b Core i9-14900K)	1.159 [1.133, 1.185]	≥ 1.07	held
b-H3-g25	bypass0/base0 ≥ 1.20 at g25 (115b Core i9-14900K)	1.238 [1.227, 1.250]	≥ 1.20	held
b-H4	foa0/base0 within [0.98, 1.10] at 11% (115b Core i9-14900K)	1.040 [1.016, 1.065]	0.98 to 1.10	held
b-H5	interaction > 0 on the CPU-only path at 11% (115b Core i9-14900K; ms per token)	-0.310 [-0.722, 0.113]	> 0	failed
b-H6	base0/base within [0.92, 1.10] at 11% (115b Core i9-14900K)	0.976 [0.958, 0.994]	0.92 to 1.10	held
b-H7a	best idle probe reading within [0.95, 1.15] of the first probe's (115b Core i9-14900K)	1.010	0.95 to 1.15	held (point)
b-H7b	the three idle runs' highest readings within 5% of each other (115b Core i9-14900K)	1.005	≤ 1.05	held (point)
c-H1a-g11	MIN-1R misses and fetches within 1% of jobs 109-114 (115c Core Ultra 9 285K, g11)	0.001	≤ 0.01 (largest relative deviation)	held (point)
c-H1a-g25	MIN-1R misses and fetches within 1% of jobs 109-114 (115c Core Ultra 9 285K, g25)	0.002	≤ 0.01 (largest relative deviation)	held (point)
c-H1b-base0	base0: no in-step fetch; misses and admissions within 1% of job 114's at 11% (115c Core Ultra 9 285K)	0.000	≤ 0.01 and no fetch	held (point)
c-H1b-bypass0	bypass0: no in-step fetch; misses and admissions within 1% of job 114's at 11% (115c Core Ultra 9 285K)	0.001	≤ 0.01 and no fetch	held (point)
c-H1c	foa0's in-step fetches ≤ 0.5x foa's at 11% (115c Core Ultra 9 285K)	0.350	≤ 0.5	held (point)
c-H2	bypass0/bypass ≥ 1.05 at 11% (115c Core Ultra 9 285K)	1.183 [1.177, 1.189]	≥ 1.05	held
c-H3-g11	bypass0/base0 ≥ 1.07 at g11 (115c Core Ultra 9 285K)	1.092 [1.083, 1.101]	≥ 1.07	held
c-H3-g25	bypass0/base0 ≥ 1.20 at g25 (115c Core Ultra 9 285K)	1.270 [1.257, 1.282]	≥ 1.20	held
c-H4	foa0/base0 within [0.98, 1.10] at 11% (115c Core Ultra 9 285K)	1.016 [1.013, 1.018]	0.98 to 1.10	held
c-H5	interaction > 0 on the CPU-only path at 11% (115c Core Ultra 9 285K; ms per token)	0.259 [0.207, 0.314]	> 0	held
c-H6	base0/base within [0.92, 1.10] at 11% (115c Core Ultra 9 285K)	1.076 [1.071, 1.081]	0.92 to 1.10	held
c-H7a	best idle probe reading within [0.95, 1.15] of the first probe's (115c Core Ultra 9 285K)	0.998	0.95 to 1.15	held (point)
c-H7b	the three idle runs' highest readings within 5% of each other (115c Core Ultra 9 285K)	1.002	≤ 1.05	held (point)
H8a	mean share of the gap closed by MIN's set alone, CPU only, 11%, over 5 machines ≥ 0.10	0.190 [0.135, 0.246]	≥ 0.10	held
H8b	its 95% t-interval over machines above zero (lower end)	0.135	> 0	held (point)
valid	at least two valid hosts (else single-machine results)	2.000	≥ 2	met

References

- Chris Fawcett and Holger H. Hoos. Analysing differences between algorithm configurations through ablation. *Journal of Heuristics*, 22:431–458, 2016.
- Raj Jain. *The Art of Computer Systems Performance Analysis*. Wiley, 1991.
- Ye Li, Jian Tan, Bin Wu, Xiao He, and Feifei Li. ShapleyIQ: Influence quantification by Shapley values for performance debugging of microservices. In *Proceedings of the 28th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, Volume 4, pages 287–323, 2024. doi: 10.1145/3623278.3624771.